LIGHTNING
AUTO SERVICE

ADAS CALIBRATION GUIDEBOOK

A COLLISION-REPAIR, GLASS, AND TECHNICIAN REFERENCE

STATIC AND DYNAMIC ADAS CALIBRATIONS

OEM PROGRAMMING GUIDELINES

DIAGNOSTICS AND TOOLS FOR SAFER REPAIRS

2026 U.S. Edition

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Produced and published by
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October 2026

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Publication Notice

Purpose & Scope

This guide is intended to support damage analysis, estimating, repair planning, ADAS calibration preparation, quality control, and documentation. It is provided for educational and informational purposes only and does not replace current OEM service procedures specific to the vehicle's VIN, equipment level, software version, market, and repair event.

Terminology

- **Required:** An operation explicitly directed by the current VIN-specific OEM service procedure or applicable position statement.
- **Investigate**: A repair condition or event that may trigger additional inspection or calibration requirements and warrants verification against current OEM procedures.
- **Not Complete:** A required diagnostic scan, calibration setup, environmental or driving condition, documentation element, or verification criterion remains outstanding.

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Produced and prepared for online publication by Lightning Auto Service
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Important Safety Disclaimer

OEM Calibration Requirements

ADAS calibration requirements vary by manufacturer, model, model year, vehicle configuration, and repair procedure. Operations such as bumper-cover removal may require calibration on certain vehicles but not others. The replacement, removal, movement, or disturbance of radar sensors, cameras, brackets, grille emblems, absorbers, reinforcements, wiring, or mounting structures may introduce additional calibration requirements.

Always consult the latest applicable OEM procedures before finalizing repairs or returning a vehicle to service.

No DTCs Does Not Mean No Calibration Is Required

The absence of Diagnostic Trouble Codes (DTCs), dashboard warning lights, or apparent system malfunctions does not establish that ADAS components are properly aligned or calibrated.

ADAS cameras and radar sensors may experience physical misalignment following collisions, repairs, component removal, or certain road impacts without generating a fault code. A diagnostic scan alone cannot establish whether all OEM-required calibrations have been performed.

Calibration requirements are determined by the vehicle's specific equipment, repair operations, and current OEM procedures—not solely by the presence or absence of warning lights or DTCs. A clean scan is not a substitute for an OEM-required calibration.

Vehicle Safety & Release

Vehicles should not be considered ready for release until all applicable OEM-required safety inspections, calibrations, and post-repair verification procedures have been satisfactorily completed.

Failure to identify or perform required ADAS calibrations may compromise the intended operation of critical safety features, including automatic emergency braking, lane departure warning, and adaptive cruise control.

This publication does not constitute authorization to deviate from manufacturer repair requirements, applicable safety standards, or established vehicle release procedures.

Curriculum Disclaimer

This curriculum is designed for individuals with little to no prior experience in Advanced Driver Assistance Systems (ADAS), providing a structured introduction to the principles, technologies, and procedures essential to safe and accurate vehicle repair.

Lessons should be completed in their intended sequence, as each builds upon foundational knowledge necessary to understand the more advanced OEM procedures, diagnostic workflows, and calibration references presented in later chapters. Each lesson introduces a core concept, explains its relevance to real-world repair operations, provides practical examples, and concludes with a knowledge check to reinforce understanding and encourage critical thinking.

This guidebook is an educational resource and does not replace current OEM service information, manufacturer-approved procedures, required certifications, or hands-on technical training. Technicians must always verify vehicle-specific requirements using the latest applicable OEM documentation and follow all prescribed safety precautions, equipment specifications, and calibration procedures before performing service.

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PART I

ADAS FOUNDATIONS

Objective: Establish a strong foundation in ADAS principles, vehicle safety systems, and OEM repair requirements before introducing scan-tool procedures. Technicians will develop the critical thinking skills needed to recognize calibration triggers, evaluate vehicle conditions, understand sensor functionality, and make informed repair decisions. The goal is to prioritize safety, accuracy, and sound technical judgment over simply memorizing diagnostic menus or following automated prompts.

1. What ADAS Means

Advanced Driver Assistance Systems (ADAS) is the collection of electronic systems that help a driver perceive hazards, maintain position or speed, park, and sometimes avoid or reduce a collision. **Assistance is not autonomous driving.**

ADAS supports the driver. It does not replace the driver.

The ADAS Repair Workflow

A STEP-BY-STEP APPROACH

1 Identify the function

Determine which ADAS features could be affected by the repair.

→

2 Locate its sensors

Find the sensors, modules, brackets, and wiring related to those functions

→

3 Find OEM Procedure

Review the VIN-specific OEM repair procedure for required steps.

→

4 Repair Prerequisites

Complete all required repairs, alignments, software updates, or initializations before calibration.

→

5 Verify operation

Perform scans, calibrations, and functional tests. Confirm the system operates as intended and document the results.

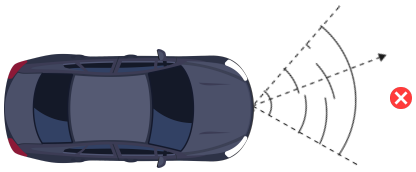
Why This Matters in a Repair Facility

A repair can restore appearance while leaving a safety function unavailable or geometricallly inaccurate. Everyone touching the vehicle must know that sensors, structure, chassis, and software operate as one system.

Practical Shop Example

IT LOOKS REPAIRED, BUT IS IT?

The front bumper **looks** repaired, but filler sits in front of the radar. The car has no warning lights. ACC may still track objects incorrectly because the signal path changed.



Even small changes to the sensor's position or the material in front of it can affect how it "sees" the road.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Assuming ADAS means only a windshield camera.
- Promising that calibration makes a vehicle self-driving.

2. Driver Assistance vs. Automation

Most passenger vehicles with ADAS are Level 1 or Level 2 systems. These are assistance features: the driver remains resposible for steering, braking, and monitoring the vehicle at all times, even when the system controls speed or lane position.

ADAS assists the driver.

The driver stays in control.

The Diagnostic Workflow

UNDERSTAND, ASSESS, VERIFY, DOCUMENT

1 Name the feature

Identify the specific driver assistance features (e.g. ACC, lane keep assist).

2 State its limits

Understand what the system can and cannot do, and under what conditions it operates.

3 Identify driver role

Confirm the driver remains responsible for monitoring the vehicle and taking over when required.

4 Check warnings

Look for warning messages, indicator lights, or DTCs that may indicate a system fault or limitation.

5 Document status

Record what was tested, the system's operating status, and any relevant notes on the repair invoice.

Why This Matters in a Repair Facility

Clear language prevents unsafe customer expectations and helps staff distinguish a convenience features from the sensors and actuators that support it.

Always communicate that ADAS features are designed to assist, not replace, attentive driving.

Practical Shop Example

LANE CENTERING + ADAPTIVE CRUISE CONTROL (ACC)

Lane centering and ACC can operate together, but the driver still supervises. A repair invoice should name the calibrated systems, instead of stating the vehicle is autonomous.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
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Common Pitfalls to Avoid

- Calling Level 2 hands-free in call conditions.
- Testing system limits on public roads.

3. Inputs, Decisions, & Outputs

Every ADAS function follows a chain: sensors collect inputs, controllers interpret them, software decides what to do, and actuators or displays produce an output. Understanding this flow helps you diagnose issues logically and avoid misdirected repairs.

Sensors see. Software thinks. Actuators act. Together, they keep drivers safer.

The ADAS Signal Flow

FROM THE ENVIRONMENT TO A VEHICLE RESPONSE

Sense environment

Sensors collect information about the car's surroundings (e.g. objects, lane markings, speed).

Transmit data

Inputs are sent to the vehicle's network and control modules via wired or wireless communication.

Build world model

The controller and software process the data, filter noise, and create a real-time model of the car's environment.

Decide response

Software compares the model to programmed logic and conditions to determine the appropriate action.

Warn or actuate

The system produces an output through a display, warning (audible/visual), or controlling an actuator like brakes.

Why This Matters in a Repair Facility

This model helps a beginning diagnose logically. A warning may originate at a sensor, a network message, controller software, vehicle geometry or an actuator. Understanding the full chain prevents unnecessary part replacements and ensures the real issue is identified.

Practical Shop Example

AUTOMATIC EMERGENCY BRAKING (AEB)

AEB needs object input, host-vehicle speed, a collision calculation, and a function brake system. All parts of the chain must work correctly.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
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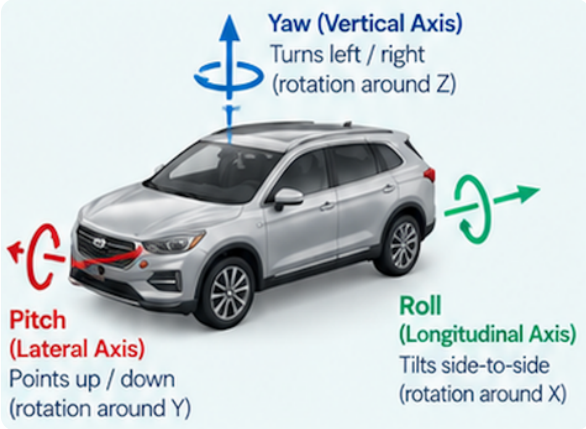
Common Pitfalls to Avoid

- Replacing a component named in a message without diagnosis.
- Confusing a sensor calibration with a brake-system repair.

4. The Vehicle Coordinate System

Vehicle systems describe locations and angles relative to longitudinal, lateral and vertical axes. ***YAW*** turns left/right, ***PITCH*** points up/down, and ***ROLL*** tilts side-to-side. Understanding this coordinate system is essentials for proper sensor alignment and calibration.



A Safe Workflow

UNDERSTAND AND VERIFY THE VEHICLE'S REFERENCE POINTS

Establish datum

Identify the car's reference points (e.g. centerline, OEM-defined datums).

Measure centerline

Determine the vehicle centerline and verify measurement points using OEM procedures.

Control yaw

Adjust and align for yaw (left/right) so the sensor is parallel to the centerline.

Control pitch

Verify pitch (up/down) angle to ensure the sensor is at the correct height and aimed at the proper distance.

Confirm result

Check measurements, complete calibration or functional test, and document the results per OEM procedures.

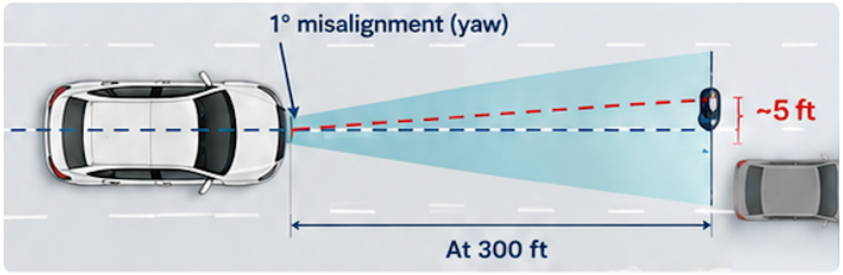
Why This Matters in a Repair Facility

A tiny angular error grows into a larger lateral error with distance. Calibration aligns the sensor view with the vehicle path and OEM datum, ensuring systems detect and respond accurately.

Just 1° of misalignment can result in several feet of error down the road.

Practical Shop Example

AIMING ERROR AT HIGHWAY DISTANCE



A radar aimed just 1° to the left can project a target several feet off-center at highway distance and may select the wrong lane. Proper alignment to the vehicle coordinate system is critical.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
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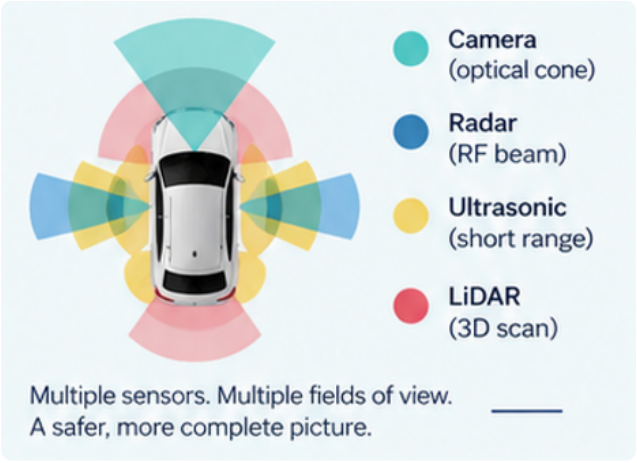
Common Pitfalls to Avoid

- Eyeballing sensor aim.
- Using body panels as datum without procedure authorization.

5. Fields of View & Sensing Zones

A field of view is the region a sensor can detect. Cameras use optical cones, radar uses radio-frequency beams, ultrasonic sensors cover short lobes, and LiDAR scans a 3-D zone. Understanding where sensors see (and what they may miss) is critical for proper diagnosis, repairs, and calibration.



Multiple sensors. Multiple fields of view.
A safer, more complete picture.

The Inspection Workflow

FROM SENSOR TO CALIBRATION

1 Locate sensor

Identify the sensor type and its mounting location.

2 Visualize its zone

Understand the sensor's field of view and sensing zone (e.g. camera cone, radar beam).

3 Inspect full path

Check for obstructions, damage, or changes anywhere within the sensor's view or beam path.

4 Check material limit

Verify that materials and objects are outside the sensor's detectable zone (e.g. no metal, repairs, or add-ons within limits)

5 Research calibration

Confirm calibration requirements and follow OEM procedures.

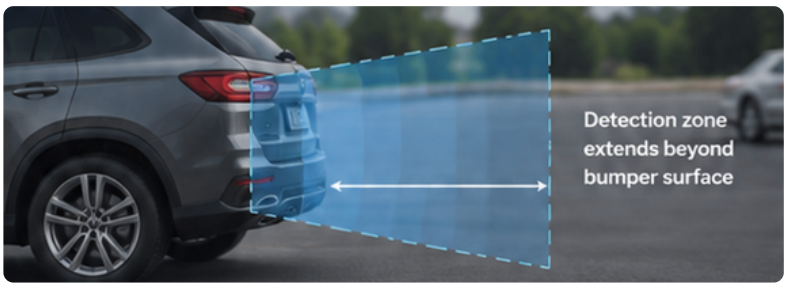
Why This Matters in a Repair Facility

Repairs outside the visible sensor housing can still change the transmission path or mounting relationship. Coverage gaps and overlaps matter, and even small changes can affect how a systems detects and responds.

Always inspect the entire field of view, not just the sensor face.

Practical Shop Example

REAR CORNER RADAR



A corner radar behind a rear bumper monitors beyond the bumper surface. Filler place several inches away may still sit inside its beam and affect performance.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

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Common Pitfalls to Avoid

- Inspecting only the sensor face.
- Ignoring bumper corners because the module is farther inward.

6. Accuracy, Precision, & Tolerance

Accuracy is closeness to the correct value. Precision is repeatability. Tolerance is the allowed range of variation. A repeatable setup can still be consistently incorrect, which is why both accuracy and precision, within OEM tolerances, are required.

Accuracy
Closeness to the correct value.

Precision
Repeatability of measurements.

Tolerance
The allowed range of variation.

The Measurement Workflow

FOLLOW OEM PROCEDURES FOR CONSISTENT, ACCURATE RESULTS

Read specification

Review OEM measurements, angles, levels, and tolerances for the specific vehicle and system.

Select tool

Use the correct calibration equipment and ensure it is calibrated per the manufacturer's recommendations.

Measure twice

Take multiple measurements to veritfy consistency and reduce the chance of error.

Record actual value

Document the measured value, conditions, and any relevant notes (e.g. ride height, trim level, setup).

Compare tolerance

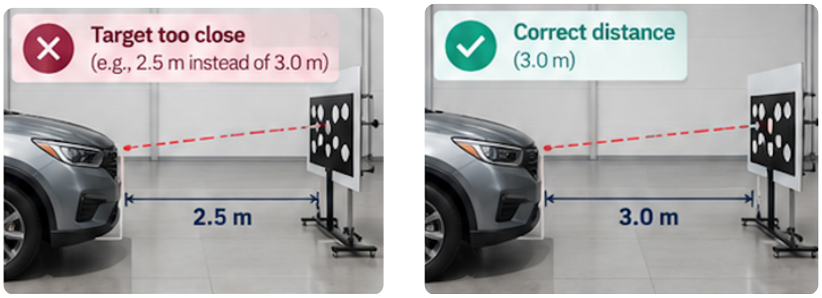
Confirm the measurement falls within the OEM-specified tolerance. If not, troubleshoot and adjust.

Why This Matters in a Repair Facility

Calibration work depends on measured distances, angles, levels, and environmental limits. "Close enough" is not a documented specification. Both accuracy and precision are essential to ensure systems perform as intended, and vehicles are repaired safely.

Practical Shop Example

INCORRECT DISTANCE, CORRECT SETUP – STILL INACCURATE



A target placed at the same wrong distance on every vehicle may be precise but inaccurate. Both equipment checks and OEM dimensions are required to ensure the setup is within tolerance.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

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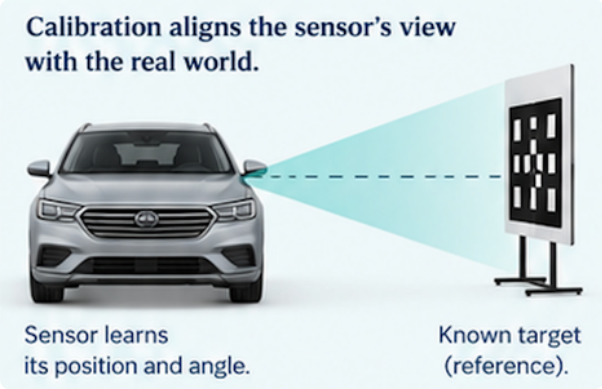
Common Pitfalls to Avoid

- Rounding measurements before checking tolerances.
- Believing repeatability alone proves accuracy.

7. What Calibration Actually Does

Calibration teaches or verifies the relationship between a sensor and the vehicle, target, road, or known reference around it. It may set an axis, zero-point, camera model, or learned value so the system can accurately perceive the world and respond correctly.



The Calibration Workflow

FROM PREPARATION TO VERIFICATION

Correct hardware

Confirm all mechanical and electrical repairs are complete. The sensor must be properly mounted, undamaged, and communicating correctly.

→

Verify Prerequisites

Check OEM requirements: software version, tire pressure, vehicle level, fuel level, target setup, environmental conditions.

→

Run routine

Follow the OEM procedure to perform the calibration, or teach-in routine.

→

Confirm completion

Verify the routine completed successfully. Check for pass/fail status and review any values or results provided by the tool.

→

Post-scan

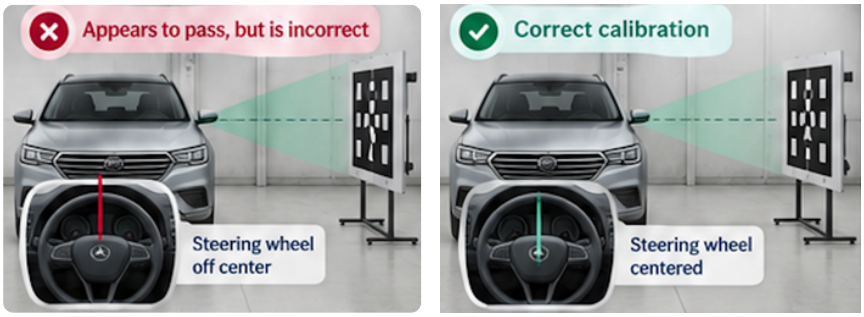
Scan for DTCs and confirm no related faults are present. Road test if required and verify the systems operates are intended.

Why This Matters in a Repair Facility

A calibration routine does not repair a bent bracket, incorrect alignment, wrong glass, low voltage, bad wiring, or incompatible software. It only adjusts the sensor's learned relationship to the vehicle and environment. Proper preparation prevents wasted time and ensures accurate results.

Practical Shop Example

CAMERA AIM PASSES BUT THE STEERING WHEEL IS OFF-CENTER



A camera aim may complete even though the steering wheel is off-center. The apparent pass must be questioned because the vehicle path reference may be wrong.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
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Common Pitfalls to Avoid

- Using calibration as a diagnostic shortcut.
- Treating "routine completed" as proof all prerequisites were correct.

8. Static Calibration

Static calibration uses a stationary vehicle and precisely-positioned targets, reflectors, patterns, or boards. The system learns the relationship between the sensor and a known reference to ensure accurate detection in the real world.

The Static Calibration Flow

FOLLOW THE STEPS FOR A SUCCESSFUL CALIBRATION

Prepare bay

Set up a clean, level calibration area with proper lighting and enough space for the required target distance.

Level vehicle

Ensure the vehicle is at the correct ride height, on a level surface, with proper tire pressure and centered.

Place targets

Position the correct target, reflector, pattern, or board at the specified distance, height and angle per OEM procedure.

Run routine

Follow the OEM procedure to perform the static calibration. Verify all conditions are met before starting.

Save evidence

Record and save calibration results, screenshots, or printouts per OEM requirements.

Why This Matters in a Repair Facility

The target represents known geometry. Floor level, centerline, distance, height, angle, lightning, and target identity directly affect the result. Even small setup errors can cause a calibration to fail or teach the wrong geometry.

If the setup is wrong, the sensor will learn the wrong information.

Practical Shop Example

CORRECT TARGET APPEARANCE, WRONG SCALE

A camera target may appear to be correct, but printed at the wrong scale. The routine may fail, or learn the incorrect geometry.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
- Repair trigger
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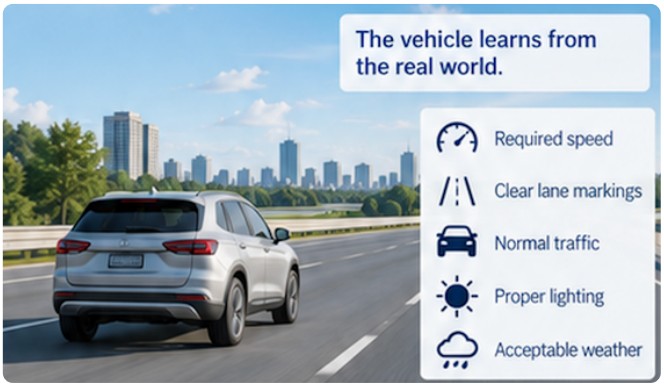
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Scaling targets without permission.
- Moving the vehicle after measurements, but before calibration.

9. Dynamic Calibration

Dynamic calibration learns while the vehicle is driven under required road, speed, lane, traffic, lighting, and weather conditions. The system uses real-world data to refine the sensor's understanding of the vehicle's movement and environment.



The Dynamic Calibration Workflow

DRIVE. MEET CONDITIONS. VERIFY. DOCUMENT.

Prepare vehicle
Verify all repairs are complete, DTCs are cleared, tires are correct and systems are ready for calibration.

→ **Choose route**
Select a route that meets OEM requirements (road type, speed range, lane markings, elevation changes).

→ **Meet conditions**
Drive the route under the required conditions (speed, steady lanes, lighting weather). Follow all OEM criteria for duration and distance.

→ **Monitor status**
Check the scan tool for calibration process. Ensure no faults are set and all conditions are being met.

→ **Document result**
Confirm completion in the scan tool, save reports or screenshots, and clear any test DTCs per OEM procedure.

Why This Matters in a Repair Facility

Completion depends on both the vehicle and the environment. A long drive doesn't prove the conditions or completion criteria were met. Following OEM requirements ensures the system learns correctly and operates as intended.

Not all miles count. The car must see the conditions the OEM specifies.

Practical Shop Example

ROUTE CONDITIONS NOT MET – INCOMPLETE CALIBRATION

A routine requires clear lane marketing at 40-60 mph. Urban traffic prevents stable speed, so the tech documents an incomplete attempt rather than a pass.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

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Common Pitfalls to Avoid

- Driving indefinitely without reading prerequisites.
- Operating a scan tool unsafely while driving.

10. Combined Calibration

Some vehicles require both static and dynamic operations or several sensor aims in a prescribed sequence. Combined calibration maps all required routines, confirms the correct order and verifies that the system is fully learned and communicating as designed.

The Combined Calibration Flow

FOLLOW THE STEPS IN THE CORRECT ORDER

Map all routines

Identify every required calibration, aim, or learn procedure for the vehicle and its systems (cameras, radar, ultrasonic).

Confirm sequence

Follow OEM prescribed order. Some routines must be completed before others for the system to learn correctly.

Complete static

Perform all require static calibrations (camera aim, radar aim, target placement). Verify successful completion.

Complete dynamic

Drive the vehicle under the required conditions (speed, traffic, lighting, weather). Ensure all dynamic routines are successfully completed.

Verify network

Check for DTCs, confirm all systems show learned status and are communicating on the network. Save and document results.

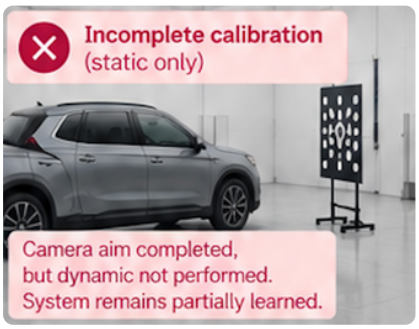
Why This Matters in a Repair Facility

Completion depends on both the vehicle and the environment. A long drive doesn't prove the conditions or completion criteria were met. Following OEM requirements ensures the system learns correctly and operates as intended.

Follow the OEM sequence and verify every system is complete.

Practical Shop Example

ONLY STATIC AIM WAS COMPLETED



An OEM procedure directs SAS initialization, static camera aim and dynamic verification. Reporting only the static aim is incomplete.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Assuming one success screen covers every required routine.
- Changing alignment between calibration steps.

11. Initialization, Relearn, & Reset

Initialization established a baseline or zero point. A relearn updates adaptive values. A reset clears learned information so a new relationship can be established. The exact routine and purpose should match the OEM procedure.

Different Purpose. Same Goal.

Initialization
Establishes a baseline or zero point.

Relearn
Updates adaptive values over time.

Reset
Clears learned information so to establish a new relationship.

The Full Workflow

FOLLOW THE OEM PROCEDURE STEP-BY-STEP

1. **Read procedure**
   Review the OEM instructions to understand whether initialization, reset, or relearn is required, and why.
2. **Select exact routine**
   Choose the specific routine for the vehicle and system. Do not use a generic or similar procedure from another model.
3. **Meet conditions**
   Ensure all required conditions are met (initiation state, vehicle level, steering position, battery voltage, temperature).
4. **Confirm value**
   Verify the system reports a successful initialization, relearn, or reset. Check for expected values or status messages.
5. **Record result**
   Save a scan report, screenshot, or printout showing the completed routine and any relevant data.

Why This Matters in a Repair Facility

These terms are often billed or described incorrectly. The exact routine or purpose must match the OEM procedure. Using the wrong routine can leave the system improperly set up, leading to inaccurate operation and repeat repairs.

The right routine for the right reason: initialization, relearn, and reset, are not interchangeable.

Practical Shop Example

CLEARING CODES DOES NOT CREATE A ZERO POINT

A steering angle zero-point procedure is required before camera aiming. Clearing DTCs alone does not create the zero-point.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Calling every routine a calibration.
- Clearing codes instead of performing the specified reset.

12. Programming & Coding

Programming installs software in a module. Coding, or configuration, tells the module which vehicle and options it belongs to. Calibration then establishes physical learned geometry. These are separate operations with different purposes and failure causes.

Programming Installs software.

Coding Configures vehicle and options (variant coding).

Calibration Teaches physical geometry.

All three may be required for a complete repair.

The Programming and Coding Flow

FOLLOW THE STEPS IN THE CORRECT ORDER

Install component

Install the new or replacement component and verify all mechanical and electrical connections are complete and secure.

Stabilize voltage

Use an approved battery support unit to maintain proper system voltage during programming and coding.

Program software

Install the latest OEM software for the vehicle. Follow the OEM procedure and ensure the programming completes successfully.

Configure variant

Code or configure the module for the correct vehicle, options and equipment. Verify the settings match the VIN and build configuration.

Calibrate

Perform the required calibration (static, dynamic or combined) to establish the physical learned geometry. Verify successful completion.

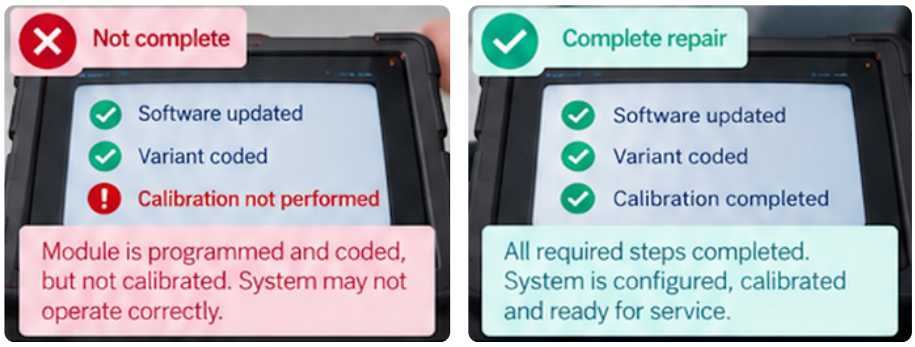
Why This Matters in a Repair Facility

A replacement sensor may communicate well but remain improperly configured. Programming, coding, and calibration are separate operations with separate failure causes. Skipping a step or using the wrong software or variant can lead to inaccurate operation, warning lights and repeat repairs.

Programming and coding must match both the vehicle and its options.

Practical Shop Example

PROGRAMMED AND CODED, BUT STILL REQUIRES CALIBRATION



A radar is installed, programmed and variant-coded, but still requires physical aiming. The invoice identifies each completed operation.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

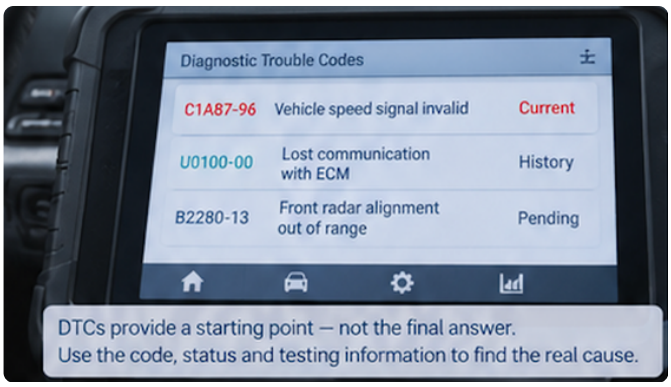
- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Using programming and calibration interchangeably.
- Programming without approved battery support.

13. Diagnostic Trouble Codes

A DTC is a module-recorded fault code with status and context. It is not a complete diagnosis and is not automatic proof that a component needs replacement.



The DTC Workflow

FOLLOW A LOGICAL PROCESS

Full scan

Scan all modules and record all DTCs, including current, history, pending and communication codes.

Record status

Note the code number, description, status (current, history, pending) and freeze-frame data.

Read description

Review the OEM description and diagnostic information. Identify which systems and components could be affected.

Follow pinpoint test

Use the OEM troubleshooting steps, wiring checks, measurements and tests to find the root cause.

Verify repair

Clear codes, perform a rescan and confirm the code does not return. Verify all related systems operate correctly.

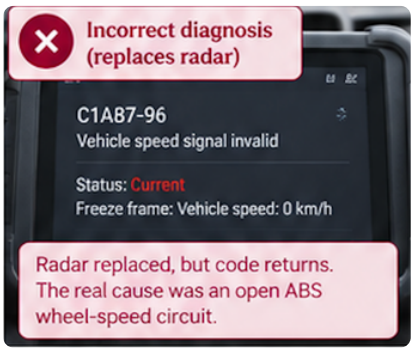
Why This Matters in a Repair Facility

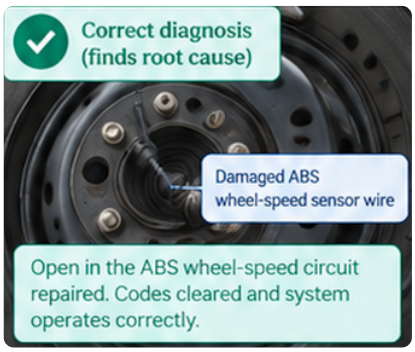
Current, history, pending and communication codes require different interpretation. ADAS faults may originate in ABS, EPS, power, network or chassis modules. Misdiagnosis can lead to unnecessary parts replacement, repeat repairs and comebacks.

Use the code, status, and testing information to uncover the real cause.

Practical Shop Example

SAME CODE, DIFFERENT CAUSES





A radar reports invalid vehicle-speed data because an ABS wheel-speed circuit is open. Replacing the radar would not solve the root cause.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Parts-swapping from a code description.
- Ignoring history and freeze-frame information.

14. Warning Lamps & Messages

The driver display communicates selected conditions, but many faults do not illuminate a dedicated lamp. Messages may be delayed, conditional, or generated by another module.

The Warning Lamp Workflow

OBSERVE. SCAN. COMPARE. REPAIR. RECHECK.

Observe cluster
Note any warning lamps or messages. Record the exact text and any conditions (ignition state, only while driving, etc.).

Scan modules
Perform a full system scan. Check for current, history and pending DCs in all related modules (camera, radar, ABS).

Compare symptoms
Match the warning lamp or message to DTCs, freeze-frame data and operating conditions. Consider related systems.

Complete repair
Follow OEM diagnostics and repairs. Clear codes only after the root cause has been corrected.

Recheck display
Verify the warning lamp or message is no longer present. Perform a post-scan and, when required, a road test to confirm the fix.

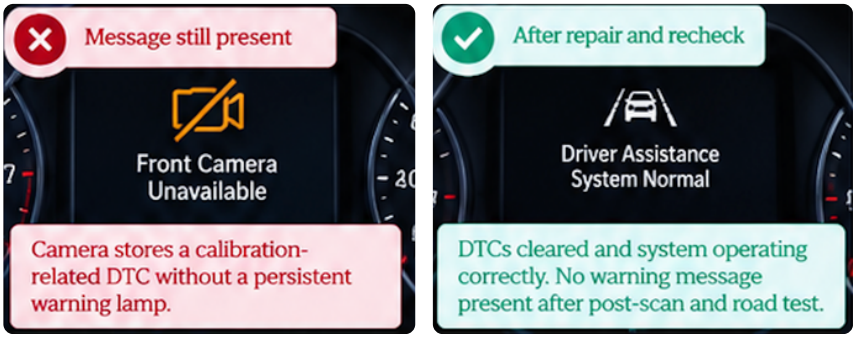
Why This Matters in a Repair Facility

A clean dashboard cannot replace a pre-scan, post-scan or OEM-required inspection. Some faults do not illuminate a lamp, and messages may be delayed, conditional or generated by another module. Always verify with scan data.

No light does not mean no problem. Confirm with a full system scan and follow OEM procedures.

Practical Shop Example

MESSAGE PRESENT, DTC STORED



A camera stores a calibration-related DTC without a persistent warning. The post-scan identifies it before delivery.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Using no warning lamp as proof of no fault.
- Photographing only the cluster instead of saving scan data.

15. Pre-Scan, In-Process Scan, & Post-Scan

A pre-scan establishes the initial electronic condition. An in-process scan supports diagnosis or verifies assembly during repair. A post-scan checks final module status and confirms the repair is complete.

Each scan answers a different question. Keep all three scan reports with the repair record.

The Scanning Workflow

EACH SCAN SERVES A PURPOSE

1. **Capture pre-state**
   Perform a full system scan before repairs. Record all DTCs (current, history, pending) and save the report.
2. **Diagnose**
   Use scan data to identify related faults across all modules. Follow OEM diagnostics and testing procedures.
3. **Verify during repair**
   Run an in-process scan after key steps (e.g., after installation, before calibration) to confirm communication and module status.
4. **Complete calibration**
   Perform all required calibrations, coding or initialization. Confirm no related DTCs remain and systems show learned status as expected.
5. **Final post-scan**
   Run a final full system scan after calibration and reassembly. Confirm all DTCs are cleared and no new codes are present. Save the report.

Why This Matters in a Repair Facility

Each scan answers a different question. A clean post-scan does not prove the vehicle was not faulty before the repair. Accurate naming (pre-scan, in-process scan, post-scan prevents insurer confusion and creates a clear, chronological record

Document the vehicle's condition at every stage: scan early, scan often, and scan last.

Practical Shop Example

ONE VEHICLE, THREE SCANS, THE COMPLETE STORY

Pre-scan (before repair)	In-process scan (during repair)	Post-scan (after repair)
DTCs (8)	DTCs (2)	DTCs (0)
C1A87-96 Camera calibration	U0100-00 Lost communication with ECM	No diagnostic trouble codes found.
U0100-00 Lost communication with ECM	C1A87-96 Camera calibration	
B2280-13 Front radar alignment		
Pre-scan shows multiple DTCs before any work.	After installation, communication is restored but calibration is still required.	Final post-scan confirms repair is complete. All systems operate correctly.

After radar installation, an in-process scan confirms communication before aiming. A final post-scan occurs after calibration and reassembly.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

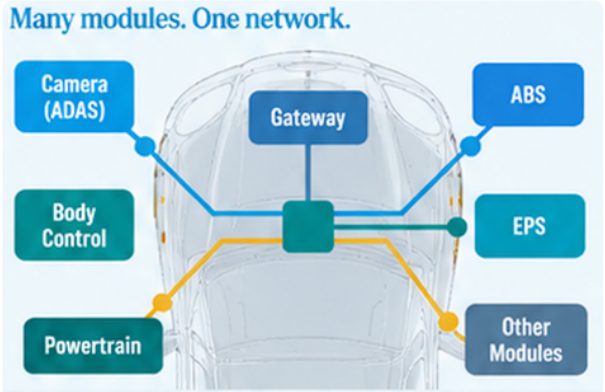
- Vehicle identity, sytem identification
- Repair trigger
- Procedure reference
- Scan results
- Prerequisite measurements
- Completion result and exceptions

Common Pitfalls to Avoid

- Renaming an in-process scan as a pre-scan.
- Stopping after codes clear without completing calibrations.

16. Vehicle Networks & Communication

Modules exchange data over networks such as the Controller Area Network (CAN). A sensor can depend on messages from ABS, EPS, body, powertrain and gateway modules. Reliable communication is required for accurate diagnosis and calibration.



The Network Diagnostic Flow

FOLLOW THE STEPS TO FIND AND FIX COMMUNICATION ISSUES

Check voltage

Verify battery voltage and charging system are within specification. Low voltage can cause communication faults.

→

Full network scan

Scan all modules for DTCs, including communication, U-codes and message-related faults.

→

Identify missing module

Check for modules that do not respond. Inspect wiring, connectors, termination and gateway status.

→

Test power/ground/ CAN

Verify power, ground and CAN signal integrity (using wiring diagrams and measurements). Repair wiring, connectors or network faults.

→

Repair, then calibrate

Restore communication and clear DTCs. Perform required calibrations and confirm all modules communicate correctly.

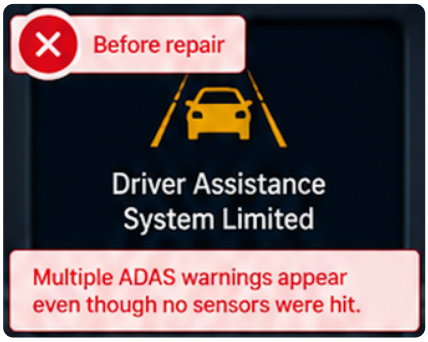
Why This Matters in a Repair Facility

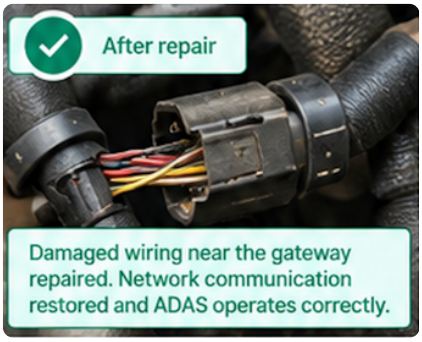
A lost-communication DTC may reflect power, ground, wiring, connector, termination, gateway or module issues. Calibration requires reliable communication across all related systems.

No communication means no confidence. Fix network issues before calibration.

Practical Shop Example

A WIRING FAULT CAUSES MULTIPLE ADAS WARNINGS





A rideshare-device removal damaged wiring near a gateway. Multiple ADAS warnings appear even though no sensors were hit.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

- Battery voltage and charging system test
- Full network scan (pre- and post-repair)
- DTC list (including U-codes)
- Wiring diagram reference
- Measurement results (power, ground, CAN)
- Photos of wiring, connectors or damage
- Completion result and exceptions

Common Pitfalls to Avoid

- Calibrating around communication faults.
- Assuming every U-code means a failed module.

17. Power Supply & Battery Support

A stable system voltage is required for scanning, programming, coding and many calibration routines. Low voltage can create false faults or corrupt programming. A charger and a programming power supply are not always the same device.

Maintain stable voltage during all diagnostic, programming and calibration procedures.

The Power Supply Workflow

KEEP VOLTAGE STABLE, AVOID PROBLEMS.

Test battery

Check battery state of health and charge level. Repair or replace if needed before proceeding.

Choose support mode

Use a charger for diagnostics and a programming power supply for software updates. Follow OEM requirements for voltage and current.

Connect safely

Connect to approved battery terminals or jump points. Ensure a secure connection and correct settings before starting.

Monitor voltage

Keep system voltage within specification (typically 12.5-14.0 V) while running procedures). Watch for voltage drops.

Save result

Record voltage, device used and any related information. Keep with the repair record.

Why This Matters in a Repair Facility

ADAS controllers and scan tools may draw power with ignition on. A charger and a programming power supply are not always the same device. Low voltage can create false faults, interrupt programming or corrupt a module.

Stable voltage, successful repair. The right power support prevents unnecessary faults and repeat repairs.

Practical Shop Example

VOLTAGE DROP DURING PROGRAMMING

A module update stops when voltage drops. The correct response is recovery per OEM process, not repeated attempts on a weak battery.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current OEM procedure require?

FOR THE EVIDENCE FILE

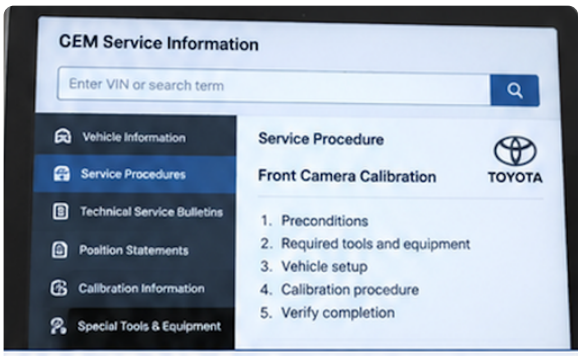
- Battery test results (before/after)
- Charger or power supply details (make, model, settings)
- System voltage logs or screen captures
- Procedure reference
- Scan results (before/after)
- Completion result and exceptions

Common Pitfalls to Avoid

- Using an uncontrolled boost charger during programming.
- Ignoring battery condition after a long repair.

18. OEM Information Hierarchy

The exact service procedure for the VIN controls the work. Position statements explain policy; calibration matrices help discovery; equipment instructions explain tool use. No single source replaces the others.



Use the VIN-specific service procedure for accurate measurements, sequence and completion criteria.

The OEM Information Flow

USE THE RIGHT SOURCE FOR THE RIGHT ANSWERS.

Identify VIN

Enter the full VIN to access the exact vehicle information and applicable procedures.

→

Open OEM Portal

Use the manufacturer's official service information website (e.g., Toyota TIS, GM SI, Ford PTS).

→

Find operation

Locate the correct service procedure, position statement, calibration matrix or equipment instruction.

→

Check revision

Confirm you are using the latest version. Check for updates, TSBs and change history.

→

Save reference

Save or print the procedure, screenshots or document ID for your records.

Why This Matters in a Repair Facility

No single source replaces the others. A public statement rarely contains all measurements, prerequisites and completion criteria. Using the correct OEM information prevents mistakes and supports a complete repair.

The VIN tells the story. Always use the VIN-specific procedure.

Practical Shop Example

DIFFERENT SOURCES, DIFFERENT PURPOSES

Service procedure (VIN-specific)	• Step-by-step instructions • Target values and dimensions • Required tools and equipment • Completion criteria
Position statement	• Explains OEM policy • Confirms when calibration may be required
Calibration matrix	• Helps identify applicable systems • Shows calibration requirements by model/option
Equipment instruction	• Explains scan tool or target setup • Provides tool-specific steps

A position statement establishes that calibration may be necessary; the service procedure supplies target number, dimensions and sequence.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets, inputs, modules or transmission zones could be affected?
- What does the current VIN-specific OEM procedure require for OEM information hierarchy?

FOR THE EVIDENCE FILE

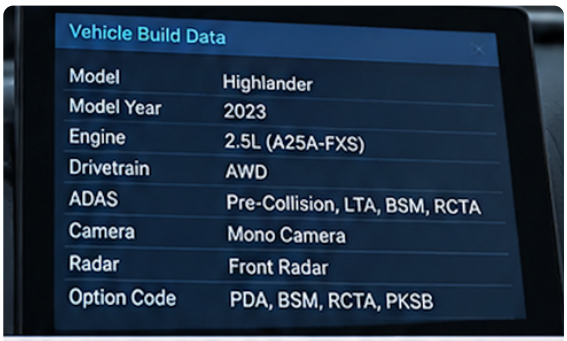
- VIN-specific service procedure
- Position statement
- Calibration matrix
- TSB information (if applicable)
- Equipment instructions
- Screenshots or document ID
- Completion result and exceptions

Common Pitfalls to Avoid

- Using a generic internet list as the final authority.
- Applying one model year
- procedure to another without verification.

19. Build Data & Option Identification

Build data reveals the installed options, but a physical inspection and scan confirm what is actually present.
Trim names alone are unreliable.



Build data shows the installed options for this vehicle. Always confirm with a scan and physical inspection.

The Build Data Workflow

IDENTIFY WHAT'S INSTALLED. CONFIRM WHAT'S PRESENT.

Decode VIN →

Enter the full VIN in the OEM system to view build data, option codes and production information.

Review options →

Identify installed safety systems, sensors, brackets and calibration requirements based on factory build data.

Inspect hardware →

Physically verify the installed components (e.g., camera, radar, brackets, wiring and part numbers).

Scan modules →

Perform a full system scan to confirm which modules respond and which options are actually present.

Match procedure

Use the correct OEM procedure for the identified options and installed hardware. Confirm calibration requirements.

Why This Matters in a Repair Facility

The same model may use different sensors, suppliers, brackets or calibration methods based on the option package and production date. Build data, a scan and a physical inspection must all agree before selecting a procedure.

Options tell the plan. Inspection confirms reality.
Never rely on trim data alone.

Practical Shop Example

SAME MODEL, DIFFERENT OPTIONS, DIFFERENT CALIBRATION.

Two SUVs share a trim badge; one has a mono camera and one has camera-radar fusion. Their calibration plans differ.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors, brackets inputs, modules or transmission zones could be affected?
- What does the current VIN-specific OEM procedure require for build data and option identification?

FOR THE EVIDENCE FILE

- VIN decode / build data report
- Option code list
- Scan results (module list)
- Photos of installed hardware (sensors, brackets, part numbers)
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Assuming every vehicle of a model has the same ADAS.
- Ordering glass or sensors from trim alone.

20. Safety Culture & Stop-Work Authority

A technician must stop work when prerequisites, procedures, tools, the environment or vehicle condition cannot support a valid calibration. Production pressure does not change sensor physics. A documented hold protects the customer, repairer and insurer.

When conditions are not met, stop work and document the reason.

The Stop-Work Workflow

KNOW THE RISK, AND TAKE THE RIGHT ACTION

Recognize hazard
Identify when prerequisites, procedures, tools, environment or vehicle condition cannot support a valid calibration.

→

Stop operation
Do not continue with scanning, programming or calibration when a risk to accuracy, safety or vehicle function exists.

→

Document reason
Record the issue, including photos and details in the repair order. Communicate with the advisor, customer or insurer as required.

→

Correct condition
Resolve the issue by meeting required conditions (e.g., proper tools, environment, repairs, or OEM procedure).

→

Resume when safe
Continue only after all conditions are met and verified. Complete the procedure and document the final result.

Why This Matters in a Repair Facility

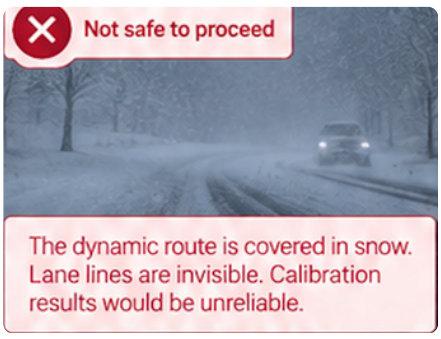
Production pressure does not change sensor physics. An incomplete or improper repair can lead to unsate vehicle operation, come-backs, liability and loss of trust. A documented hold protects the customer, repairer and insurer.

YIELD

It's better to pause than to deliver an unsafe repair. When in doubt, stop work.

Practical Shop Example

SAME VEHICLE, DIFFERENT OUTCOME.



The dynamic route is covered in snow. Lane lines are invisible. Calibration results would be unreliable.

Road is clear and lane lines are visible. Required conditions are met and calibration can be completed.

The dynamic route is covered in snow and lane lines are invisible. The technician marks the procedure incomplete and reschedules.

A TECH/ESTIMATOR SHOULD ASK

- Which sensors, brackets, inputs, modules, or transmission zones could be affected?
- What does the current VIN-specific OEM procedure require for safety culture and stop-work authority?

FOR THE EVIDENCE FILE

- Repair order notes (reason for hold)
- Photos (environment/conditions)
- EM procedure reference
- Communication with advisor/ customer/insurer
- Completion result and exceptions

Common Pitfalls to Avoid

- Calling a failed or incomplete routine successful.
- Allowing delivery with an undisclosed ADAS limitation.

Knowledge Check | Test your understanding of this section's content.

PART I
ADAS FOUNDATIONS

To view and complete the **Knowledge Check** on this section, please visit:

Link to online assessment: https://www.classmarker.com/online-test/start/?quiz=nt76ad90a33c04f8

Quiz Password: adasfoundations

You may also access the online assessment through this QR code:

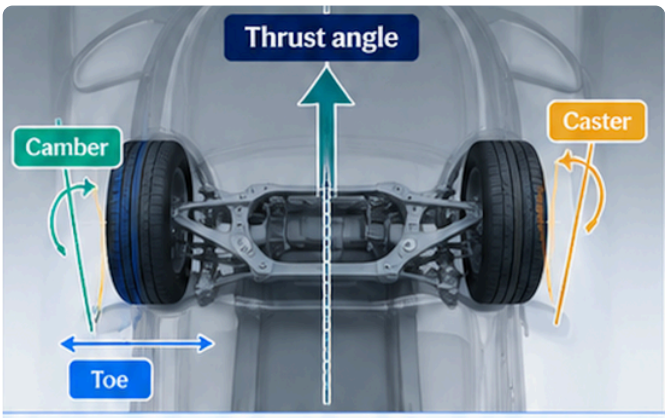
PART II

VEHICLE, SENSOR, & SHOP BASICS

Objective: Develop a comprehensive understanding of vehicle systems, ADAS sensor technologies, and essential shop conditions before beginning diagnostic, repair, or calibration procedures. Technicians will learn to identify sensor locations and functions, evaluate vehicle readiness, recognize environmental and equipment requirements, and verify critical prerequisites using OEM service information. The goal is to establish safe, consistent service practices that minimize errors, protect vehicle safety systems, and ensure accurate, reliable repair and calibration outcomes.

21. Basic Wheel Alignment

Wheel alignment sets the wheel angles such as toe, camber and caster. The thrust angle describes the rear axle direction relative to the vehicle centerline.



Proper wheel alignment keeps the vehicle centered and on track. ADAS relies on correct geometry.

The Alignment Workflow

FOLLOW THE STEPS FOR SUCCESSFUL ALIGNMENT

Inspect suspension

Check for worn or damaged parts.

→

Measure alignment

Read current angles (toe, camber, caster, thrust angle).

→

Correct geometry

Make adjustments to OEM specifications.

→

Initialize SAS

Reset the steering angle sensor after alignment.

→

Calibrate ADAS

Perform required calibrations after alignment.

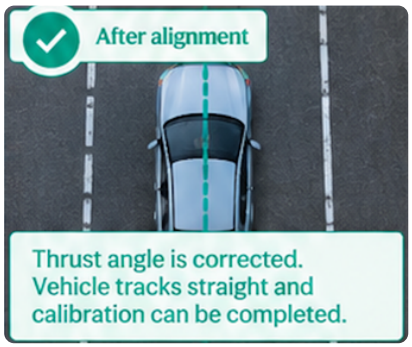
Why This Matters in a Repair Facility

ADAS predicts vehicle path from chassis geometry. A centered target cannot compensate for a vehicle that travels at an angle.

Alignment comes first. Correct geometry is required before an ADAS calibration.

Practical Shop Example

STRAIGHT BODY, THRUST ANGLE FAULT



A TECH/ESTIMATOR SHOULD ASK

- Are there suspension or steering issues?
- What are the current alignment angles?
- What does the OEM procedure require before ADAS calibration?

FOR THE EVIDENCE FILE

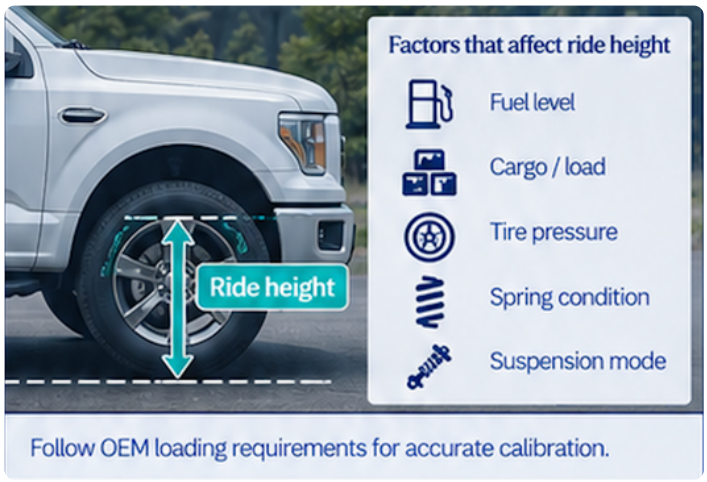
- Pre- and post-alignment printouts
- SAS reset confirmation
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Calibrating before final alignment.
- Centering the steering wheel without correcting geometry.

22. Ride Height & Vehicle Loading

Ride height is the body or suspension position relative to specified datums. Fuel, cargo, tire pressure, springs and air suspension change pitch and sensor height.



The Loading Workflow

PREPARE THE VEHICLE PER OEM SPECIFICATIONS.

Unload vehicle
Remove cargo and accessories as required.

→

Set tire pressure
Adjust to OEM specifications (cold).

→

Set suspension mode
Use the correct mode (normal, access, transport, etc.).

→

Measure height
Check ride height at specified datums.

→

Compare specification
Verify measurements are within OEM range.

Why This Matters in a Repair Facility

Camera and radar axes move with the body. OEM procedures may specify fuel, ballast, cargo removal or suspension mode. The vehicle must match these conditions for accurate calibration.

Calibrate the vehicle, not the condition. Follow OEM loading requirements.

Practical Shop Example

DIFFERENT LOADING, DIFFERENT RESULTS

A pickup calibrated with heavy bed cargo points differently after unloading. The procedure's loading condition must be followed.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- What are the OEM loading requirements (fuel, cargo, ballast)?
- What suspension mode should be used?
- What does the current VIN-specific OEM procedure require for ride height and vehicle loading?

FOR THE EVIDENCE FILE

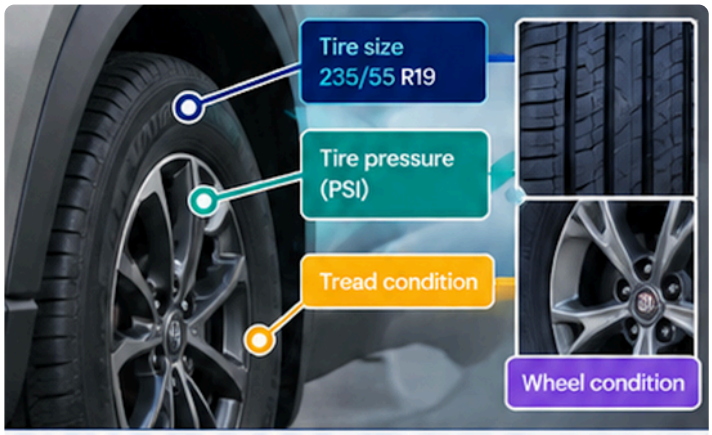
- Photos (before/after)
- Tire pressure record
- Suspension mode confirmation
- Ride height measurements
- (at specified datums)
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Ignoring cargo and fuel requirements.
- Calibrating an air-suspension vehicle in the wrong mode.

23. Tires & Wheels

Tire size, inflation pressure, tread wear, and wheel condition directly affect vehicle ride height, wheel-speed readings, alignment, and overall vehicle geometry. Even minor variations can compromise ADAS sensor accuracy and calibration results. Before calibration, technicians must verify OEM tire specifications, inspect tire and wheel condition, and confirm proper inflation, alignment, and ride height.



Correct tire size, pressure, tread and wheel condition are required for accurate ADAS calibration.

The Tires and Wheels Flow

PREPARE THE CAR FOR ACCURATE RESULTS.

Confirm size

Verify tire size matches OEM specification on all four corners.

→

Inspect condition

Check for uneven wear, damage or mismatched tires/ wheels.

→

Set pressures

Adjust to OEM specified cold pressures.

→

Check wheel runout

Inspect for bent wheels or excessive runout.

→

Record values

Document tire size, pressures and condition in the repair order.

Why This Matters in a Repair Facility

Mismatched rolling circumference can create wheel-speed plausibility issues and a tilted or pitched vehicle can alter sensor aim.

All four corners count. Tires and wheel must match OEM requirements.

Practical Shop Example

DIFFERENT TIRE PRESSURES, DIFFERENT RESULTS

One low front tire lowers the vehicle and can change camera geometry.

Tire pressure set to OEM specification. Vehicle height and sensor aim are correct.

One low front tire lowers a camera corner and changes static setup geometry. Pressure is corrected before measuring targets.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Are the tire sizes, pressures and
- conditions correct?
- What does the current VIN-specific OEM procedure require for tires and wheels?

FOR THE EVIDENCE FILE

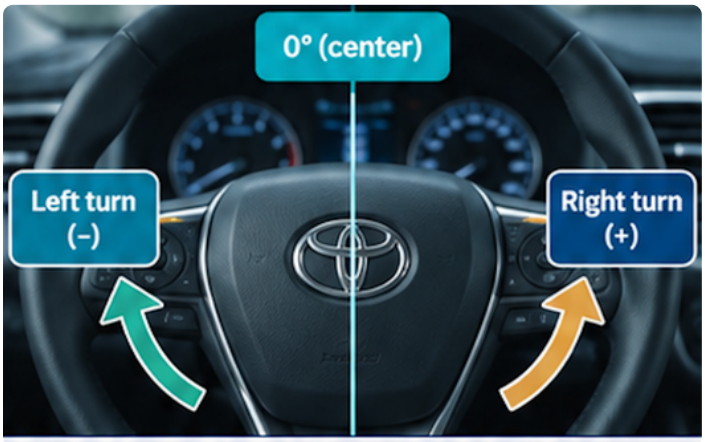
- Tire size (all four corners)
- Tire pressure record (cold
- Photos of tire/wheel condition (if needed)
- OtM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Using door-jamb pressure without checking procedure conditions.
- Ignoring mismatched aftermarket wheel/tire sizes.

24. Steering Angle Sensor

The SAS reports steering position to ABS, ESC, EPS and ADAS. Zero usually corresponds to the true straight-ahead vehicle path, not merely the wheel spokes.



The steering angle sensor measures steering position. Zero is the true straight-ahead vehicle path.

The SAS Workflow

CENTER. VERIFY. PROCEED.

Center mechanically
Set the steering to the true straight-ahead position.

→ **Align vehicle**
Complete wheel alignment and ensure vehicle is on a level surface.

→ **Run zero point**
Use the scan tool to initialize or calibrate the SAS to zero.

→ **Verify live data**
Check that steering angle reads near 0° when straight ahead and turns smoothly left and right.

→ **Continue sequence**
Proceed with next required calibrations (camera, radar, etc.)

Why This Matters in a Repair Facility

A wrong zero shifts the predicted path and can affect lane, radar and stability functions.

Zero to the road, not the wheel. The steering wheel may not be visually straight when centered.

Practical Shop Example

SAME REPAIR, DIFFERENT RESULT.

After tie-rod replacement, the SAS is not initialized. Live data shows -5.3° when driving straight.

SAS is initialized. Live data reads 0.0° and the vehicle tracks straight. Proceed with camera calibration.

After tie-rod replacement, alignment and SAS initialization are completed before the forward camera routine.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Is the steering mechanically centered?
- What does the current VIN-specific OEM procedure require for the steering angle sensor?

FOR THE EVIDENCE FILE

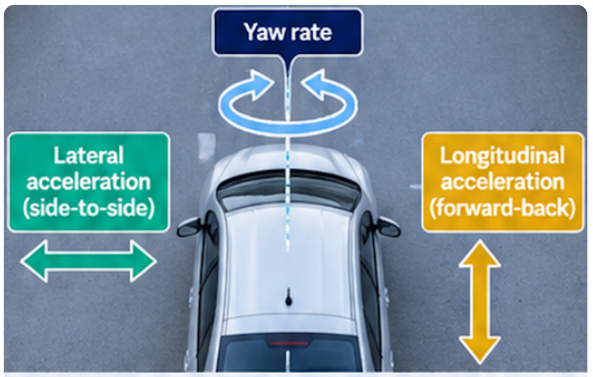
- Pre- and post-SAS live data (0°)
- Scan tool report
- Alignment printout
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Initializing SAS before alignment.
- Assuming a straight steering wheel equals zero degrees.

25. Yaw & Acceleration Sensors

Yaw rate sensors measure how quickly a vehicle rotates around its vertical axis, helping determine its direction and stability during turns. Acceleration sensors measure lateral (side-to-side) and longitudinal (forward-and-backward) acceleration, detecting changes in vehicle motion. Together, these sensors provide critical data to stability control and ADAS systems, helping them interpret vehicle movement and respond accurately.



Yaw and acceleration sensors measure how the vehicle moves in all directions.

The SAS Workflow

FOLLOW THE STEPS FOR ACCURATE RESULTS.

Level vehicle → **Stabilize movement** → **Run routine** → **Verify data** → **Post-scan**

- **Level vehicle:** Set up the vehicle on a level, stationary surface.
- **Stabilize movement:** Ensure the vehicle is stable and not moving.
- **Run routine:** Follow the scan tool procedure to initialize or calibrate the sensors.
- **Verify data:** Check that live data is within expected range.
- **Post-scan:** Confirm no faults and save results.

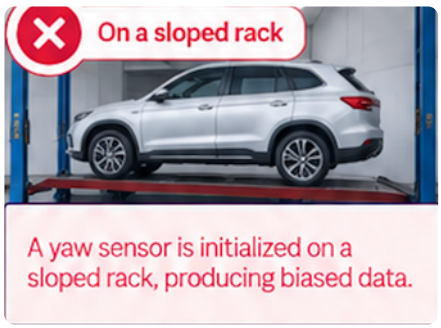
Why This Matters in a Repair Facility

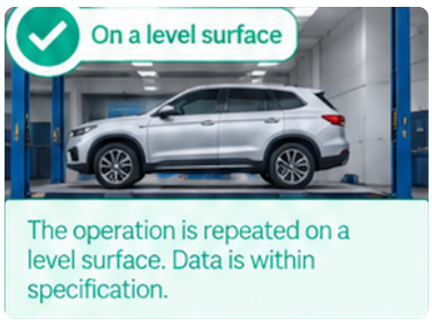
ADAS and stability controllers compare expected motion with actual motion. Zero-point calibration may require a level, stationary surface.

A stable and level vehicle leads to accurate data. Follow OEM conditions.

Practical Shop Example

SAME SENSOR, DIFFERENT SURFACE, DIFFERENT RESULT.





A yaw sensor is initialized on a sloped rack, producing biased data. The operation is repeated on a compliant surface.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Does the procedure require a level, stationary surface?
- What does the current VIN-specific OEM procedure require for yaw and acceleration sensors?

FOR THE EVIDENCE FILE

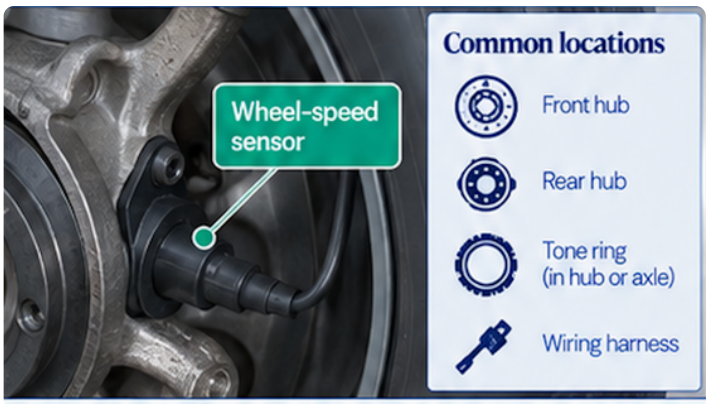
- Scan tool report (zero-point or calibration result)
- Photos (vehicle position/surface)
- OEM procedure reference
- Live data (before/after)
- Completion result and exceptions

Common Pitfalls to Avoid

- Calibrating on a lift or slope without permission.
- Moving the vehicle during intialization.

26. Wheel-Speed Sensors

Wheel-speed sensors report individual wheel rotation to the ABS and other controllers. Accurate wheel-speed data is required for ABS, stability, traction and advanced driver assistance systems.



Wheel-speed sensors measure individual wheel speed and send the signal to ABS and other controllers.

The Wheel-Speed Workflow

CHECK. TEST. VERIFY.

Inspect harness

Check the sensor, wiring and connectors for damage or corrosion.

→ **Read live data**

Verify each wheel speed reads correctly and is stable.

→ **Compare wheels**

Check all four wheels rotate equally at the same speed.

→ **Repair root cause**

Fix wiring, tone ring, bearing or connector issues.

→ **Verify all modules**

Confirm no DTCs and all related systems operate normally.

Why This Matters in a Repair Facility

ACC, AEB, stability and path calculations depend on reliable wheel-speed data. A single faulty signal can disable multiple ADAS functions.

All four wheels should agree. Consistent wheel-speed data is essential for safe operation.

Practical Shop Example

SAME VEHICLE. DIFFERENT HARNESS. DIFFERENT RESULT.

A torn right-front harness sets an ABS DTC and a radar limited-assist message.

Harness is repaired. All wheel-speed data is normal and no DTCs are present.

A torn right-front harness sets an ABS DTC and radar limited-assist message. Repair the speed signal before radar calibration.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which wheel(s) are affected?
- Are all four wheel speeds reading correctly?
- What does the current VIN-specific OEM procedure require for wheel-speed sensors?

FOR THE EVIDENCE FILE

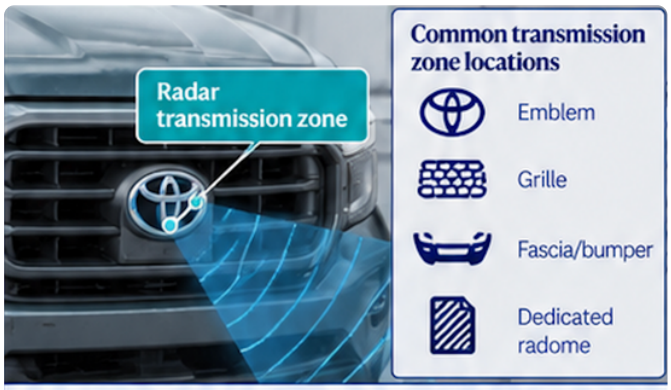
- Scan report (before/after)
- Live data (all four wheel speeds)
- Photos (sensor, harness, tone ring)
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Replacing ADAS sensors for an
- ABS-source fault.
- Ignoring intermittent dropouts during turns.

27. Radar Transmission Zones

A transmission zone is the material and space through which radar energy passes. It may include an emblem, grille, fascia or dedicated radome.



Radar energy passes through the transmission zone. It must be OEM approved and free of damage or non-approved materials.

The Radar Inspection Flow

CHECK THE PATH. VERIFY CONDITION. RESTORE AS NEEDED.

Identify beam path

Locate the radar transmission zone on the vehicle.

→

Find material limits

Check OEM information for approved materials, thickness and coatings.

→

Inspect repairs

Look for damage, non-approved materials or excessive coatings.

→

Replace prohibited damage

Use only OEM approved parts and materials.

→

Aim and verify

Complete setup and confirm proper operation with a scan.

Why This Matters in a Repair Facility

Material type, thickness, repairs, coatings, metallic content and geometry can attenuate or reflect radar energy.

Keep the path clear for a clear picture. Use OEM approved materials and repair methods.

Practical Shop Example

SAME VEHICLE. DIFFERENT TRANSMISSION ZONE.

An insurer writes "buff radar." The actual radar housing/radome cannot be cosmetically repaired unless the manufacturer expressly permits it.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors are affected?
- What are the OEM material limits and repair methods?
- What does the current VIN-specific OEM procedure require for radar transmission zones?

FOR THE EVIDENCE FILE

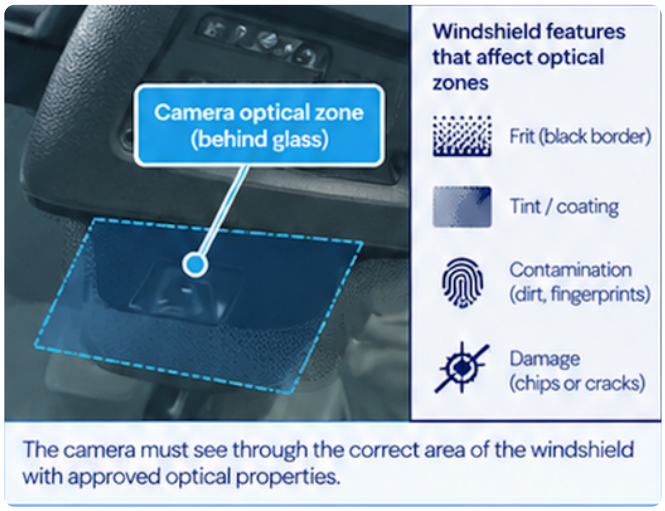
- Photos (before/after)
- OEM material/repair information
- Scan tool results
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Polishing or filling radar hardware.
- Using non-approved coatings, films or emblem covers.

28. Optical Zones & Windshield Glass

The camera looks through specific areas of the windshield with controlled optical properties.
Bracket geometry, tint and contamination can affect image quality.



The camera must see through the correct area of the windshield with approved optical properties.

The Optical Zone Workflow

CHECK THE GLASS. PREPARE THE CAMERA.

Decode glass option

Verify the correct windshield glass is specified for the vehicle.

→

Inspect bracket

Check camera bracket position, condition and alignment.

→

Clean optical zone

Remove dirt, adhesive residue, haze or fingerprints from the camera view area.

→

Install covers

Install any required cosmetic covers or shrouds.

→

Calibrate camera

Complete camera calibration and verify proper operation.

Why This Matters in a Repair Facility

Distortion, tint, contamination or wrong bracket position changes the image before software sees it.

The camera sees what's through the glass. Use OEM-approved glass and keep the optical zone clean.

Practical Shop Example

SAME VEHICLE. DIFFERENT GLASS. DIFFERENT RESULT.

This replacement windshield has the wrong camera option. The tint and frit pattern block the camera view.

OEM-specified glass with the correct optical zone is installed. The camera has a clear view for calibration.

A replacement windshield fits physically but has the wrong camera option. Correct glass is installed before aiming.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Does the replacement glass match the OEM specification?
- Which sensors are affected?
- What does the current VIN-specific OEM procedure require for glass?

FOR THE EVIDENCE FILE

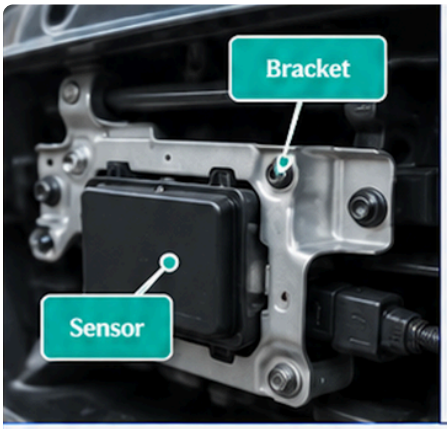
- OEM glass part number
- Photos (glass, frit, camera area)
- Scan tool results
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Choosing glass only by body style.
- Leaving adhesive, haze or fingerprints in the camera window.

29. Sensor Brackets & Fasteners

The bracket establishes sensor position and angle. Fasteners, clips, shims and torque preserve that geometry.



The bracket, fasteners and mounting features keep the sensor at the correct position and angle.

The Bracket Inspection Flow

CHECK. COMPARE. SECURE. VERIFY.

Expose mounting

Remove covers or trim to access the sensor bracket.

Inspect datum

Check the bracket for damage, correct part, and proper mounting points.

Compare part

Verify the bracket matches OEM design, including clips, shims and hardware.

Torque correctly

Tighten fasteners OEM specifications and install any required shims or clips.

Measure orientation

Confirm the sensor is at the correct angle using scan tool or target procedure.

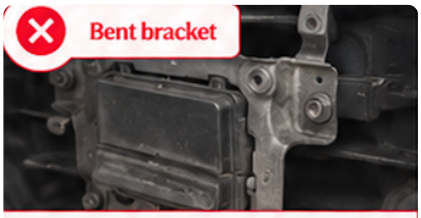
Why This Matters in a Repair Facility

A sensor housing can look perfect while a bracket is slightly twisted. Calibration cannot legitimize damaged mounting.

The bracket holds the aim. Secure mounting preserves sensor geometry.

Practical Shop Example

SAME VEHICLE. DIFFERENT BRACKET CONDITION.



A front impact bends a radar bracket behind an undamaged cover. The sensor aim is out of specification.

The bracket is replaced with the correct OEM part and torqued to specification. The sensor measures within specification.

A front impact bends a radar bracket behind an undamaged cover. Measurement and replacement occur before aiming.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Is the bracket or mounting affected?
- Does the OEM procedure require specific brackets, shims or fasteners?
- What are the correct torque specifications?

FOR THE EVIDENCE FILE

- • Photos (before/after)
- OEM part number comparison
- Torque specification record
- Scan tool results (sensor angle)
- OEM procedure reference
- Completion result and exceptions

- Straightening a non-repairable precision bracket.
- Reusing distorted retainers or wrong bolts.

30. Bumper Covers & Refinishing

Bumper covers may carry sensors or sit directly in radar and ultrasonic sensing zones. Repair materials and coatings can affect sensor performance.

Bumper cover design, materials and coatings can affect radar and ultrasonic sensor performance. Follow OEM repair procedures.

The Bumper Cover Flow

FOLLOW THE STEPS FOR A SUCCESSFUL REPAIR.

Map sensor zones

Identify sensors and check OEM refinish and repair restrictions for the bumper cover.

Review OEM limits

Verify approved materials, thickness and coating requirements.

Control repair thickness

Use approved repair methods and keep materials within OEM limits.

Install correctly

Reinstall the cover with proper alignment and hardware.

Test / calibrate

Verify sensor operation with a scan tool or calibration procedure.

Why This Matters in a Repair Facility

Filler, welds, metallic paint, excessive film or thickness can change signal behavior. Sensor orientation can change after cover repair.

Follow OEM procedures for materials and thickness. The right repair methods keep sensors seeing clearly.

Practical Shop Example

SAME VEHICLE. DIFFERENT REPAIR APPROACH.

A rear bumper repair crosses the BSM zone with excessive filler. The sensor range is reduced.

The bumper is repaired using OEM-approved materials and thickness. The sensor operates correctly after calibration.

A rear bumper repair crosses the BSM zone. The planner retrieves refinishing restrictions and adds sensor inspection/calibration research.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Are there thickness limits?
- Does the current VIN-specific procedure require sensor testing or calibration?

FOR THE EVIDENCE FILE

- Photos (before/after)
- OEM repair and refinish information
- Thickness measurements
- Scan tool results (sensor operation)
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Testing all bumper areas alike.
- Using non-approved paints, films or fillers.
- Exceeding OEM thickness limits or skipping sensor verification.

31. Mirrors, Doors, & Camera Pose

Side cameras are often mounted in mirrors or doors. Mirror detection, door alignment and camera clip position establish the camera's image pose for surround-view stitching.

Side cameras are mounted in mirrors, doors or body panels. Correct mounting and alignment are required for accurate image pose.

The Camera Pose Workflow

FIT. VERIFY. INSTALL. CHECK. CALIBRATE.

Fit door

Complete door installation and verify proper fit and alignment.

→ **View mirror detent**

Confirm the mirror adjusts and detents correctly with no binding or looseness.

→ **Install camera**

Install the camera in the correct clip and orientation.

→ **Inspect image**

Check camera view for correct aim, coverage and obstructions.

→ **Calibrate surround-view**

Perform surround-view calibration with all cameras in known locations and angles.

Why This Matters in a Repair Facility

Surround-view stitching assumes every camera is in a known location and angle. Door alignment, mirror seating and camera clip position establish the correct image pose.

Proper fit keeps the view true. Correct installation and alignment support accurate camera pose.

Practical Shop Example

SAME VEHICLE. DIFFERENT DOOR CONDITION..

A mirror camera is transferred after a door replacement. The door is not fully aligned, causing an incorrect image angle.

The door is correctly aligned and the mirror is fully seated. The camera view is correct after calibration.

A mirror camera is transferred after a door replacement. Door fit and mirror seating are completed before surround-view calibration.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which cameras are affected?
- Does the current VIN-specific
- OEM procedure require mirror/door alignment checks?
- What are the correct camera clip positions and orientations?

FOR THE EVIDENCE FILE

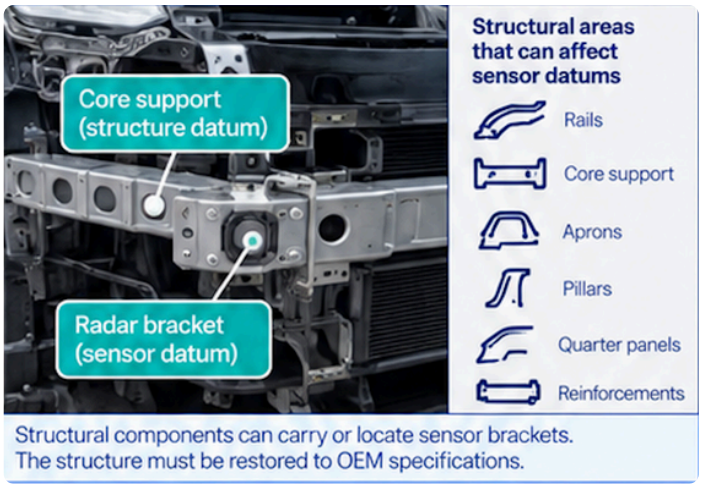
- Photos (before/after)
- Scan tool results (camera views)
- OEM procedure reference
- Door/mirror alignment records
- Calibration report
- Completion result and exceptions

Common Pitfalls to Avoid

- Calibrating before hinge adjustment.
- Ignoring a loose folding-mirror mechanism.
- Installing the camera in the wrong clip or orientation.

32. Structure & Sensor Datums

Rails, core supports, aprons, roofs, pillars, quarters and reinforcements may carry or locate sensor brackets.



Structural components can carry or locate sensor brackets. The structure must be restored to OEM specifications.

The Structure / Datum Flow

MEASURE. COMPARE. REPAIR. VERIFY. CALIBRATE.

Measure structure

Check key dimensions at OEM datum points using a measuring system.

Compare OEM Dimensions

Verify measurements against OEM specifications.

Correct damage

Repair and align structural components to OEM specifications before handling sensor brackets.

Verify bracket datum

Confirm sensor bracket position, angle and datum match OEM specifications.

Calibrate last

Perform calibration after structure and bracket datums are verified

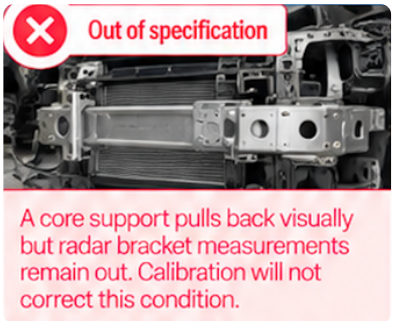
Why This Matters in a Repair Facility

A structural error becomes a sensor-position error. Cosmetic panel gaps alone do not prove datum restoration.

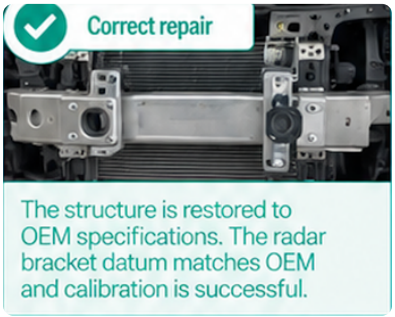
Measure structure, not just appearance. OEM dimensions ensure correct sensor location.

Practical Shop Example

SAME VEHICLE. DIFFERENT STRUCTURE. DIFFERENT RESULT.



A core support pulls back visually but radar bracket measurements remain out. Calibration will not correct this condition.



The structure is restored to OEM specifications. The radar bracket datum matches OEM and calibration is successful.

A core support pulls back visually but radar bracket measurements remain out. Structure is corrected before calibration.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which structural components are affected?
- What are the OEM dimensions and measuring points?
- What does the current VIN-specific OEM procedure require?

FOR THE EVIDENCE FILE

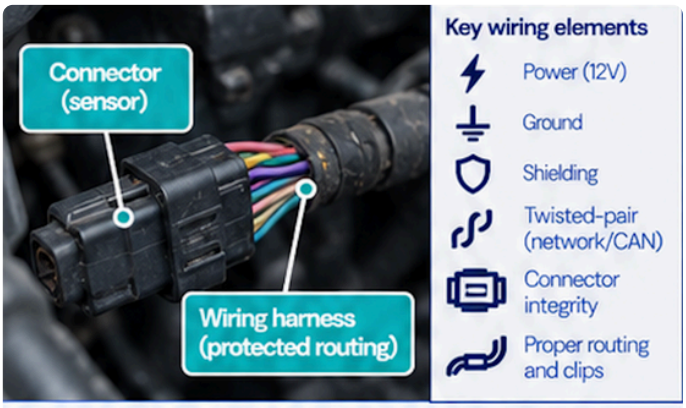
- Before/after measurement reports
- OEM body dimension printouts
- Photos (structure and brackets)
- Scan tool results
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Using calibration to hide structural error.
- Skipping final measurements after pulls or sectioning-
- Relying on cosmetic panel gaps alone.

33. Wiring, Connectors, & Grounds

Sensors require correct power, ground, shielding, twisted-pair network wiring and connector integrity for reliable operation.



Clean, secure connectors and properly routed wiring keep sensors communicating and operating correctly.

The Wiring Inspection Flow

INSPECT. TEST. REPAIR. VERIFY.

Inspect routing

Check wiring route, protection, clips and for signs of damage or repair.

Load-test power / ground

Verify voltage supply and ground integrity under load.

Check terminals

Inspect for bent pins, backed-out terminals, corrosion or water intrusion.

Test network

Check communication on CAN or other networks for proper operation.

Verify under conditions

Recheck after repair with a road test, wiggle test or scan tool while moving components.

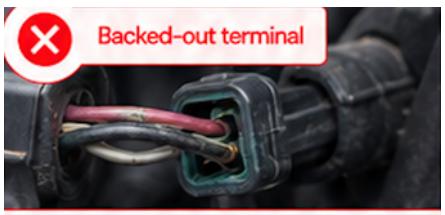
Why This Matters in a Repair Facility

High resistance, water intrusion, pin fit or repaired routing can create intermittent faults and corrupted data. Electrical integrity is critical for reliable ADAS operation.

Good power, ground and clean connections keep sensors online.

Practical Shop Example

SAME VEHICLE, DIFFERENT CONNECTION.



A camera loses communication only over bumps. A backed-out terminal is found after wiggle testing.

The terminal is fully seated and the connector is secure. Communication is stable and no faults return.

A camera loses communication only over bumps. A backed-out terminal is found after wiggle testing.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which sensors are affected?
- Is there power and ground at the sensor?
- Are connectors and terminals intact?
- What does the current VIN-specific OEM procedure require for wiring and connector checks?

FOR THE EVIDENCE FILE

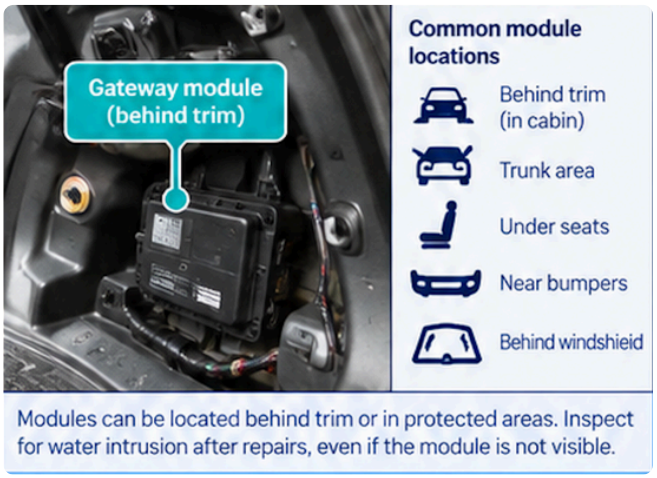
- Photos (routing, connectors)
- Voltage and ground test results
- Network scan tool results
- OEM procedure reference
- Repair documentation
- Completion result and exceptions

Common Pitfalls to Avoid

- Judging wiring by appearance only.
- Probing or repairing prohibited shielded circuits incorrectly.
- Ignoring water intrusion, corrosion or loose terminals.

34. Module Location & Water Intrusion

ADAS and gateway modules can be located behind trim, in trunks, under seats, near bumpers or behind the windshield. Water intrusion can affect modules far from the visible impact.



Modules can be located behind trim or in protected areas. Inspect for water intrusion after repairs, even if the module is not visible.

The Module Inspection Flow

FIND. INSPECT. CHECK. REPAIR. VERIFY.

Locate modules

Identify module locations using OEM information and vehicle inspection.

→

Inspect moisture path

Check for water intrusion, damaged seals, missing plugs and open paths.

→

Check connectors

Inspect for corrosion, proper seating and damage at module connectors.

→

Repair leak / damage

Repair the source of the leak and any damaged components or connectors.

→

Scan and verify

Scan for DTCs, verify module operation and confirm no water-related faults remain.

Why This Matters in a Repair Facility

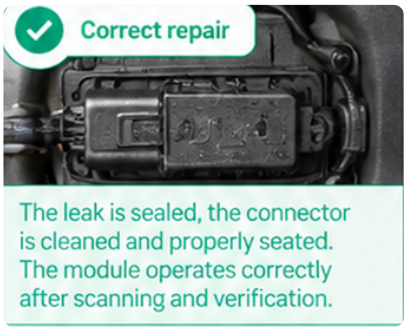
Collision damage, glass leaks and cleaning water can affect modules far from the visible impact. Fix the leak first.

Water intrusion can cause intermittent faults and corrupted data.

Practical Shop Example

SAME VEHICLE, DIFFERENT CONDITION.

A rear quarter repair leaves a body plug open. Water reaches the side-radar connector and causes intermittent warnings.



The leak is sealed, the connector is cleaned and properly seated. The module operates correctly after scanning and verification.

A rear quarter repair leaves a body plug open; water reaches the side-radar connector and causes intermittent warnings.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Where are the modules located on this vehicle?
- What could allow water intrusion?
- Does the OEM procedure require module inspection after glass or body repairs?

FOR THE EVIDENCE FILE

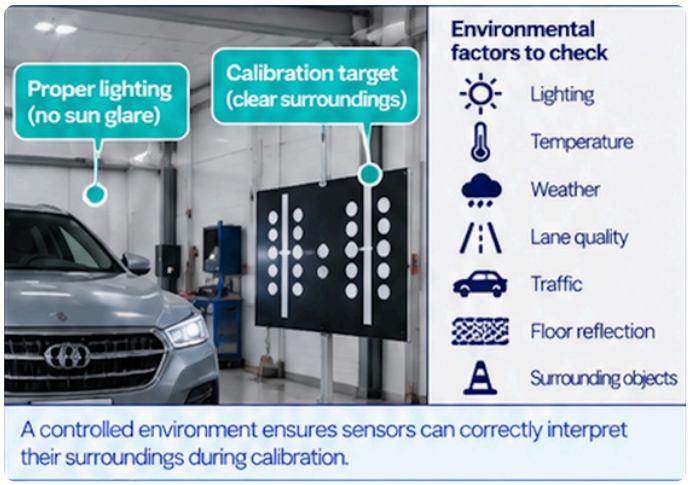
- Photos (before/after)
- Scan tool results (DTCs)
- OEM module location reference
- Repair documentation
- Water leak repair records
- Completion result and exceptions

Common Pitfalls to Avoid

- Replacing electronics before correcting the leak.
- Failing to inspect after glass or quarter work.
- Ignoring missing plugs, seals or trim.

35. Environmental Limits

Static and dynamic procedures may be controlled by light, temperature, weather, lane quality, traffic, floor reflection and surrounding objects.



A controlled environment ensures sensors can correctly interpret their surroundings during calibration.

The Environmental Check

EVALUATE. PREPARE. CONTROL. PERFORM. DOCUMENT.

Read limits

Review OEM requirements for lighting, temperature, weather and other environmental limits.

Study environment

Check the shop or drive route for potential interference (glare, traffic, reflective surfaces, obstacles).

Remove interference

Adjust lighting, move obstacles, clean floors or choose a better route/time to eliminate interference.

Perform routine

Complete the static or dynamic procedure under compliant environmental conditions.

Record conditions

Document lighting, weather, temperature, route and any special notes from the procedure.

Why This Matters in a Repair Facility

Sensors interpret their environment during calibration. Noncompliant conditions can prevent completion or teach unreliable references.

Control the conditions. A stable environment leads to a successful calibration.

Practical Shop Example

SAME VEHICLE, DIFFERENT ENVIRONMENT.

Sun glare washes out a camera target. Calibration fails or produces unreliable results.

The bay lighting is controlled and the target is clearly visible. Calibration completes successfully.

Sun glare washes out a camera target. The bay lighting and target orientation are corrected before retrying.

A TECH/ESTIMATOR SHOULD ASK

- Are the environmental conditions within OEM limits?
- Could lighting, weather, traffic or surroundings affect the sensors?
- Does the current OEM procedure require special conditions?

FOR THE EVIDENCE FILE

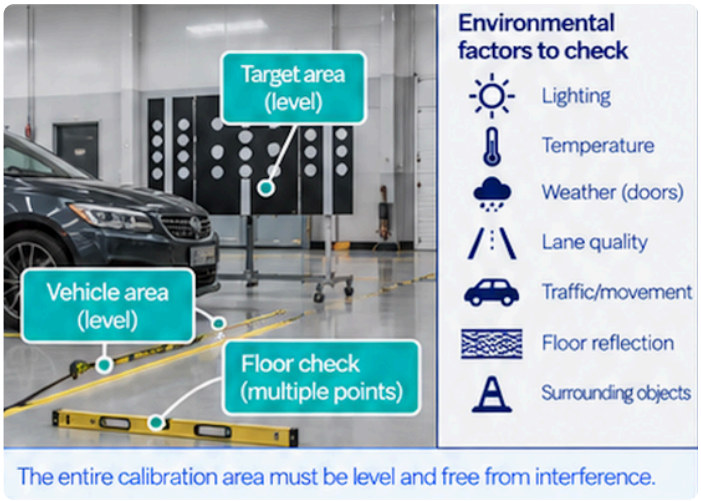
- Photos (setup and environment)
- Scan tool results
- OEM procedure reference
- Environmental condition notes (lighting, weather, temperature, route)
- Completion result and exceptions

Common Pitfalls to Avoid

- Treating environmental requirements as suggestions.
- Failing to document why a dynamic routine could not complete.
- Ignoring reflections or nearby moving objects.

36. Floor Level & Bay Validation

A calibration floor must meet the equipment and OEM tolerance over the vehicle and target footprint.



The Bay Validation Flow

DEFINE. SURVEY. RECORD. MARK. REVALIDATE.

Define footprint

Review OEM requirements for the vehicle and target footprint.

→

Survey grid

Check floor level at multiple points across the vehicle and target area.

→

Record elevations

Document measurements and identify any areas outside tolerance.

→

Mark approved zone

Mark the usable footprint and keep it clear of obstructions.

→

Revalidate schedule

Recheck regularly after any shop changes or surface work.

Why This Matters in a Repair Facility

Slope alters target height/angle and vehicle attitude. A bubble level at one spot does not characterize the full area. A controlled environment helps the sensor see the world correctly.

Level surfaces lead to reliable results.

Practical Shop Example

SAME VEHICLE, DIFFERENT FLOOR CONDITION..

The bay passes near the vehicle but falls outside tolerance where the target stands. The usable footprint is marked and controlled.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Does the floor meet OM level
- tolerances?
- Are there slopes, drains or transitions in the target area?
- Does the current procedure require floor level and bay validation?

FOR THE EVIDENCE FILE

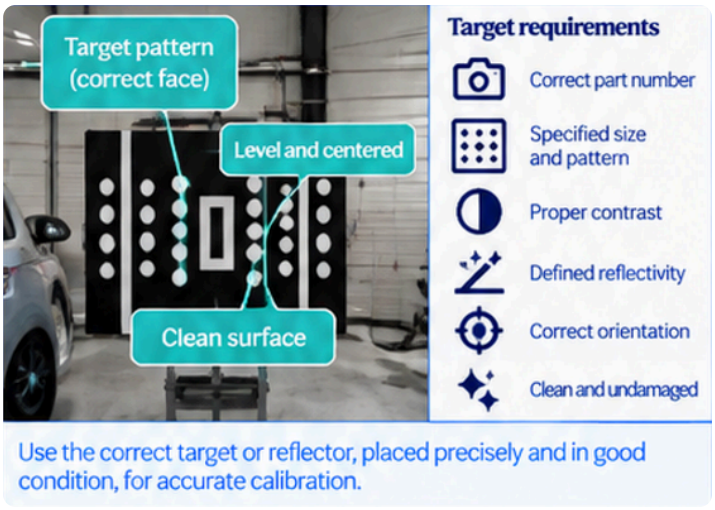
- Floor level measurements
- Photos (grid and elevations)
- OEM procedure reference
- Marked calibration area
- Revalidation records
- Completion result and exceptions

Common Pitfalls to Avoid

- Checking only the vehicle pad.
- Assuming a newly poured floor is level.
- Ignoring slopes, drains or surface transitions
- Failing to revalidate after shop changes.

37. Targets & Reflectors

Targets are engineered patterns or reflectors with a specified size, contrast, reflectivity and orientation.



Use the correct target or reflector, placed precisely and in good condition, for accurate calibration.

The Target Setup Flow

READ. INSPECT. SELECT. PLACE. PHOTOGRAPH.

Read target ID

Verify the correct target or reflector part number from the OEM procedure.

→

Inspect condition

Check for damage, dirt, warping or fading that could affect the pattern.

→

Select correct face

Use the specified face, side or configuration for the system.

→

Place precisely

Set distance, height, angle and orientation to OEM specifications. Ensure it is level and stable.

→

Photograph ID

Capture a clear photo of the target/reflector and its ID before calibration.

Why This Matters in a Repair Facility

The scan tool expects a known reference. Wrong revision, face, scale, damage or glare changes what the sensor detects.

The right target makes the difference. A correct, clean and precisely placed target helps the sensor see the pattern as intended.

Practical Shop Example

SAME VEHICLE, DIFFERENT TARGET CONDITION..

A dirty or warped board produces an incorrect reference and can cause calibration to fail.

The correct target part number is used, it is clean and placed to OEM specifications. Calibration completes successfully.

Two similar boards belong to different camera systems. The target part number is verified before setup.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which target or reflector is required?
- What does the OEM procedure specify for size, contrast, distance and height?
- Is the target clean and undamaged?
- What are the correct placement and orientation requirements?

FOR THE EVIDENCE FILE

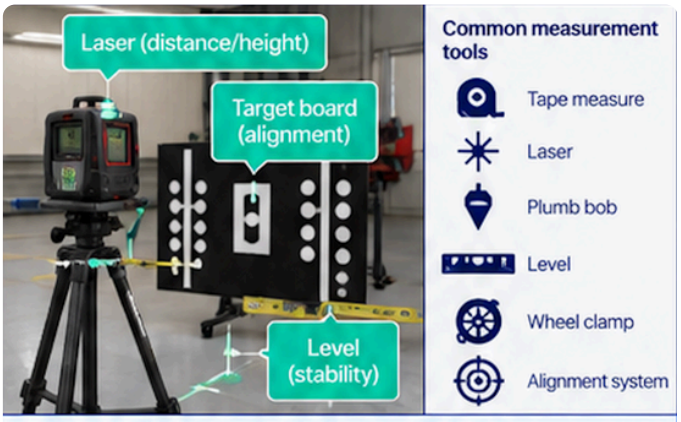
- Target part number and specification
- Photos (setup and close-up of ID)
- Distance, height and angle measurements
- Scan tool results
- OEM procedure reference
- Completion result and exceptions

Common Pitfalls to Avoid

- Printing an unofficial target.
- Using a dirty, warped or faded board
- Selecting the wrong target face or revision.
- Incorrect placement, height, angle or orientation.

38. Measurement Tools

Tape measures, lasers, plumb bobs, levels, wheel clamps and alignment systems convert OEM dimensions into a physical setup.



Accurate, calibrated tools help convert OEM dimensions into a physical setup.

The Measurement Workflow

INSPECT. VERIFY. APPLY. MEASURE. CROSS-CHECK.

Inspect tool

Check condition, calibration date and proper operation before use.

Verify calibration

Confirm the tool is current and measuring correctly per OEM requirements.

Apply compensation

Account for tool mounting, runout or setup offsets as required by the OEM procedure.

Measure datum

Capture the required dimensions, heights and angles at the specified points.

Cross-check

Verify measurements with a second method or tool to confirm accuracy.

Why This Matters in a Repair Facility

Tool uncertainty and user technique become part of the calibration result. Calibrated and properly used tools help reproduce OEM dimensions.

A good setup starts with accurate tools.

Practical Shop Example

SAME VEHICLE, DIFFERENT TOOL CONDITION..

A laser is accurate but mounted on a wheel clamp with runout. The measurement is off and the target is placed incorrectly.

The laser is calibrated and properly mounted. Compensation is applied and the target is placed at the correct height and distance.

A laser is accurate but mounted on a wheel clamp with runout. Compensation is completed before establishing centerline.

A TECH/ESTIMATOR SHOULD ASK

- What changed?
- Which tools are required?
- Are the tools calibrated and in good condition?
- Does the OEM procedure require tool-specific steps or compensation?
- How are the measurements verified?

FOR THE EVIDENCE FILE

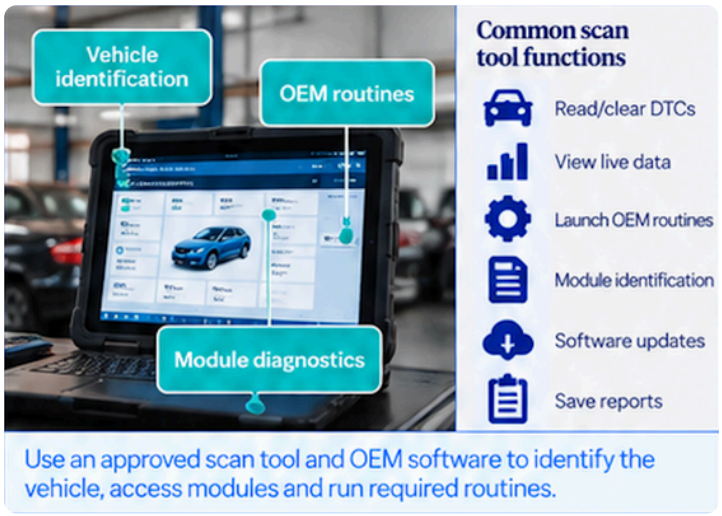
- Tool calibration records
- Photos of tool setup
- Measurement values (distance, height, angle)
- OEM procedure reference
- Cross-check documentation
- Completion result and exceptions

Common Pitfalls to Avoid

- Measuring from trim instead of the specified datum.
- Using damaged or out-of-calibration tools.
- Skipping required compensation.
- Not cross-checking measurements.

39. Scan Tools & OEM Software

A scan tool communicates with modules, reads DTCs and data, and launches routines. OEM software may be required for complete coverage or official workflows.



Use an approved scan tool and OEM software to identify the vehicle, access modules and run required routines.

The Scan Tool Workflow

CONFIRM. UPDATE. IDENTIFY. SCAN. SAVE.

Confirm coverage

Verify the tool and software support the vehicle, modules and required routines.

Update software

Use the latest software version and required OEM subscriptions.

Identify vehicle

Confirm VIN, model, trim and options in the scan tool or OEM platform.

Run full scan

Check DTCs, view data, and launch required OEM routines (e.g., calibrations, tests).

Save native report

Document scan results, routines performed and completion status in the native format

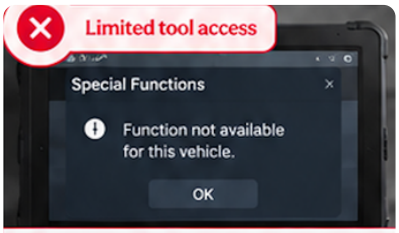
Why This Matters in a Repair Facility

The label "OEM compatible" does not prove every module, test or report is supported. Using the correct tool and software ensures complete and accurate procedures.

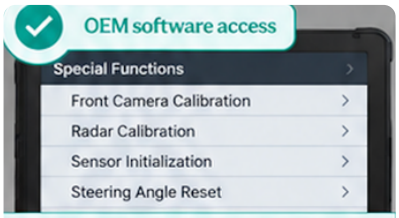
The right software unlocks the right information. OEM software provides the complete and official workflow.

Practical Shop Example

SAME VEHICLE, DIFFERENT TOOL AND ACCESS..



A generic tool cannot access a camera self-test. Required OEM software is used instead.



The OEM platform provides the required routine. The calibration completes successfully and is recorded in the native report.

A generic tool cannot access a camera self-test. The OEM platform is used as directed.

A TECH/ESTIMATOR SHOULD ASK

- Does the tool support this vehicle and the required routines?
- Which modules, tests or calibrations are available?
- Is OEM software or a subscription required?
- Are software versions up to date?
- What reports must be saved?

FOR THE EVIDENCE FILE

- Scan tool vehicle identification
- Software version and subscription status
- Pre- and post-scan reports (native file)
- OEM routine or calibration records
- Completion result and exceptions
- Screenshot or printout from the OEM platform

Common Pitfalls to Avoid

- Assuming no displayed codes means all modules were scanned
- Using outdated software or wrong vehicle selection.
- Relying only on a generic tool when OEM software is required
- Not saving the native report.

40. Cybersecurity & Gateway Access

Secure gateways restrict diagnostic commands and programming. Authorized credentials or approved access may be required.

A secure, authenticated connection is required to access protected modules and functions.

The Gateway Access Flow

CONFIRM. AUTHENTICATE. MAINTAIN. RUN. LOG.

Confirm account

Verify you have a valid OEM account| with the correct permissions.

→ **Authenticate**

Log in and complete any multi-factor authentication or approval steps required.

→ **Maintain connection**

Keep a stable internet connection during the entire session.

→ **Run authorized routine**

Access and perform the required diagnostic or calibration routine through the OEM platform.

→ **Log result**

Save the session log or completion report in the repair record.

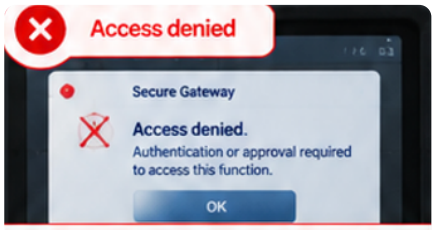
Why This Matters in a Repair Facility

A blocked routine is not necessarily a vehicle fault. Access must be legitimate and documented. Using authorized access protects the vehicle, the data and the repair process.

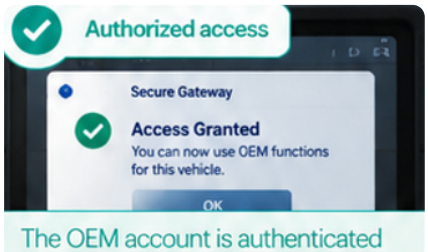
Access enables accurate diagnostics and repairs

Practical Shop Example

SAME VEHICLE, DIFFERENT ACCESS..



The tool reads codes but cannot execute calibration until secure-gateway authentication is completed.



The OEM account is authenticated and access is approved. The required routine is available and completes successfully.

A TECH/ESTIMATOR SHOULD ASK

- Does the shop have an active
- OEM account?
- Are the required access approvals in place?
- Is the internet connection stable?
- Does the current VIN require secure-gateway access?
- What records must be saved?

FOR THE EVIDENCE FILE

- Screen captures of authentication and access approval
- OEM account details (as required by shop policy)
- Scan tool session log or report
- Completed routine results
- OEM procedure reference
- Completion result and axceptions

Common Pitfalls to Avoid

- Bypassing security controls
- Misdiagnosing an access denial as a failed module.
- Using an outdated OEM account or software version.
- Letting the internet connection drop during the routine.

Knowledge Check | Test your understanding of this section's content.

PART II
VEHICLE, SENSOR, & SHOP BASICS

To view and complete the **Knowledge Check** on this section, please visit:

Link to online assessment: https://www.classmarker.com/online-test/start/?quiz=e3t6ad90fce35215

Quiz Password: vehiclebasics

You may also access the online assessment through this QR code:

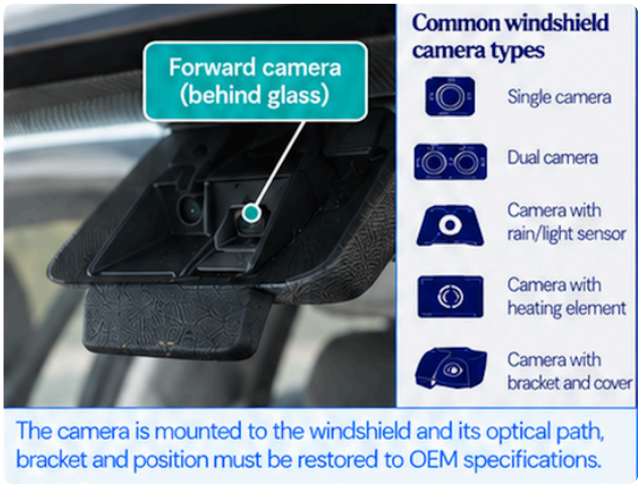
PART III

REPAIR-EVENT PLAYBOOKS

Objective: Equip technicians and repair professionals with practical, repair-event-specific workflows and quick-reference tools that reinforce foundational ADAS knowledge. Through structured checklists, decision-making guides, and real-world repair scenarios, technicians will learn to identify calibration triggers, verify OEM requirements, document critical procedures, and navigate common service challenges. The goal is to translate technical knowledge into consistent, efficient, and safety-focused repair practices that support accurate diagnostics, proper calibration, and confident decision-making throughout the repair process.

41. Windshield Replacement Playbook

Replacing glass can change the camera's optical path, bracket relationship and vehicle-to-camera geometry. Follow the OEM procedure for the correct windshield option, installation and calibration.



The camera is mounted to the windshield and its optical path, bracket and position must be restored to OEM specifications.

The Windshield Replacement Flow

IDENTIFY. VERIFY. INSPECT. INSTALL. CALIBRATE.

Identify camera

Confirm the vehicle's forward-camera system, features and requirements using OEM information. and the scan tool.

Verify glass option

Use the correct windshield option with the proper fit, coating and camera provisions per OEM specifications.

Inspect bracket

Check the bracket for damage, verify proper hardware and ensure the camera will sit correctly.

Install and cure

Install the new windshield, transfer or replace the bracket as required and allow the adhesive to cure per OEM specs.

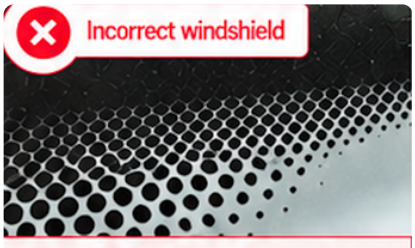
Calibrate / verify

Perform required camera aiming or learning, confirm no DTCs and verify system operation.

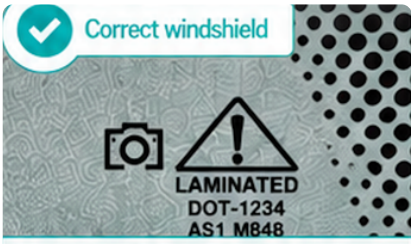
Why This Matters in a Repair Facility

Many forward-camera systems require aiming or learning after windshield replacement; some also specify glass inspection, bracket checks and DTC procedures. The right glass and a proper install keep the camera seeing what it should. Follow OEM requirements for glass, bracket and calibration.

Practical Shop Example



A windshield without the correct camera option is installed. The camera cannot see through the glass correctly, causing calibration to fail.



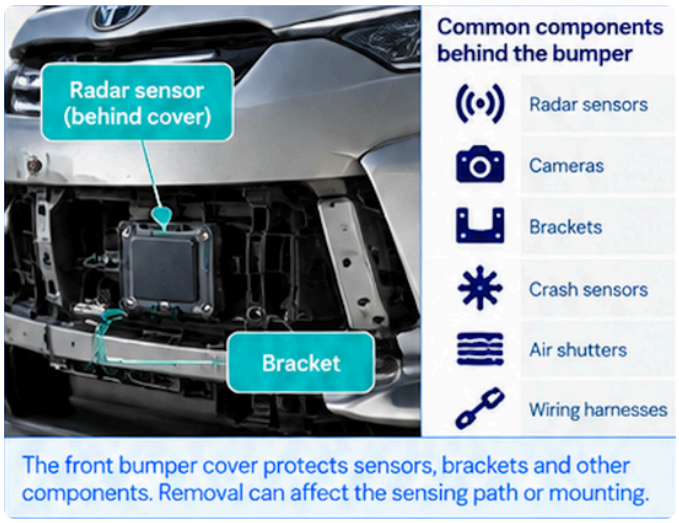
The correct OEM windshield option with the proper camera provision is installed. The camera calibrates successfully and operates as intended.

Common Pitfalls to Avoid

- Sending the vehicle before adhesive cure or full reassembly.
- Assuming a reused camera means no calibration.
- Using an incorrect windshield option or aftermarket glass without the required camera features
- Skipping bracket inspection or using damaged hardware.

42. Front Bumper Removal Playbook

Front bumper removal can affect sensors, radar, brackets and transmission zones. Follow OEM procedures to avoid damage and maintain calibration requirements.



The front bumper cover protects sensors, brackets and other components. Removal can affect the sensing path or mounting.

The Front Bumper Flow

MAP HARDWARE. RECORD. PROTECT. INSPECT. FOLLOW OEM.

Map hardware

Identify and document sensors, brackets, wiring and other components before removal using OEM information.

Record pre-state

Take photos of sensor locations, bracket positions, connector routing and any adjustments before removal.

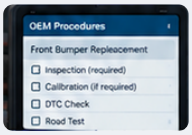
Protect sensor

Follow OEM guidance to protect sensors and surrounding components during removal.

Inspect after install

Inspect brackets, sensors, connectors and wiring after installation for damage, correct fit and proper routing

Follow OEM trigger



Complete any required inspection, calibration or verification steps per OEM procedure.

Why This Matters in a Repair Facility

Bumper removal is not always a calibration requirement, but it is a common research trigger when the sensing path or mounting can be affected. Remove with a plan. Reinstall with purpose. Proper documentation and inspection prevent unnecessary calibration and repeat repairs.

Practical Shop Example

The bumper was removed without documenting sensor and bracket positions. The radar bracket was bent during removal, requiring replacement and calibration.

Hardware was mapped and photographed before removal. Components were protected, reinstalled correctly and inspected. No damage or calibration required per OEM procedure.

Common Pitfalls to Avoid

- Blanket billing every bumper R&R without procedure support.
- Connector or wiring damage during removal or reinstallation.
- Failing to inspect brackets and mounting hardware
- Assuming bumper removal always requires calibration.
- Leaving the radar powered or connected when specified by OEM to disconnect.

43. Front Bumper Repair & Refinish

Repairs add heat, filler, weld material, primer, basecoat and clearcoat that may change radar or ultrasonic performance. Follow OEM restrictions for the specific zone, material and repair method.

Certain areas of the front bumper cover are within the radar or ultrasonic sensing zone. Repair materials and finish thickness can affect sensor performance.

The Front Bumper Flow

LOCATE ZONE. RETRIEVE LIMITS. CHOOSE REPAIR. CONTROL FINISH. INSPECT.

Locate zone

Identify radar or ultrasonic zones using OEM information. Mark the area before repair.

Retrieve limits

Review OEM repair procedures for allowed materials, maximum coating thickness and repair methods.

Choose repair/replace

Determine if the bumper can be repaired or if replacement is required based on damage type, location and OEM restrictions.

Inspect after install

Use approved materials and follow OEM coating and thickness limits. Avoid metallic films or unauthorized products.

Follow OEM trigger

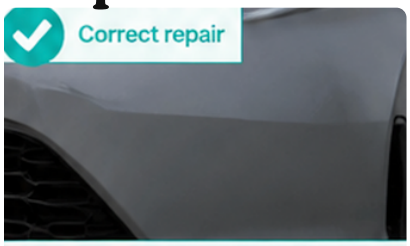
Inspect the repair and finish thickness. Verify sensor performance and calibrate if required per OEM procedure.

Why This Matters in a Repair Facility

OEM restrictions can apply to specific zones, maximum coating, damage type or required replacement. Improper materials or excessive coating thickness can affect radar or ultrasonic performance. Follow the OEM limits for materials and coating thickness. A correct repair protects sensor performance.

Practical Shop Example

Excessive filler and coating thickness in the radar zone can reduce sensor performance. The repair does not follow OEM limits.



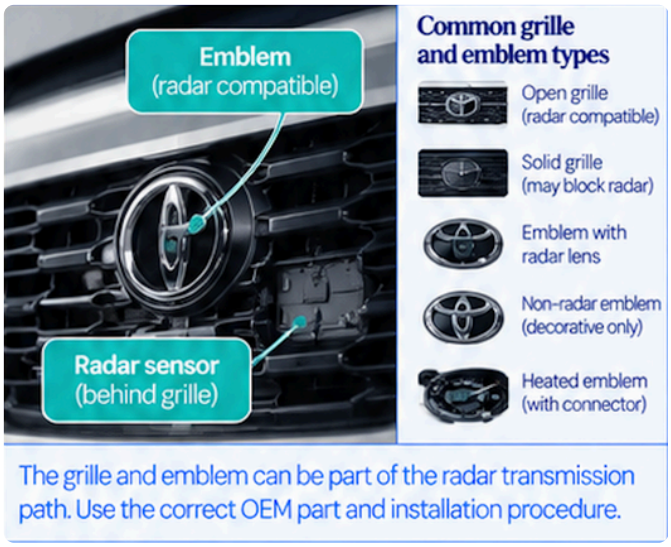
The bumper is repaired using OEM-approved materials and thickness limits. The finish is even and within specification, maintaining radar performance.

Common Pitfalls to Avoid

- Sanding or buffing a sensor/radar dome.
- Applying film or metallic coating without OEM approval.
- Exceeding maximum coating thickness.
- Using non-approved repair materials or methods.
- Repairing a bumper that must be replaced per CEM procedure.

44. Grille or Emblem Replacement

Grilles and emblems may be part of the radar transmission path. Visually similar parts can have different radar-compatible construction. Follow OEM procedures and verity the part number before installation.



The grille and emblem can be part of the radar transmission path. Use the correct OEM part and installation procedure.

The Grille Replacement Flow

IDENTIFY. VERIFY. INSPECT. INSTALL. AIM IF REQUIRED.

Identify radar path

Confirm the vehicle uses a front radar and identify the radar transmission path through the grille or emblem using OEM information.

Verify part number

Confirm the correct OEM grille or emblem part number. Ensure the replacement part is radar-compatible.

Inspect heater/ connector

Inspect the emblem or grille for heating elements, connectors and mounting features. Verify hardware and clips are included and undamaged.

Install / torque

Install the grille or emblem using OEM procedure and torque specifications. Ensure proper alignment and secure all clips and fasteners.

Aim if required

Perform radar aiming or calibration if required by the OEM procedure after installation.

Why This Matters in a Repair Facility

Visually identical grilles or emblems may have different radar-compatible construction. Using the wrong part can block or distort the radar signal and affect ADAS performance. Looks the same does not mean it works the same. Always verify OEM part numbers and compatibility.

Practical Shop Example

A non-radar grille was installed. The solid pattern blocks the radar signal, causing reduced or no radar function and DTCs.

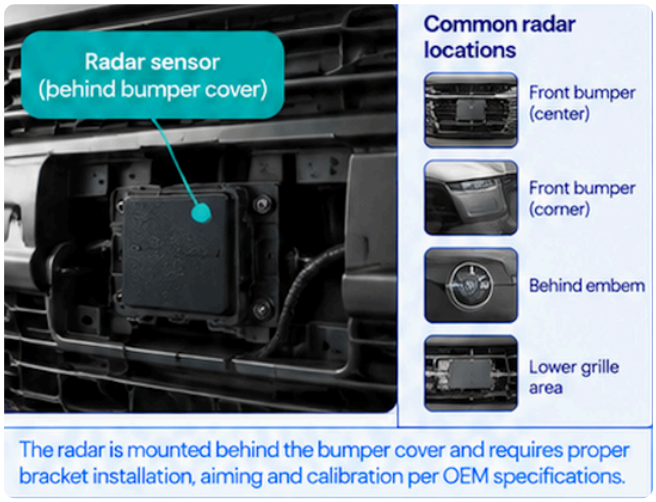
The correct OEM grille with radar-compatible construction is installed. The radar signal passes through and the system operates as intended.

Common Pitfalls to Avoid

- Using appearance as proof of radar compatibility.
- Ignoring clips or features that determine emblem angle.
- Installing a non-radar grille or decorative emblem.
- Not verifying the OEM part number.
- Skipping radar aiming when required.

45. Radar Replacement

Replacing a radar can require programming, coding, mounting inspection and calibration. Follow OEM procedures to ensure the correct part, installation and setup.



The radar is mounted behind the bumper cover and requires proper bracket installation, aiming and calibration per OEM specifications.

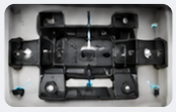
The Radar Replacement Flow

VERIFY PART POLICY. INSPECT BRACKET. INSTALL. PROGRAM. CALIBRATE.

Verify part policy

Confirm the correct OEM radar part number and verify if used or aftermerket components are allowed per OEM policy.

Inspect bracket



Inspect the bracket, mounting points and surrounding structure for damage. Verify the radar will sit correctly and is secure.

Install / torque

Install the radar and mounting hardware per OEM procedure. Torque fasteners to specification and ensure proper alignment.

Program / configure

Complete any required programming, coding or configuration after installation using OEM software.

Calibrate

Perform static or dynamic calibration per OEM procedure. Verify radar operation and check for DTCs after calibration.

Why This Matters in a Repair Facility

Component identity and physical aim are separate. Used or aftermarket ADAS components may be prohibited by OEM policy. Proper installation, programming and calibration ensure the radar functions as intended. The right part, installed and calibrated correctly, keeps ADAS working as designed.
Always follow OEM requirements for parts, installation and setup

Practical Shop Example

The radar was installed with a damaged bracket and not torqued to specification. The radar is misaligned and cannot be calibrated, resulting in DTCs and limited system function.

The radar is installed with a new bracket, torqued to specification and properly aligned. The radar is programmed and calibrated successfully with no DTCs.

Common Pitfalls to Avoid

- Installing a used radar without checking OEM prohibition.
- Aiming before coding or structural repair.
- Using the wrong part number or non-OEM component.
- Not inspecting or replacing a damaged bracket.
- Skipping required programming or calibration steps.

46. Rear Bumper Replacement

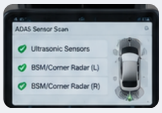
Rear covers often conceal BSM/corner radars and ultrasonic sensors. Follow OEM procedures to preserve sensor side, orientation, brackets, transmission zones and refinishing restrictions.

Rear bumpers may contain BSM/corner radars and ultrasonic sensors. Sensor side, orientation and mounting position must be maintained.

The Rear Bumper Flow

SCAN/MAP SENSORS. INSPECT BRACKETS. VERIFY COVER. TRANSFER CORRECTLY. CALIBRATE.

Scan/map sensors



Scan the vehicle and map all rear sensors, Document sensor type, location, side and part numbers using OEM information.

Inspect bracket

Inspect sensor brackets, mounting points and surrounding structure for damage. Verify the correct brackets are used and are secure.

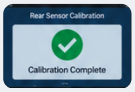
Verify cover

Verify the new or repaired cover has the correct sensor provisions, hole size and orientation per OEM specifications.

Transfer correctly

Transfer sensors, brackets and wiring correctly. Maintain sensor side (left/right) and orientation. Ensure connectors and clips are properly seated and undamaged.

Calibrate/ learn



Perform required calibration or learning procedures per OEM specifications. Verify sensor operation and check for DTCs after installation.

Why This Matters in a Repair Facility

Sensor side, orientation, brackets, transmission zones and refinishing restrictions must be preserved Visually identical parts may not be interchangeable, and incorrect installation can affect ADAS function. Always verify OEM information for sensor location, side and part numbers.

Practical Shop Example

Left and right BSM sensors were installed on the wrong sides. The system did not detect vehicles correctly, resulting in DTCs and reduced functionality.

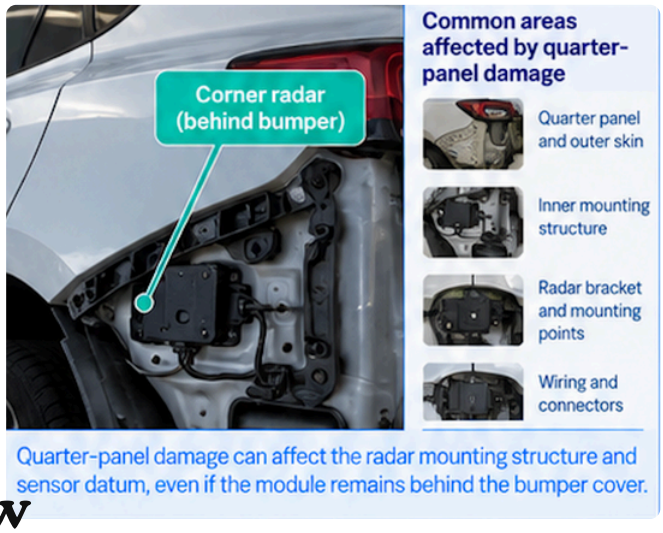
Sensors were mapped, transferred in the correct positions and installed using proper brackets. The system calibrates successfully and operates as intended.

Common Pitfalls to Avoid

- Swapping left and right side sensors.
- Failing to inspect quarter mounting points.
- Reusing damaged brackets or clips.
- Incorrect sensor orientation or connector routing.
- Skipping required calibration or learning steps.

47. Quarter-Panel Repair

Quarter-panel damage can move the corner radar mounting structure, even when the module sits behind the bumper. Measure, repair and verify the radar datum before final assembly and calibration.



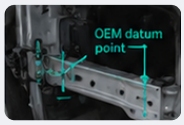
The Quarter-Panel Repair Flow

TRACE IMPACT. MEASURE STRUCTURE. INSPECT DATUM. FINISH REPAIR. CALIBRATE.

Trace impact

Assess the impact and extent of damage to the quarter panel and adjacent structures. Identify potential effects on the radar mounting area.

Measure structure



Measure the mounting structure and datum points per OEM specifications. Compare to pre-repair or reference dimensions.

Inspect radar datum

Inspect the radar mounting bracket and surrounding structure for damage or movement. Verify mounting points, alignment and integrity before proceeding.

Finish repair

Complete structural repairs per OEM procedure and ensure mounting points are restored to specification and free of distortion.

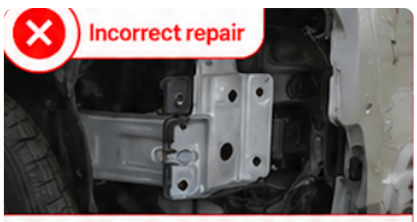
Calibrate/ learn

Perform required calibration or verification after structural repair and reassembly. Confirm proper radar operation and check for DTCs.

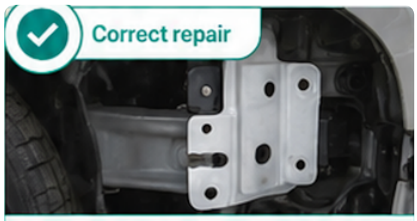
Why This Matters in a Repair Facility

Side or rear impacts can alter the sensor datum and transmission path, even if the radar module is not directly damaged. Proper measurement, structural repair and verification are essential to maintain radar performance and vehicle safety. Structural integrity protects ADAS function. Even small shifts in mounting structure can affect radar accuracy.

Practical Shop Example



The quarter panel was repaired without verifying the radar mounting structure. The bracket is misaligned, causing an inaccurate radar datum and calibration failure.



The quarter panel was repaired per OEM specifications. Mounting points were measured and restored to the correct position. The radar calibrates successfully with no DTCs.

Common Pitfalls to Avoid

- Checking only bumper cover damage.
- Aiming or calibrating before structural dimensions are final.
- Missing damage to mounting brackets or datum points.
- Not following OEM measurement and repair procedures.
- Skipping post-repair calibration or verification.

48. Door Replacement & Hinge Adjustment

Door position and fitment can influence mirror-mounted cameras and side sensors. Follow OEM procedures to ensure proper fit, gap, flushness and hinge torque, then calibrate camera(s) as required.

Door fit, gap and position can affect the field of view and alignment of mirror-mounted cameras and side sensors.

The Door Replacement Flow

FIT HINGES. SET GAPS. ROUTE HARNESS. SEAT STRIKER. CALIBRATE.

Fit hinges

Install the door and hinges per OEM procedure. Torque hinge bolts to specification and check for smooth operation.

Set gaps

Adjust door position to achieve OEM-specified gap and flushness at all reference points (top, bottom, front and rear edges).

Route harness

Inspect and properly route the wiring harness through the door and body. Ensure connectors are seated and secured, with no pinching or damage.

Seat striker

Adjust the door striker so the door closes correctly and matches OEM specifications. Verify consistent latching and operation.

Calibrate camera

Perform camera calibration as required by OEM procedure after final adjustment and verification.

Why This Matters in a Repair Facility

Door position and fitment can affect the viewing angle and operation of mirror-mounted cameras and side sensors. Proper hinge installation, gap and flushness, and wiring routing are essential to ensure accurate sensor performance and vehicle safety. A properly fitted door keeps ADAS working as designed. Always follow OEM specifications for fit, gap and hinge torque.

Practical Shop Example

The door is misaligned with an uneven gap, changing the mirror camera's viewing angle. This can cause detection errors and require recalibration after adjustment.

The door is properly aligned with even gap and flushness per OEM specifications. The mirror camera maintains the correct field of view and operates as intended.

Common Pitfalls to Avoid

- Calibrating before final door adjustment.
- Pinching camera/network wiring in the hinge area.
- Incorrect gap or flushness outside OEM specifications.
- Not checking for camera calibration requirements after door replacement or adjustment.

49. Mirror Replacement

Mirror assemblies can include cameras, BSM lamps, heating, folding memory and control modules. Follow OEM procedures to ensure proper installation, initialization and camera calibration.

Mirror assemblies may include cameras, BSM lamps, heating, folding and control modules. Always follow OEM procedures.

The Mirror Replacement Flow

VERIFY OPTIONS. INSTALL ASSEMBLY. INITIALIZE FOLD. CHECK IMAGE. CALIBRATE.

Verify options

Confirm the correct OEM mirror assembly based on vehicle options (camera, BSM, heating, folding, memory, control module). Verify part number and supersessions.

Install assembly

Install the new mirror assembly per OEM procedure. Ensure connectors are properly seated and the harness is routed and secured without pinching.

Initialize fold

Initialize the folding function and confirm all electrical features operate correctly (heating, turn signal, puddle lamp, memory, BSM).

Check image

Confirm the camera image displays correctly with no distortion, blank areas or error messages.

Calibrate

Perform camera calibration per OEM procedure after installation and initialization. Verify proper operation and check for DTCs.

Why This Matters in a Repair Facility

Mirror assemblies play a critical role in driver assistance systems. A new assembly may require coding and camera calibration, and mechanical detent accuracy affects image quality and system performance. A properly installed and calibrated mirror keeps ADAS working as intended. Follow OEM requirements for parts, installation and setup.

Practical Shop Example

The mirror was installed without proper seating of the camera or harness routing. The surround-view image was distorted and required recalibration after correction.

The mirror was installed per OEM procedure with correct harness routing and camera position. The surround-view image displays correctly and calibrates successfully with no DTCs.

Common Pitfalls to Avoid

- Ordering by paint/shape only.
- Ignoring a clicking or incomplete fold mechanism.
- Not seating connectors or pinching the harness.
- Skipping required camera calibration.
- Failing to initialize folding, heating or memory functions.

50. Liftgate or Tailgate Repair

Rear cameras and surround-view cameras are mounted to liftgates and tailgates whose alignment affects view and guidelines. Ensure proper repair, fit and adjustment before installation, programming and calibration.

The Tailgate Repair Flow

REPAIR CLOSURE. SET FIT/LATCH. INSTALL CAMERA. PROGRAM IF NEEDED. CALIBRATE.

Repair closure

Complete structural and cosmetic repairs. Ensure the liftgate or tallgate closes correctly with even gaps and proper alignment.

Set fit/latch

Adjust hinges, strikers and latch to achieve proper fit, flushness and gap measurements per OEM specifications.

Install camera

Install the camera and related components (handle, switch, wiring harness) per OEM procedure. Ensure proper seating and routing.

Program if needed

Program or initialize the camera and any related modules (e.g., liftgate control module) if required by OEM procedure.

Calibrate

Perform camera calibration or verification per OEM procedure. Confirm proper image, guidelines and system operation.

Why This Matters in a Repair Facility

Panel position, latch adjustment and camera bracket alignment must be final before calibration. An improperly aligned liftgate or tailgate can cause inaccurate camera views and guidelines, affecting parking assistance and overall safety. Proper fit and alignment keep ADAS working as designed. Follow OEM specifications for repair, adjustment and calibration.

Practical Shop Example

The liftgate was misaligned and the camera bracket was not adjusted. The rear camera view was off-center and guidelines were inaccurate, requiring readjustment and recalibration.

The liftgate was properly aligned with correct gaps and latch adjustment. The rear camera was installed and calibrated per OEM procedure, displaying accurate guidelines with no DTCs.

Common Pitfalls to Avoid

- Aiming with trim or handle loose.
- Confusing rear-camera guidelines with full surround calibration.
- Not adjusting hinges, strikers or latch to OEM specifications.
- Skipping required programming or calibration steps.
- Improperly routing or pinching the wiring harness.

51. Wheel Alignment Playbook

Proper wheel alignment is essential for ADAS performance. The final thrust line and steering angle determine the vehicle's path reference for many satety systems.

Common ADAS systems affected by alignment

Lane-keeping assist (LKA)

Adaptive cruise control (ACC)

Automatic emergency braking (AEB)

Blind spot monitoring (BSM)

Proper wheel alignment ensures the vehicle's thrust line and steering angle match OEM specifications for accurate ADAS performance.

The Wheel Alignment Flow

INSPECT CHASSIS. ALIGN WHEELS. CENTER STEERING. INITIALIZE SAS. CALIBRATE.

Inspect chassis

Check for suspension damage, bent components or worn parts that could affect alignment. Repair as needed before proceeding.

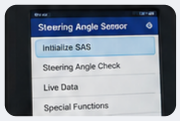
Align wheels

Perform a complete 4-wheel alignment to OEM specifications, ensuring correct camber, caster, toe and thrust angle.

Center steering

Center the steering wheel per OEM procedure to ensure the steering angle sensor (SAS) reads correctly.

Initialize SAS



Initialize or relearn the steering angle sensor (SAS) after alignment, following OEM sequence and requirements.

Calibrate

Perform required ADAS calibrations (static or dynamic) per OEM procedure after alignment and SAS initialization.

Why This Matters in a Repair Facility

The final thrust line and steering angle determine the vehicle's path reference for many ADAS systems. An incorrect alignment can lead to inaccurate sensor readings, degraded system performance and reduced vehicle safety. Proper alignment keeps ADAS systems working as intended. Even small change in thrust angle or steering angle can affect system accuracy.

Practical Shop Example

The vehicle was aligned to OEM specifications, the steering wheel was centered and the SAS was initialized. ADAS systems calibrated successfully with no DTCs.

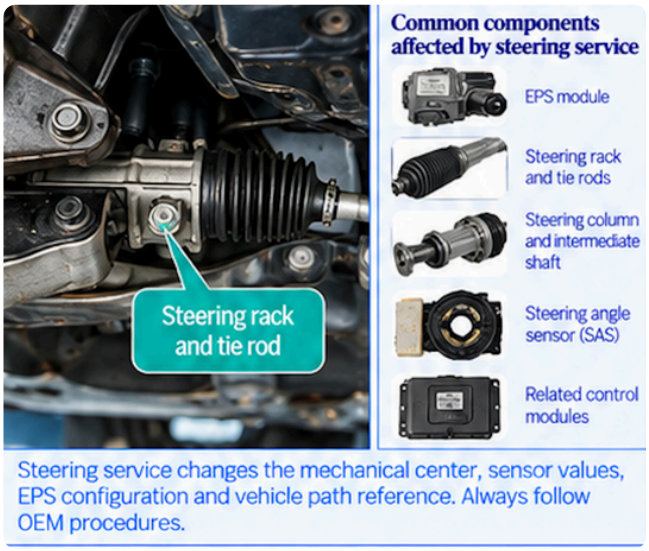
The vehicle was aligned outside of OEM specifications and the steering wheel was off-center. ADAS systems reported warning messages and required recalibration after correction.

Common Pitfalls to Avoid

- Calibrating before alignment is complete.
- Changing alignment settings after calibration.
- Not centering the steering wheel before initializing SAS.
- Ignoring chassis or suspension damage.
- Using non-OEM alignment specifications.

52. Steering Rack or Column Replacement

Steering service changes the mechanical center, sensor values, EPS contiguration and path reference. Follow OEM procedures for programming, alignment, zero point setup and ADAS calibration.



Steering service changes the mechanical center, sensor values, EPS configuration and vehicle path reference. Always follow OEM procedures.

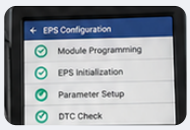
The Steering Rack Flow

INSTALL/TORQUE. PROGRAM EPS. ALIGN VEHICLE. SET ZERO POINTS. CALIBRATE.

Install/torque

Install the new steering rack or column per OEM procedure. Torque all fasteners to specification and inspect related components (tie rods, intermediate shaft, mounts and boots).

Program EPS



Program or initialize the EPS system per OEM procedure. Verify no DTCs are present and confirm proper configuration.

Align vehicle

Perform a complete 4-wheel alignment to OEM specifications, ensuring correct thrust angle, steering center and vehicle geometry.

Set zero points

Initialize or relearn the steering angle sensor (SAS) and set the zero point per OEM sequence.

Calibrate

Perform required ADAS calibrations (static or dynamic) per OEM procedure after alignment and SAS initialization. Verify proper operation and check for DTCs.

Why This Matters in a Repair Facility

Steering service affects the vehicle's mechanical center, sensor values, EPS configuration and path reference. Proper programming, alignment and zero point setup are essential to ensure accurate ADAS performance and vehicle safety. A correctly aligned steering system keeps ADAS working as intended. Always follow OEM specifications for programming, alignment and calibration.

Practical Shop Example

The steering angle sensor was not initialized after rack replacement. The steering wheel was off-center and ADAS systems reported faults, requiring recalibration.

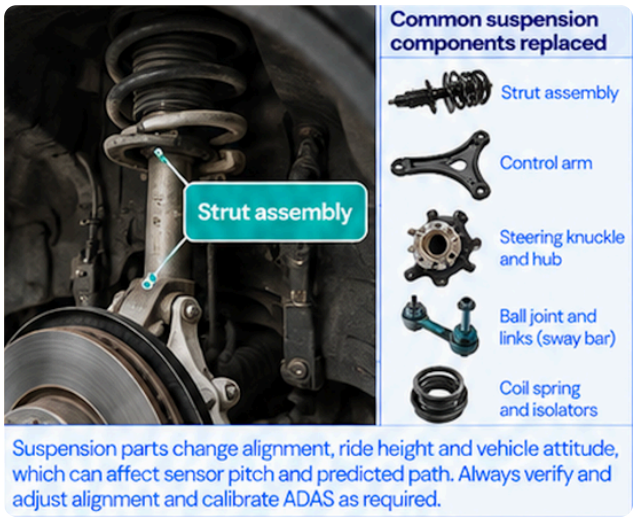
The steering angle sensor was initialized per OEM procedure. The steering wheel was centered and ADAS systems calibrated successfully with no DTCs.

Common Pitfalls to Avoid

- Skipping EPS configuration.
- Setting SAS zero with the rack off center.
- Not performing a full 4-wheel alignment.
- Ignoring intermediate shaft or tie rod inspection.
- Calibrating ADAS before completing alignment and SAS initialization.

53. Strut, Control Arm, or Knuckle Replacement

Suspension parts affect alignment, ride height and vehicle attitude. Even non-ADAS components can change sensor pitch or predicted path, so proper measurement, alignment and calibration may be required.



Suspension parts change alignment, ride height and vehicle attitude, which can affect sensor pitch and predicted path. Always verify and adjust alignment and calibrate ADAS as required.

The Replacement Workflow

COMPLETE MECHANICAL REPAIR. SETTLE SUSPENSION. MEASURE HEIGHT. ALIGN. RESEARCH/CALIBRATE.

Complete Mechanical Repair

Install the new strut, control arm or knuckle per OEM procedure. Torque all fasteners to specification and inspect related components (bushings, ball joints, links).

Settle suspension

Settle the suspension per OEM procedure (e.g., lower vehicle, roll or bounce, then re-measure) to ensure accurate ride height and alignment measurements.

Measure height

Measure ride height at all specified points and verify the vehicle is within OEM specifications before alignment and calibration.

Align

Perform a complete wheel alignment to OEM specifications, including thrust angle and steering angle as required.

Calibrate

Review OEM procedures to determine required initializations, sensor calibration (static or dynamic), and any SAS reset after alignment.

Why This Matters in a Repair Facility

Suspension components change vehicle alignment, ride height and attitude, which can affect sensor pitch or predicted path. Proper measurement, alignment and ADAS calibration may be required to maintain system accuracy and vehicle safety. Proper suspension repair and alignment keep ADAS working as intended. Always follow OEM specifications for repair, ride height, alignment and calibration.

Practical Shop Example

The strut was replaced but the vehicle ride height was not verified and alignment was not performed. The camera's predicted path was off, requiring recalibration and additional adjustments.

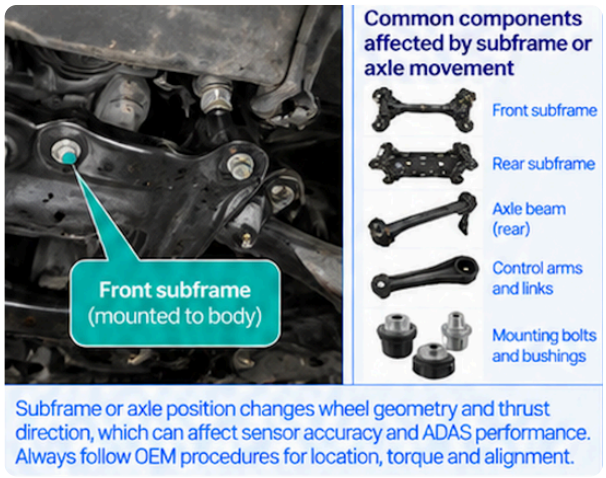
The strut was replaced, the suspension was properly settled, ride height was verified and a complete alignment was performed. ADAS systems operated correctly with no DTCs.

Common Pitfalls to Avoid

- Using pre-repair alignment values.
- Ignoring spring seating or ride-height difference.
- Skipping ride height measurement before alignment.
- Not following OEM procedures for sensor calibration.
- Overlooking related components (bushings, ball joints, links).

54. Subframe or Axle Movement

Subframe or axle position changes wheel geometry and thrust direction, which can affect ADAS performance. Proper location, torque, alignment and calibration are required.



The Subframe/Axle Movement Flow

LOCATE SUBFRAME. TORQUE AT CONDITION. MEASURE STRUCTURE. ALIGN THRUST. INITIALIZE/CALIBRATE.

Locate subframe

Verify subframe or axle position and mounting points per OEM procedure. Inspect for damage and defective bushings, hardware or frame attachments.

Torque at condition

Torque all fasteners to OEM specification at the required ride height or loaded condition. Follow the correct sequence and use new hardware as specified.

Measure structure

Measure subframe/axle position and verify structural dimensions against OEM specifications to confirm correct location.

Align thrust

Perform a complete four-wheel alignment and verify the thrust angle is within OEM specifications.

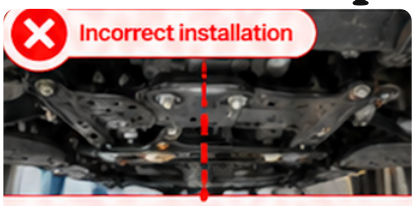
Calibrate

Initialize or calibrate ADAS systems per OEM procedure after subframe or axle movement and alignment. Verify proper operation and check for DTCs.

Why This Matters in a Repair Facility

Subframe or axle position changes wheel geometry and thrust direction, which can affect sensor alignment and ADAS performance. The body can be dimensionally correct while the chassis path is not. A properly positioned subframe or axle keeps ADAS working as intended. Follow OEM specifications for location, torque, alignment and calibration.

Practical Shop Example



The subframe was installed off center and not torqued at the required ride height. The thrust angle was incorrect and ADAS systems reported faults, requiring realignment and recalibration.

The subframe was installed per OEM procedure, torqued at the required ride height and correctly aligned. The thrust angle was within specifications and ADAS systems calibrated successfully with no DTCs.

Common Pitfalls to Avoid

- Performing only front toe adjustment.
- Calibrating before final torque/settling.
- Not torquing at specified ride height or loaded condition.
- Ignoring structural measurement requirements.
- Skipping alignment or thrust angle verification.

55. Tire or Wheel Changes

Changes in tire size, pressure or rolling circumference can alter ride height and wheel-speed relationships, which may affect ADAS performance. Always follow OEM specifications.

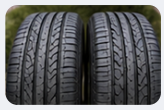
The Tire/Wheel Change Flow

VERIFY APPROVED SIZE. MATCH AXLE TIRES. SET PRESSURE. CHECK RIDE HEIGHT. RESEARCH PROCEDURE.

Verify approved size

Confirm the tire size, load rating and type meet OEM specifications. Check for any restrictions related to ADAS or vehicle systems.

Match axle tires



Ensure both tires on the same axle match in size, brand, model and tread pattern per OEM specifications.

Set pressure

Set tire pressure to the OEM specification (consider loaded conditions if required) and reset TPMS as needed.

Check ride height

Verify the vehicle's ride height is within OEM specifications after tire installation and pressure adjustment.

Research procedure

Review OEM procedures for the specific vehicle. Some vehicles require additional steps after tire or wheel changes (e.g., TPMS reset, ADAS calibration).

Why This Matters in a Repair Facility

Changes in tire size, pressure or rolling circumference can alter ride height and wheel-speed relationships, which may affect ADAS performance. OEM calibration triggers vary, but an abnormal setup can invalidate prerequisites and lead to inaccurate system operation. Even small changes in rolling circumference can affect system accuracy.

Practical Shop Example

Oversize tires were installed and pressure was set incorrectly. The vehicle's ride height was outside OEM specifications, which affected wheel-speed sensors and ADAS performance, requiring re-adjustment and calibration.

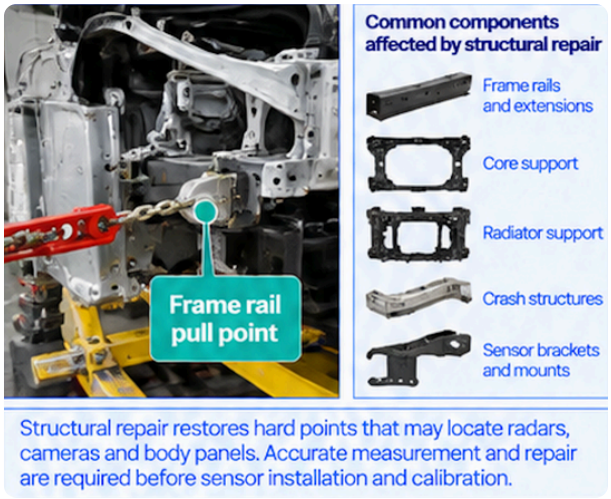
OEM-approved tires were installed, both axle tires matched, pressure was set to specification and ride height was verified. ADAS systems operated correctly with no DTCs after the tire change.

Common Pitfalls to Avoid

- Calibrating with temporary spare installed.
- Ignoring uneven tread depth.
- Mixing tire sizes or brands on the same axle.
- Incorrect tire pressure (especially loaded vs. unloaded).
- Not verifying ride height after tire installation.

56. Structural Pull or Rail Repair

Structural correction restores hard points that may locate radars, cameras and body panels. Accurate measurement and repair are required before sensor installation and calibration.



Structural repair restores hard points that may locate radars, cameras and body panels. Accurate measurement and repair are required before sensor installation and calibration.

The Structural Repair Flow

MEASURE BEFORE. PLAN PULL. CORRECT STRUCTURE. MEASURE AFTER. INSTALL/CALIBRATE.

Measure before

Perform a complete pre-repair measurement to document structural damage and identify affected hard points for sensors, brackets and body panels.

Plan pull

Plan the pulling procedure using OEM repair information, including pull points, direction, fixtures and target measurements.

Correct structure

Perform the structural pull and restore the frame/rail or core support to OEM specifications. Follow proper sequence and verify alignment during the repair.

Measure after

Complete a post-repair measurement to confirm all structural dimensions are within OEM specifications and hard points are correctly located.

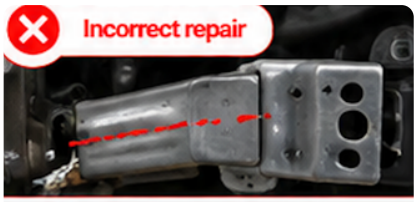
Install/ calibrate

Reinstall sensor brackets and components, then perform required ADAS calibrations per OEM procedure after structural repair is complete and verified.

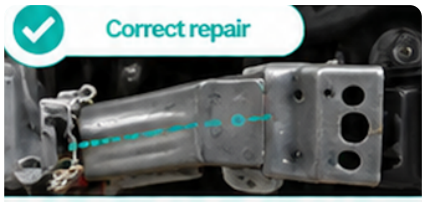
Why This Matters in a Repair Facility

Structural correction restores hard points that may locate radars, cameras and body panels. A successful ADAS calibration cannot certify structural dimensions, so proper measurement and repair are required before sensor installation and calibration. Follow OEM procedures for measurement, pulling and verification.

Practical Shop Example



The frame rail was not pulled to OEM specifications. Hard points were mislocated, causing the radar bracket to be out of position and resulting in calibration faults.



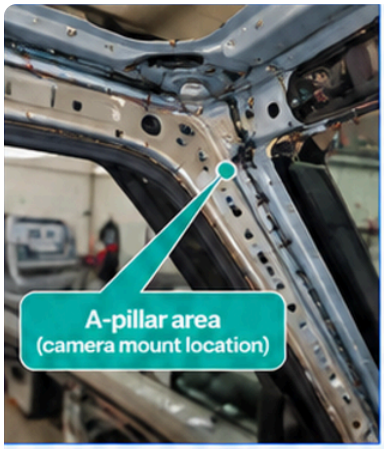
The frame rail was pulled to OEM specifications. Hard points were correctly restored, the sensor bracket was properly located and ADAS systems calibrated successfully with no DTCs.

Common Pitfalls to Avoid

- Using sensor aim as a structural measurement.
- Discarding final dimension evidence.
- Pulling without OEM repair information.
- Not verifying post-repair measurements.
- Calibrating before structural repair is complete and verified.

57. Roof, Header, or A-Pillar Repair

These areas locate the windshield, cameras, brackets, and other sensors, and can change glass opening geometry. Repairs and measurements are required before sensor installation and calibration.



Common components affected by roof, header or A-pillar repair

Windshield and glass opening

Roof panel and header

Camera brackets and mounts

A-pillar inner and outer panels

Forward-facing camera

Roof, header or A-pillar repairs can change the windshield location and glass opening geometry, which may affect camera position and aim. Always verify measurements and inspect brackets before calibration.

The Roof Repair Flow

MEASURE OPENING. REPAIR STRUCTURE. VERIFY GLASS FIT. INSPECT BRACKET. CALIBRATE.

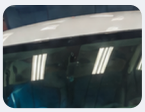
Measure opening

Measure the windshield glass opening at specified points per OEM procedure. Document dimensions before repairs.

Repair structure

Complete the structural repair to OEM specifications, including replacement panels, welds and corrosion protection.

Verify glass fit



Test fit the windshield and verify proper fit, gaps and alignment per OEM specifications before installing sensors or brackets.

Inspect bracket

Inspect camera brackets and mounts for damage, proper location and secure attachment. Replace or adjust as required.

Calibrate

Install components and perform required ADAS calibrations per OEM procedure after structural repair is complete and verified.

Why This Matters in a Repair Facility

The roof, header and A-pillar areas locate the windshield, cameras, brackets and other sensors. Camera position and pitch/yaw can shift even if the windshield installs without visible issues. Proper measurements and repairs are required to ensure accurate ADAS performance and vehicle safety. Accurate measurements and proper structural repair keep ADAS working as intended.

Practical Shop Example

Incorrect repair

The roof was repaired but the glass opening was out of specification. The forward camera's pitch was incorrect and ADAS systems reported faults, requiring recalibration and additional repair.

Correct repair

The roof was repaired per OEM specifications, the glass opening was within tolerance and the camera bracket was properly installed. ADAS systems calibrated successfully with no DTCs.

Common Pitfalls to Avoid

- Focusing only on leak prevention.
- Assuming a new windshield corrects roof geometry.
- Not verifying glass opening measurements.
- Reusing a damaged or misaligned camera bracket.
- Calibrating before structural repair is complete and verified.

58. Airbag Deployment & Collision Events

Deployment indicates significant crash energy and can create faults across restraints, brake, steering and ADAS networks. A complete diagnosis and required repairs must be performed before sensor aiming and calibration.

The Collision Event Flow

SECURE VEHICLE. FULL SCAN. REPAIR STRUCTURE/SRS. RESTORE NETWORKS. CALIBRATE/VERIFY.

Secure vehicle

Ensure the vehicle is safe to work on. Follow OEM procedures for post-collision handling, battery disconnect (if required) and restraint system precautions.

Full scan

Perform a complete vehicle scan of all systems (SRS, brake, steering. ADAS and related modules) and document all DTCs, freeze frame data and scan results.

Repair structure/SRS

Complete all required structural repairs and replace deployed SRS components (airbags, pretensioners, sensors, control module) per OEM specifications.

Restore networks

Clear crash-related codes after repairs, perform required initializations and restore network communication for all affected systems.

Calibrate

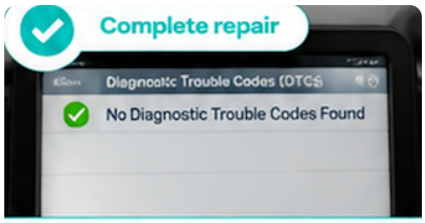
Perform required sensor calibrations (cameras, radars, steering angle, brake) and verify proper system operation with no DTCs.

Why This Matters in a Repair Facility

Airbag deployment indicates significant crash energy and can create faults across restraints, brake, steering and ADAS networks. ADAS calibration planning occurs after all required structural, restraint and electronic repairs are completed. Complete diagnosis and repairs are required before sensor aiming and calibration to ensure safe and accurate ADAS performance.

Practical Shop Example

The vehicle was repaired and calibrated with unresolved SRS and crash-related codes. ADAS systems reported faults and did not operate correctly, requiring additional diagnostics and repairs.



All required structural and SRS repairs were completed, crash-related codes were cleared, networks were restored and ADAS systems calibrated successfully with no DTCs.

Common Pitfalls to Avoid

- Calibrating early to speed estimating.
- Clearing crash-related codes without required repairs.
- Not following OEM post-collision procedures.
- Overlooking related system faults (brake, steering, ADAS).
- Assuming calibration will correct SRS or network faults.

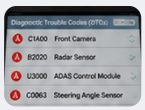
59. Module Replacement

Replacing an electronic control module may require initialization, software programming, immobilizer or security authorization, vehicle-specific configuration, and calibration to restore proper functionality. Each procedure must follow the OEM-specified sequence to ensure the module communicates correctly with other vehicle systems and retains the appropriate settings. Skipping or performing steps out of order can result in incomplete setup, diagnostic trouble codes (DTCs), communication failures, or malfunctioning ADAS features.

The Module Replacement Flow

CONFIRM DIAGNOSIS. VERIFY PART. SUPPORT VOLTAGE. PROGRAM/CONFIGURE. CALIBRATE/POST-SCAN.

Confirm diagnosis



Verify system faults and confirm the module needs replacement. Check for related DTCs and follow OEM diagnostics.

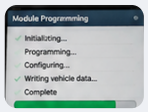
Verify part

Confirm the replacement module matches the OEM part number, software level and vehicle options (VIN, build data).

Support voltage

Maintain stable battery support voltage per OEM specifications during programming and initialization.

Program / configure



Program, configure and initialize the module using the OEM scan tool. Follow required security access or immobilizer procedures.

Calibrate

Perform required ADAS calibrations (camera, radar, steering angle, etc.) and complete a post-scan to verify proper operation with no DTCs.

Why This Matters in a Repair Facility

The correct order prevents incomplete setup and misloading DTCs. A new module may require initialization, programming, immobilizer/security access, configuration and calibration to operate correctly with the vehicle's ADAS systems. Follow the OEM procedure and correct sequence to ensure the new module is recognized, configured and calibrated.

Practical Shop Example

A module with the wrong part number was installed and programmed without verifying vehicle options. The system set multiple DTCs and calibration could not be completed, requiring reprogramming and the correct part.

The correct OEM module was installed, programming and configuration were completed per OEM procedure, and all required calibrations were performed. The system operated correctly with no DTCs.

Common Pitfalls to Avoid

- Replacing module without root-cause testing.
- Interrupting programming or skipping variant coding.
- Using the wrong part number or software level.
- Not maintaining proper battery support voltage.
- Skipping required calibrations after module replacement.

60. Battery Disconnect or Low Voltage

Battery events can cause cameras to learned values, set faults or interrupt network communication. Always follow OEM procedures when disconnecting the battery and after restoring power.

The Battery / Voltage Flow

TEST BATTERY. STABILIZE VOLTAGE. SCAN ALL MODULES. PERFORM LISTED RELEARNS. VERIFY AFTER DRIVE.

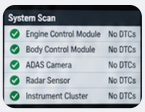
Test battery

Test battery condition and replace if needed. Ensure battery meets OEM specifications for the vehicle.

Stabilize voltage

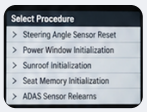
Use a battery maintainer to stabilize voltage during repairs, scans, programming or calibrations per OEM requirements.

Scan all modules



Perform a complete scan of all modules. Record and address any DTCs before continuing.

Perform listed relearns



Perform all OEM-specified relearns, initializations or calibrations that are required after a battery disconnect or low voltage event.

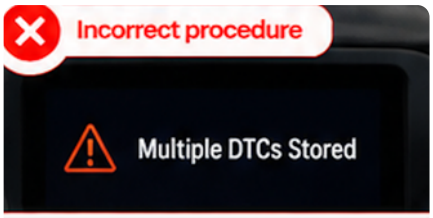
Calibrate

Verify proper operation after module sleep cycle and/or drive as required by the Ot.M. Confirm no DTCs return and systems operate as intended.

Why This Matters in a Repair Facility

A battery disconnect is not a universal calibration trigger, but it is a procedure and scan research trigger. Low voltage or loss of power can cause cameras to lose learned values, set faults or interrupt network communication. Maintain proper battery voltage during repairs, scans and programming to prevent module faults and ensure correct operation.

Practical Shop Example



The battery was disconnected without following OEM procedures. Multiple modules set DTCs, cameras lost learned values and required reinitialization and calibrations.

The battery was maintained per OEM procedure, all required initializations were completed, and a post-sleep drive verified proper operation with no DTCs.

Common Pitfalls to Avoid

- Automatically billing every calibration after disconnect.
- Ignoring low-voltage root cause.
- Not using a battery maintainer during programming.
- Skipping required initializations or relearns.
- Not verifying system operation after sleep/drive.

61. Software Update or Campaign

Software updates can change controller logic, configuration, calibration prerequisites or feature availability. Always follow OEM procedures and verify required initialization or calibration after updates.

The Software Update Flow

CHECK CAMPAIGN. READ BULLETIN. SUPPORT VOLTAGE. PROGRAM. PERFORM DIRECTED ROUTINES.

Check campaign

Check for open campaigns or software updates using the OEM portal. Review applicable bulletins and verify the update applies to the vehicle.

Read bulletin

Read the full bulletin and understand the requirements, including battery support, required tools, software level, and post-programming procedures.

Support voltage

Connect an approved battery maintainer and stabilize voltage per OEM specifications before programming to prevent interruptions.

Program

Program the module using the OEM scan tool and follow on-screen instructions. Verify the software level after programming is complete.

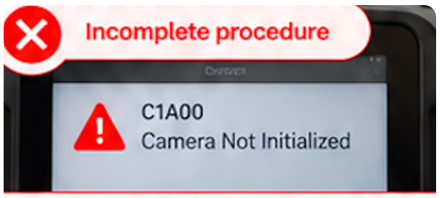
Perform directed routines

Perform all directed routines, such as initialization, calibration or adaptive leam. Clear DCs and verify proper system operation.

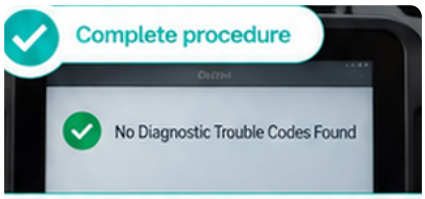
Why This Matters in a Repair Facility

A hardware-perfect vehicle may require post-programming initialization or calibration. Software updates can change controller logic, configuration, calibration prerequisites or feature availability, so following the correct OEM procedure ensures proper system operation. Always follow OEM procedures and verify required initialization or calibration after software updates to ensure correct system operation.

Practical Shop Example



The camera software was updated but the required initialization was not performed. The system set DTCs and ADAS features were disabled until the proper procedure was completed.



The software update was completed per OEM procedure, including initialization and calibration. No DTCs were present and ADAS features operated correctly.

Common Pitfalls to Avoid

- Assuming programming preserves every learned value.
- Using outdated procedure after software change.
- Not using battery support during programming.
- Skipping required initialization or calibration.
- Not verifying software level and system operation after update.

Knowledge Check | Test your understanding of this section's content.

PART III
REPAIR-EVENT PLAYBOOKS

To view and complete the **Knowledge Check** on this section, please visit:

Link to online assessment: https://www.classmarker.com/online-test/start/?quiz=4xj6ad91c58d64a9

Quiz Password: repairevents

You may also access the online assessment through this QR code:

PART IV

CALIBRATION, QUALITY, & DOCUMENTATION

Objective: Provide technicians and repair professionals with standardized calibration procedures, quality-control measures, and documentation practices that support safe, accurate, and efficient day-to-day operations. Through practical checklists, verification workflows, and OEM-aligned guidelines, technicians will learn to confirm calibration prerequisites, evaluate results, identify potential errors, and maintain complete, defensible service records. The goal is to establish consistent quality standards, improve operational efficiency, and ensure every vehicle is properly inspected, calibrated, documented, and verified before returning to service.

62. Writing a Complete Work Order

A complete work order connects the customer or insurer request, vehicle identity, repair trigger, requested operations and authorization. It ensures the technician has the correct information to perform and document all required ADAS procedures.

A complete work order should include:

Vehicle Identification
Year, make, model, VIN, mileage and drivetrain (gas, hybrid, EV).

Repair Event
What was repaired or replaced and why (collision, mechanical, glass, etc).

Requested operations
Scans, calibrations, programming or resets per OEM.

Authorization
Customer/insurer approval, line items and reference (RO, claim or estimate).

The Work Order Workflow

CAPTURE VIN. DESCRIBE REPAIR. LIST SYMPTOMS. DEFINE AUTHORIZATION. ASSIGN TECHNICIAN.

Capture VIN

Verify year, make, model, trim, drivetrain and mileage. Confirm the vehicle is the same as the one being repaired.

Describe repair

List all parts replaced repaired or adjusted (e.g., bumper, windshield, radar bracket, suspension). Include the reason for repair (collision, mechanical, glass, etc.).

List symptoms

Document any warning lights, DTCs, customer concerns or inoperative features reported before and after the repair.

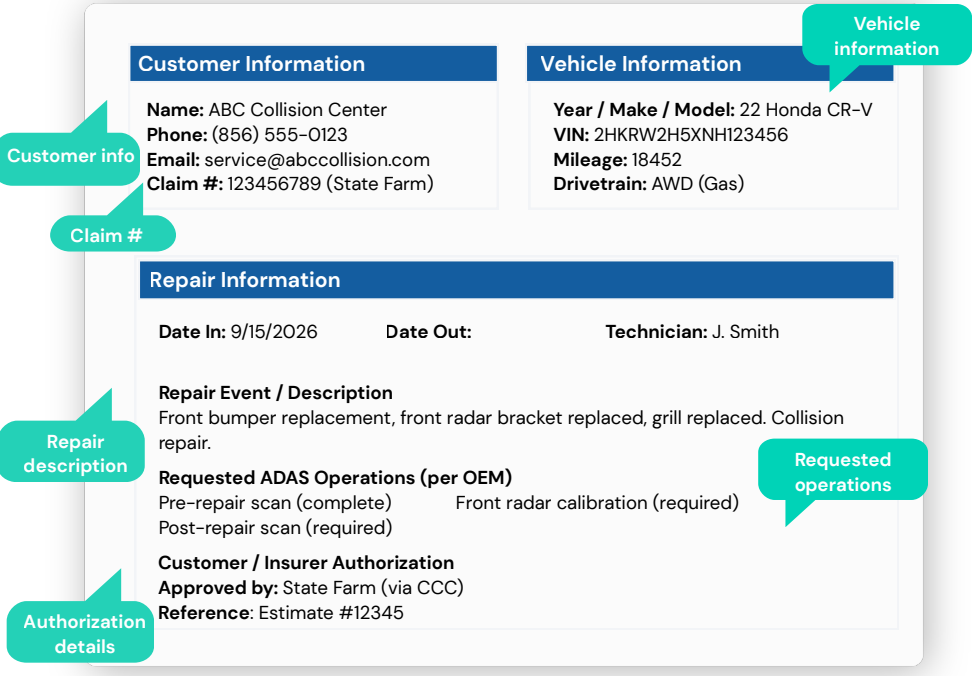
Define authorization

Include customer or insurer approval, claim number (if applicable), line items for required scans, calibrations, programming and vendor documentation.

Assign tech(s)

Confirm the tech's qualifications, assign the work, and ensure all required procedures and documentation are completed.

Work Order Example



Why This Matters in a Repair Facility

Technicians cannot research or document correctly from a vague "calibrate car" request.

A complete work order ensures:

- The correct systems are identified
- OEM procedures are followed
- All required scans, calibrations, and programming are performed
- Documentation supports reimbursement
- Miscommunication and repeat repairs are avoided.

Common Pitfalls to Avoid

- Missing mileage or exact repair event.
- Using only an insurer line description like "calibrate car."
- Not including required scans or programming.

62. Writing a Complete Work Order

Knowledge Check

Explain how you would write a complete work order for a **2023 Toyota RAV4** that received a windshield replacement. Identify one document that proves the work was completed correctly.

Customer Information

Name:
Phone:
Email:
Claim #:

Vehicle Information

Year / Make / Model:
VIN:
Mileage:
Drivetrain:

Repair Information

Date In: **Date Out:** **Technician:**

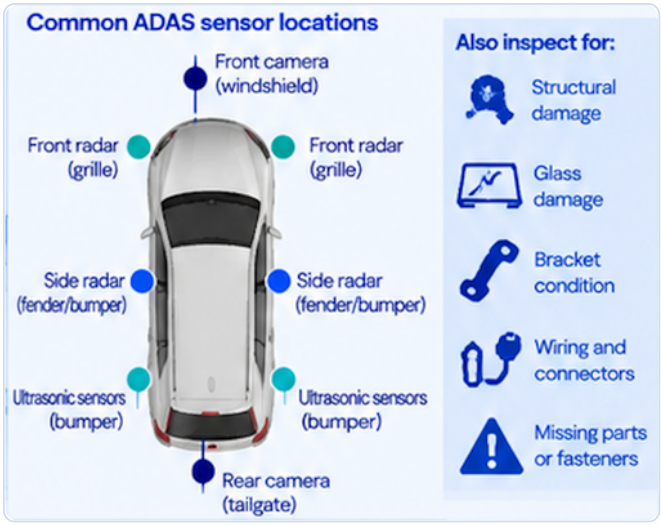
Repair Event / Description

Requested ADAS Operations (per OEM)

Customer / Insurer Authorization
Approved by:
Reference:

63. Damage-Analysis Walkarounds

A deliberate walkaround identities sensor locations, impact paths, missing parts, warnings and incomplete assembly before any calibrations are pertormed.



The Walkaround Workflow

REVIEW ESTIMATE. WALK IMPACT PATH. LOCATE SENSORS. PHOTOGRAPH CONDITION. COMPARE PROCEDURE.

Review estimate

Understand the reported damage, customer concerns and initial repair plan.

Walk impact path

Follow the direction of impact. Look for primary and secondary damage, even if not listed on the estimate.

Locate sensors

Identify all ADAS sensors, brackets and related components in the affected area and adjacent zones.

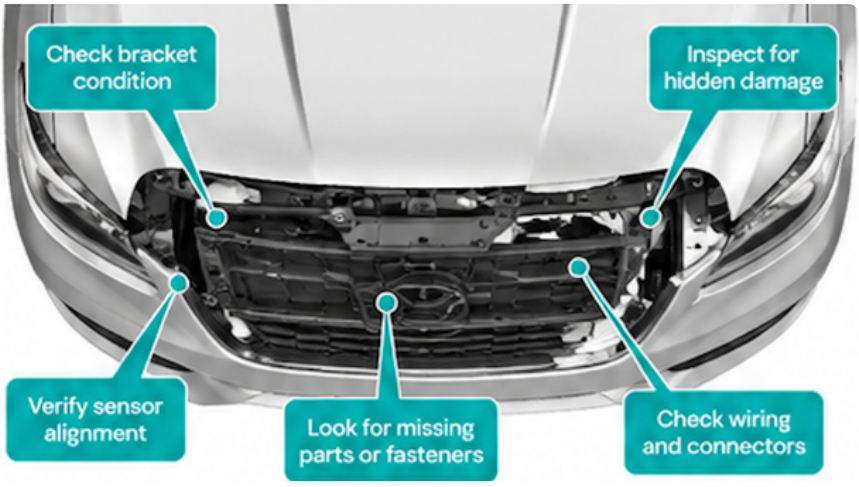
Photograph condition

Take clear photos of each sensor area, brackets, mounts, wiring, glass and surrounding parts before disassembly.

Compare procedure

Review OEM procedures for required inspections, replacements, calibrations or initialization based on the damage found.

Practical Example



In this example, the bumper cover was damaged and the front radar bracket was bent. The impact also shifted the grille, which affected radar alignment. These issues were not identified in the initial estimate.

Why This Matters in a Repair Facility

The scan does not show bent brackets, wrong glass or unfinished body work. A walkaround ensures:

- All sensor locations are inspected
- Hidden and secondary damage is identified.
- Missing parts and incorrect installations are found early.
- Proper OEM procedures are followed.
- Incomplete assemblies and repeat repairs are avoided

Common Pitfalls to Avoid

- Starting calibration from the scan report alone.
- Failing to inspect the opposite or secondary impact.
- Missing damaged brackets, wiring or fasteners.
- Not photographing sensor areas before disassembly.

63. Damage-Analysis Walkarounds

Knowledge Check

Explain, in your own words, the appropriate answers to each of the questions below.

1. Describe the steps you would take to perform a damage-analysis walkaround on a vehicle with front-end collision damage.

2. Identify one condition that could make the information inaccurate and explain how to prevent it.

3. Name one document or piece of evidence that proves the walkaround was completed correctly.

64. Reading an OEM Calibration Procedure

Before moving the vehicle, always read the OEM calibration procedure. It provides applicability, warnings, required tools, preliminary conditions, measurements, sequence and completion criteria.

Finding OEM procedures:

OEM service information portals
(Toyota TIS, Honda Service Express, Ford PTS, GM SPS, Stellantis, etc.

Vehicle-specific procedures
Usually based on VIN, model, system, and repair performed.

Always use the latest intel
Confirm any applicable service bulletins or campaigns.

The OEM Calibration Workflow

REVIEW ESTIMATE. WALK IMPACT PATH. LOCATE SENSORS. PHOTOGRAPH CONDITION. COMPARE PROCEDURE.

Confirm applicability
Understand the reported damage, customer concerns and initial repair plan.

→

Read warnings
Follow the direction of impact. Look for primary and secondary damage, even if not listed on the estimate.

→

List prerequisites
Identify all ADAS sensors, brackets and related components in the affected area and adjacent zones.

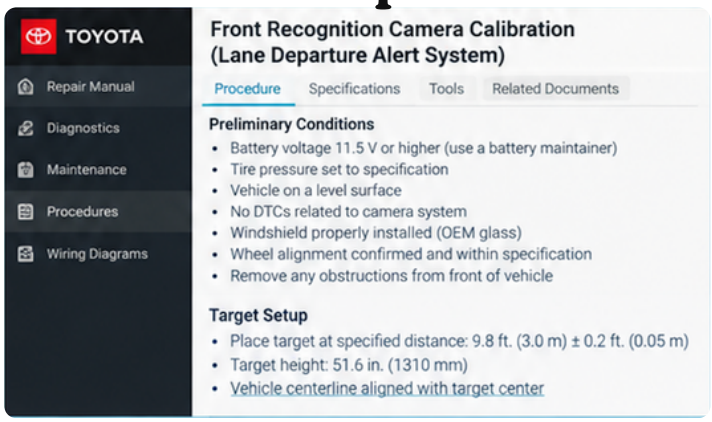
→

Extract dimensions
Take clear photos of each sensor area, brackets, mounts, wiring, glass and surrounding parts before disassembly.

→

Define completion
Review OEM procedures for required inspections, replacements, calibrations or initialization based on the damage found.

Practical Example



OEM Procedure Highlights (Toyota Front Camera)

Why This Matters in a Repair Facility

A single missed note can invalidate the entire setup.

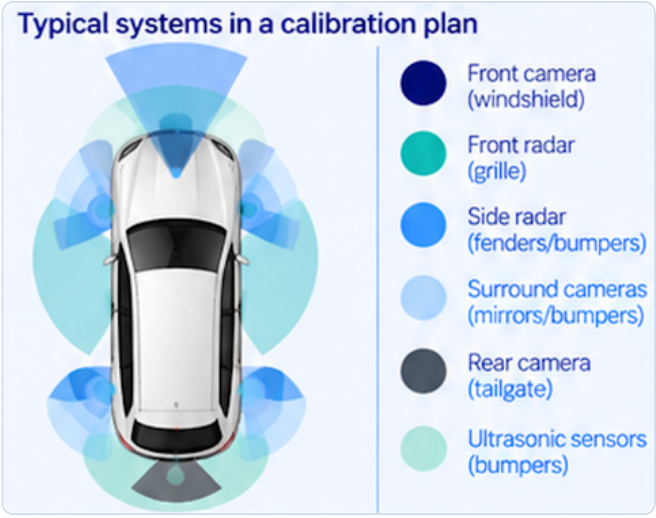
Reading the OEM procedure ensures the correct preconditions, measurements and completion criteria are followed, which prevents repeat repairs and keeps the vehicles safety systems operating as intended.

Key Information to Review

- **Vehicle applicability:** Year, make, model, trim, drivetrain, system, and VIN range.
- **Warnings and cautions:** Battery support, environmental conditions, vehicle setup and restrictions.
- **Required tools and equipment:** Targets, scan tool, reflectors, special tools.
- **Preliminary conditions:** Tire pressure, fuel level, ride height, alignment, software level, DTCs.
- **Measurements and setup:** Target placement, distances, vehicle positioning, height specifications.
- **Completion criteria:** On-screen result, scan report, road test (if required) and no related DTCs.

65. Building the Calibration Plan

The calibration plan lists every sensor, initialization, programming step, static/dynamic routine and required sequence. It ensures all connected operations are completed and documented.



The Calibration Plan

IDENTIFY SYSTEMS. MAP TRIGGERS. RETRIEVE PROCEDURES. ORDER OPERATIONS. ASSIGN EVIDENCE.

Identify systems

List all ADAS systems affected by the repair based on RO, scan results and physical damage inspection.

Map triggers

Determine which operations are required based on what was repaired, replaced, recalibrated or rpepogrammed,

Retrieve procedures

Access the OEM service information for each operation, including prerequisites, special tools and setup requirements.

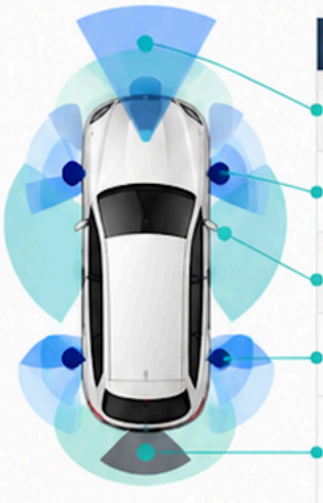
Order operations

Plan the correct sequence (e.g., programming before calibration, static before dynamic).
Account for dependencies and shared steps.

Assign evidence

Define what documentation is required for each operation (scan reports, calibration reports, photos, notes) and who is responsible for capturing it.

Practical Example



System	Trigger	Required Operation
Front camera (windshield)	Windshield replacement	Camera calibration (static)
Front radar (grille)	Grille replaced	Radar calibration (static or dynamic)
Left side radar (fender)	Fender replaced	Radar calibration (static)
Right surround camera (mirror)	Mirror replaced	Camera calibration (static)
Ultrasonic sensors (front bumper)	Bumper replaced	ADAS initialization and calibration (if required)

A clear calibration plan turns a complex repair into a structured process, ensuring every system is calibrated and documented correctly.

Why This Matters in a Repair Facility

Complex repairs often need several connected operations. A complete calibration plan ensures:

- No required calibrations, initializations or programming steps are missed.
- Operations are completed in the correct order.
- All applicable sensors and modules are addressed Proper documentation is collected for quality
- control and reimbusrement.
- Repeat repairs and comebacks are avoided.

Common Pitfalls to Avoid

- Planning only the line requested by the shop.
- Missing companion sensors or chassis inputs.
- Incorrect sequence (e.g., calibrating before programming).
- Not accounting for prerequisites or special tools.

65. Building the Calibration Plan

Knowledge Check

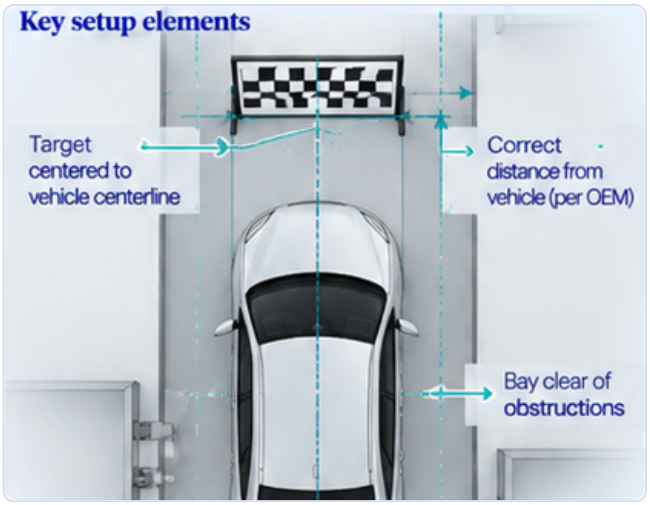
Explain, in your own words, the appropriate answers to each of the questions below.

1. Build a calibration plan for a vehicle with a front bumper replacement and windshield replacement. Circle the area/areas that are mostly likely to be affected.

2. Identify one condition that could make the plan inaccurate and explain how to prevent it.

66. Static-Bay Setup

The bay setup translates OEM dimensions into a controlled physical geometry. Accurate target placement is essential for a successful static calibration.



The Static-Bay Setup

VALIDATE FLOOR. POSITION VEHICLE. ESTABLISH DATUM. PLACE TARGET. CROSS-CHECK.

Validate floor

Ensure the bay surface is level, flat and free of obstructions. Confirm lighting conditions meet OEM requirements.

Position vehicle

Place vehicle on level surface, straight and centered. Verify tire pressure, ride height and vehicle condition as required.

Establish datum

Set vehicle centerline and reference points using OEM-specified measurements (e.g., centerline, axle measurements, target height).

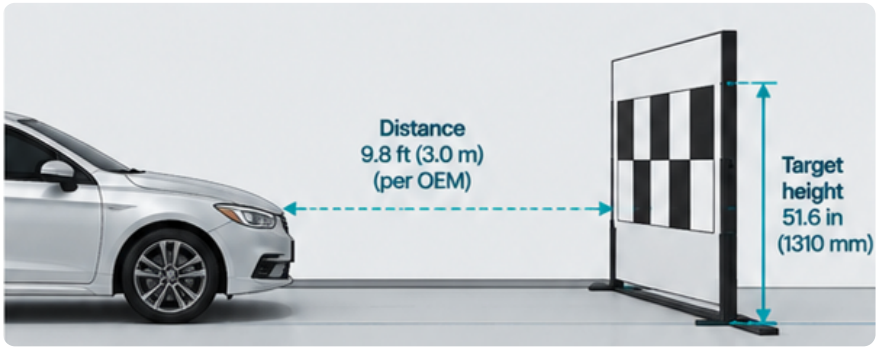
Place target

Position the calibration target at the correct distance, height and alignment per OEM procedure. Confirm target type and orientation.

Cross-check

Double-check all dimensions, alignment and stability. Verify no equipment, doors or personnel will move during the calibration routine.

Practical Example



OEM procedure highlights:

- Vehicle on level surface with correct ride height and tire pressure.
- Target centered to vehicle centerline.
- Specific distance and height measurements.
- Follow OEM alignment and lighting requirements.

Why This Matters in a Repair Facility

Target placement errors can lead to an incorrect reference, even if the scan tool reports completion.

A proper static-bay setup ensures:

- Accurate target placement and alignment.
- Valid scan and calibration results.
- Reduced repeat repairs and comebacks.
- Proper documentation for quality control
- and reimbursement.
- A safe, controlled environment for consistent
- results.

Common Pitfalls to Avoid

- Using one measurement only.
- Allowing doors, lift or equipment to move during the routine.
- Incorrect target height, distance or centerline alignment.
- Not verifying tire pressure or ride height.

67. Dynamic Route Planning

A compliant route is selected for speed, road markings, curvature, traffic, weather and safe tool use to complete dynamic calibrations accurately and safely.

Key route requirements

- **Compliant speed range** — Maintain OEM-specified speeds (e.g., 25–70 mph).
- **Clear lane markings** — Well-defined lane lines on both sides.
- **Appropriate road type** — Includes straights and gentle curves as required by the OEM.
- **Favorable weather** — Avoid heavy rain, snow, fog or high winds.
- **Safe tool use** — Mount and secure the scan tool per OEM and operator safety guidelines.

The Dynamic Route Workflow

READ CONDITIONS. CHECK WEATHER. SELECT ROUTE. MOUNT TOOL SAFELY. RECORD DRIVE.

Read conditions

Review OEM procedure for required speed range, distance, road type (urban, highway), lane markings and number of drive cycles.

Check weather

Confirm safe weather conditions (no heavy rain, snow, fog or high winds) and plan for light traffic.

Select route

Choose a route with clear lane markings, appropriate speed limits, and required road features (straights, gentle curves, merges/exits if required).

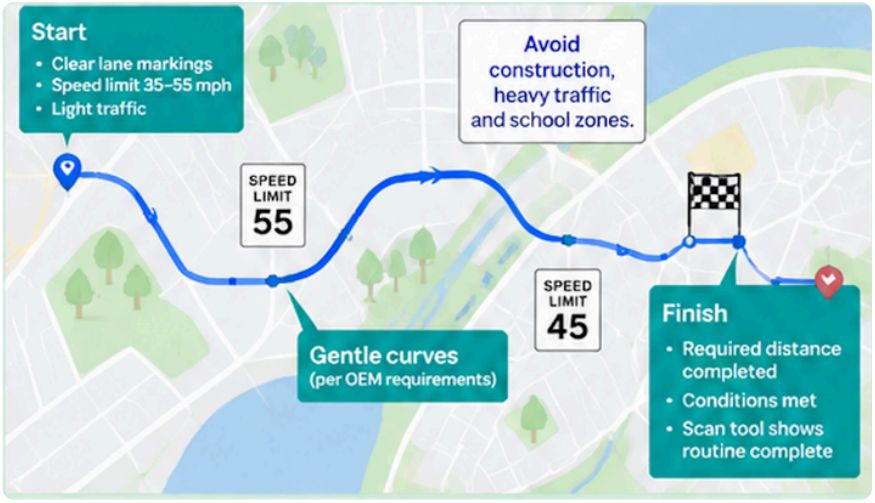
Mount tool safely

Secure the scan tool using an approved mount. Verify cable routing, screen visibility and that it does not interfere with airbags, controls or driving.

Record drive

Follow the OEM sequence, maintain required speeds and road conditions, monitor for completion status and save the results.

Practical Example



A well-planned route with the right conditions is the key to a successful dynamic calibration.

Why This Matters in a Repair Facility

Repeated failed drives waste time and can pressure unsafe decisions.

A compliant route ensures:

- The OEM driving conditions are met.
- Accurate and complete calibration results.
- Reduced repeat repairs and come-backs.
- Technician and public safety.
- Efficient use of time and resources.

Common Pitfalls to Avoid

- Choosing the route after starting the routine.
- Forcing speed or following distance in traffic.
- Using a route with poor or faded lane markings.
- Attempting the drive in heavy rain, snow, fog or high winds.
- Not securely mounting the scan tool.

68. Handling Calibration Failures

A failed calibration provides valuable information. Follow a structured process to identify the cause, correct the condition and successfully complete the calibration.

Common causes of calibration failure

Vehicle condition
Misaligned components, incorrect repairs, aftermarket parts.

Setup and environment
Incorrect target placement, lighting. floor surface, interference.

Tool or software
Outdated software, missing diagnostic steps, incorrect settings.

Component damage
Faulty sensor, wiring issue, bracket damage, or internal fault.

The Calibration Failure Flow

SAVE FAILURE DATA. STOP RETRIES. CATEGORIZE CAUSE. CORRECT CONDITION. RESTART FROM PROCEDURE.

Save failure data

Record and save all scan results, error codes and calibration logs before making any changes.

Stop retries

Avoid repeating the calibration without diagnosing the cause. Multiple failed attempts can create additional DTCs and wasted time.

Categorize cause

Determine if the issue is vehicle condition, setup/environment, tool/software or component damage.

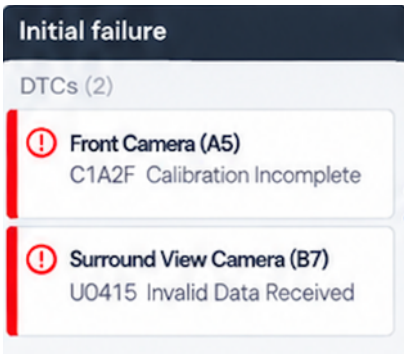
Correct condition

Make the necessary repairs, adjustments or setup changes based on the root cause (e.g., realign bracket, reposition target, update software).

Restart from procedure

Follow the OEM calibration procedure from the beginning and confirm a successful result with a clear scan report.

Practical Example



After diagnosis and correction

- ✓ Windshield bracket was partially seated, causing camera misalignment.
- ✓ Bracket realigned and torque verified to OEM spec.
- ✓ Target height adjusted by 10 mm to match OEM procedure.
- ✓ Both cameras calibrated successfully. No related DTCs.

Key takeaway: The initial failures indicated "calibration incomplete" and "invalid data received." After inspecting the windshield bracket and correcting target height, both cameras calibrated successfully and cleared all DTCs.

Why This Matters in a Repair Facility

A structure approach to calibration failure ensures:

- The real cause is identified and corrected.
- Repeat failures and unnecessary DTCs are avoided.
- Repairs are completed to OEM standards.
- Accurate documentation is available for quality control and reimbursement.
- Time and resources are used efficiently.

Common Pitfalls to Avoid

- Choosing the route after starting the routine.
- Forcing speed or following distance in traffic.
- Using a route with poor or faded lane markings.
- Attempting the drive in bad weather.
- Not securely mounting the scan tool.

69. Handling an Incomplete Dynamic Learn

An incomplete dynamic learn occurs when the required driving conditions or learning thresholds are not achieved. Follow a structured process to identify the cause, correct the condition and successfully complete the procedure.

Common causes of dynamic learn failure

Weather conditions
Rain, snow, fog or high winds can prevent leaming thresholds.

Driving conditions
Insufficient speed, missing road types, lack of lane markings or traffic.

Interruption
Drive stopped, ignition off, too disconnected or user exit.

Vehicle/system status
Active DTCs, incorrect calibration, incomplete repairs or software issues.

The Dynamic Learn Flow

RECORD START. MONITOR CRITERIA. DOCUMENT INTERRUPTION. RESCHEDULE SAFELY. COMPLETE LATER.

Record start

Document the start time, mileage, route and intended conditions (e.g., road type, speed range, weather).

Monitor criteria

Watch live progress in the scan tool. Verify all required conditions are being met (speed distance, lane markings, road type, etc.).

Document interruption

If the learn does not complete, record what happened (time, mileage, conditions, reason) and save the scan report.

Reschedule safely

Correct the issue (e.g., choose a better route or weather window), confirm no DTCs and restart following OEM requirements.

Complete later

Repeat the drive until all learning criteria are met.
Save the final report showing a successful dynamic learn.

Practical Example

Incomplete dynamic learn

Vehicle	2023 Honda CR-V
System	Forward camera (lane keeping assist)
Start time	9:12 AM
Mileage	12,435 mi
Conditions	Heavy rain, low visibility
Result	Incomplete Learning thresholds not met (insufficient lane marking detection)

Successful dynamic learn

Vehicle	2023 Honda CR-V
System	Forward camera (lane keeping assist)
Start time	11:04 AM
Mileage	12,462 mi
Conditions	Clear, dry roads Visible lane markings 45–65 mph mixed routes
Result	Complete All learning criteria met No related DTCs

An incomplete dynamic learn is not a failure, it's information. Capture the details, correct the condition and complete it under the right conditions.

Why This Matters in a Repair Facility

The file must distinguish attempted time/miles from a completed calibration. Proper handling ensures:

- Accurate repair documentation.
- Reduced repeat drives and wasted time.
- Proper scheduling based on weather and traffic.
- Clear record of attempted vs. completed results.
- Customer is informed of status and timeline.

Common Pitfalls to Avoid

- Billing as completed because miles were driven.
- Leaving the customer unaware of system status.
- Attempting the drive in poor weather or traffic.
- Not documenting the reason for the interruption.
- Ignoring active DTCs or incomplete repairs.

70. Post-Scan Interpretation

A post-scan identifies remaining current and history DTCs, module communication and the final electronic condition of the vehicle.

Post-Scan information includes:

DTC status
Current, history or pending codes in all modules.

Module communication
Confirms all modules are online and communicating.

System condition
Identifies active faults, intermittent issues and final electronic health.

Documentation
Provides proof of repair completion and system status.

The Post-Scan Flow

SCAN ALL MODULES. COMPARE PRE/POST. CLASSIFY STATUS. DIAGNOSE ACTIVE FAULTS. ATTACH REPORT.

Scan all modules

Perform a full vehicle scan. Include all ADAS, body, chassis, powertrain and network modules.

Compare pre/post

Review and compare the current scan to the pre-repair scan. Note codes that have cleared, remain or are new.

Classify status

Identify each DTC as current, history (past) or pending. Focus on current codes that affect ADAS or related systems.

Diagnose active faults

Follow OEM diagnostics for any remaining current codes. Do not clear codes until the cause is corrected and verified.

Attach report

Save and attach the final post-scan report to the work order. Include notes on remaining codes and next steps if needed.

Practical Example

Pre-scan (before repair)

Module	DTC	Status
Front Camera (A5)	C1A2F	Current
ABS	U0415	Current
SRS	B1C54	History
HVAC	B10D9	History

Post-scan (after calibration)

Module	DTC	Status
Front Camera (A5)	—	None
ABS	—	None
SRS	B1C54	History
HVAC	B10D9	History

RESULT: Camera and related ADAS codes cleared after successful calibration. Only non-related history codes remain and do not affect vehicle delivery.

Why This Matters in a Repair Facility

Clearing DTCs is not the same as resolving them, and some faults return only after a drive cycle. A proper post-scan ensures:

- All modules are communicating.
- Repairs and calibrations were successful.
- Remaining codes are identified and addressed.
- Accurate documentation for quality control and reimbursement.
- Prevents customer comebacks and safety risks.

Common Pitfalls to Avoid

- Deleting codes without preserving the report.
- Calling all history codes unrelated without analysis.
- Not scanning all modules.
- Assuming no codes means no further action is needed.
- Failing to compare with the pre-scan.

71. Functional Verification

Functional verification confirms OEM completion messages, data values, warning status and permitted road/function checks. It closes the gap between performing a routine and proving the repaired system is ready.

KEY VERIFICATION CHECKS

Completion message
Verify the scan tool shows successful calibration and no outstanding actions.

Ignition / sleep cycle
Perform required key cycle or sleep cycle per OEM procedure.

Data values
Check live data for expected values (e.g., camera aim, sensor status).

Permitted road/function checks
Confirm all required functions operate as intended (e.g., lane keeping, AEB).

The Functional Check Flow

CHECK SUCCESS MESSAGE. CYCLE IGNITION/SLEEP. READ DATA. RUN PERMITTED CHECK. DOCUMENT RESULT.

Check success message
Verify the scan tool shows OEM calibration complete messages and no required actions remain.

→ **Cycle ignition / sleep**
Perform the required ignition cycle or allow the vehicle to enter sleep mode per OEM instructions.

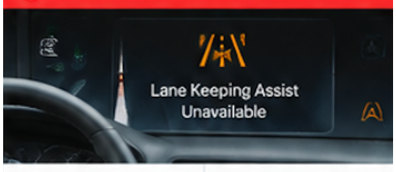
→ **Read data**
Check relevant live data values (e.g., camera aim, sensor status, system health) and confirm they are within OEM specifications.

→ **Run permitted check**
Perform OEM-specified road or function checks (e.g., AEB, lane keeping, blind spot, adaptive cruise) to confirm proper operation.

→ **Document result**
Record and save verification results, including scan reports, data screenshots and road test notes.

Practical Example

Before verification (issue found)



Scan status	Calibration complete
Lane keeping	Not available
Camera aim (live data)	-1.8° (out of spec)

Although calibration reported complete, the live data showed the camera aim was out of specification and lane keeping was not operating.

After verification (success)

Scan status	Calibration complete
Lane keeping	Available
Camera aim (live data)	0.1° (in spec)

After adjusting the camera aim and completing the permitted road check, all systems operated normally and data values were within specification.

Why This Matters in a Repair Facility

Functional verification closes the gap between performing a routine and proving the repaired system is ready.
It ensures:

- OEM completion messages are true.
- All relevant data values are correct.
- Driver assistance functions operate as intended Safer vehicles for customers and the public.
- Proper documentation for quality control and reimbursement.

Common Pitfalls to Avoid

- Testing AEB by driving at an object.
- Relying only on the absence of cluster warnings.
- Skipping the ignition/sleep cycle.
- Not checking relevant live data values.
- Failing to perform OEM-specified road/function checks.

72. Calibration Photo Documentation

Photos should provide identity, condition, sensor/target selection, measurements, environment and result without exposing unsafe shortcuts. A complete photo set allows a reviewer to reconstruct the operation.

ESSENTIAL PHOTOS TO INCLUDE

Vehicle identification: Clear VIN and vehicle.

Sensor/target: Show target used for calibration.

Full setup: Target placement, distance, alignment.

Measurements: Height, distance, level, etc

Result screen: Clear final calibration result.

The Photo Documentation Flow

PHOTOGRAPH VIN. PHOTOGRAPH SENSOR. SHOW FULL SETUP. SHOW MEASUREMENTS. CAPTURE RESULT.

Photograph VIN

Capture a clear photo of the VIN and an overall vehicle shot (showing make, model and condition).

Photograph sensor

Take close-up photos of the calibrated sensors) or target setup, including part number/ID if visible.

Show full setup

Capture wide bay photos showing the complete setup, including target placement, distance and alignment.

Show measurements

Include key measurements such as target height, distance, level and any OEM-specific dimensions required for the procedure.

Capture result

Take a clear photo of the final calibration result screen showing a successful completion, including date and time.

Practical Example

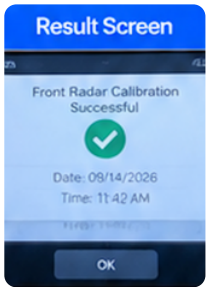
FRONT RADAR STATIC CALIBRATION

Full vehicle view with clear VIN.

Front radar sensor and target ID.

Wide bay shot showing target placement and alignment.

Target height and distance measurements per OEM procedure.



Final result screen with date and time.

Why This Matters in a Repair Facility

A complete and clear photo set allows a reviewer to reconstruct the operation and confirms the calibration was performed correctly.

It ensures:

- Proper sensor/target identification.
- Correct setup and measurements.
- Accurate, verifiable results.
- Supports quality control and reimbursement.
- Reduces questions and repeat work.

Common Pitfalls to Avoid

- Close-up photos with no context.
- Missing measurements or unclear target placement.
- Photographing measurements after equipment was moved.
- Not showing the final result screen.
- Blurry or incomplete photos.

73. Writing the Calibration Invoice

The calibration invoice should clearly describe the actual operation, system, method and result.

Diagnostic, programming and calibration are distinct and should be billed separately.

KEY ELEMENTS TO INCLUDE

System Name: e.g. forward radar, windshield camera

Operation performed: e.g. static/dynamic calibration

Result: e.g. no DTCs, calibraiton successful

Reference attachment: Include scan report, photos, and/or OEM procedure reference.

The Calibration Invoice Flow

NAME SYSTEM. NAME OPERATION. STATE TRIGGER. STATE RESULT. REFERENCE ATTACHMENTS.

Name system

Identify the specific ADAS system involved (e.g., forward radar, front camera, side radar).

Name operation

State the exact procedure performed (e.g., static calibration, dynamic calibration, programming, diagnostics).

State trigger

Include why the procedure was required (e.g., bumper replacement, windshield replacement, DTCs present).

State result

Document the outcome (e.g., calibration successful, no DTCs, software updated).

Reference attachment

Attach supporting documentation such as scan reports, photos, and OEM procedure references.

Practical Example

FRONT RADAR STATIC CALIBRATION

System	Forward Radar (ACC)
Operation	Static Calibration
Trigger	Front bumper replacement
Method	Performed per OEM procedure (Manufacturer service information)
Result	Calibration successful, no related DTCs
Attachments	Pre/post scan reports, target setup, final result screenshot
Notes	Calibration performed using OEM procedure for target placement and vehicle preparation.

Target setup photo included with invoice.

Why This Matters in a Repair Facility

Accurate wording reduces disputes and avoids duplicate or misleading charges. It ensures:

- Clear communication with insurers and customers.
- Proper reimbursement.
- Distinguishes calibration, diagnostics and programming.
- Provides a complete record of the work performed
- Supports quality control and liability protection.

Common Pitfalls to Avoid

- Calling an in-process scan a pre-scan.
- Billing programming when only calibration occurred
- Using vague descriptions (e.g., "ADAS work").
- Not attaching supporting documentation.
- Combining multiple systems into one line item.

73. Writing the Calibration Invoice

Real-World Example

BLIND SPOT MONITOR STATIC CALIBRATION

LIGHTNING AUTO SERVICE

Invoice:
RO:
Issued:
Due:
Total: $420.00

Vehicle: 2026 Toyota
VIN:

Bill From:
Lightning Auto Service

Tax ID:
Phone:
Email:

Bill To:

Phone:
Email:

ADAS OPERATIONS	CODE	PRICE	QTY/HR	DIS. (%)	TOTAL
Blind Spot Radar Procedure Type: Check/Adjust Installation Angle / Inspection (Reflection Power) / Static or Dynamic Calibration / Programming with Replacement Notes: Calibrate the vehicle's blind spot radar sensors to ensure accurate detection of adjacent vehicles and proper system operation according to OEM specifications.	-	$600.00	1	30	$420.00

System Operation Trigger(s) Method

Subtotal: $420.00
Sales Tax (0%): $0.00
Total Price $420.00

AUTEL

Shop Name: Lightning Auto Service

2026 Toyota Post-scan Report

Vehicle Information
2026/Toyota/

Odometer Reading: 4943.63miles
License Plate: --

Customer Information
Name: --

Tel: --

Device information
Scanner: MaxiSys Ultra&ADAS
Version: V4.40-7.50

System Scanned(38)

Pre-scan Report | Post-scan Report

System | Calibration | Status/DTC | Calibrated | Status/DTC

Post-Scan Report

*NOTE: Vehicle photos (including target setup) not included to protect owner privacy.

74. Insurer Support Package

The support package links the repair event to OEM authority and provides clear, organized evidence of completion. Adjusters need a concise reason, not an unorganized data dump.

KEY ELEMENTS TO INCLUDE

State trigger: Explain what was repaired and why calibration was required.
Cite Authority: Reference OEM procedures, TIS documents or service information.
Show prerequisites: Include pre-scan, setup photos and any required measurements.
Show completion: Provide post-scan results and a summary of outcomes.

The Insurer Support Package

STATE TRIGGER. CITE AUTHORITY. SHOW PREREQUISITES. SHOW COMPLETION. ATTACH POST-SCAN.

State trigger

Clearly state what was repaired and which ADAS systems) required calibration, diagnostic or programming.

Cite authority

Reference the specific OEM procedure, TIS document or service information that requires the service.

Show prerequisites

Include pre-scan results, setup photos, tool information and any required measurements.

Show completion

Include post-scan results, calibration status and confirmation that all required functions are operating as intended.

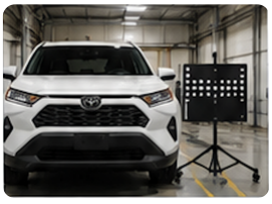
Attach post-scan

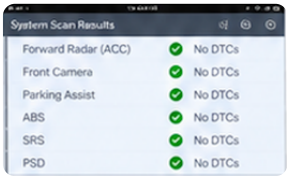
Attach the final post-scan report along with photos and a brief summary for the claim file.

Practical Example

FRONT BUMPER REPLACEMENT

Repair performed	Front bumper replacement
Systems affected	Forward radar (ACC), front camera (lane keep), parking sensors
OEM reference	AllData / Repair Logic: Front Radar Adjustment Procedure (TIS-ADAS-0214)
Prerequisites	Pre-scan results, target setup photos, ride height measured and within spec.
Completion	Static calibration completed successfully. No related DTCs.
Attachments	Pre/post scan reports, target setup, measurement photos, OEM procedure screenshot
Notes	Calibration performed using OEM procedure. All ADAS functions verified and operating as intended.





Why This Matters in a Repair Facility

A clear, complete support package helps adjusters understand why the calibration was required and confirms it was completed correctly.
It ensures:

- Faster claim approvals.
- Fewer questions and delays. v Proper reimbursement.
- A complete record of work performed.
- Supports compliance with OEM procedures and industry standards.

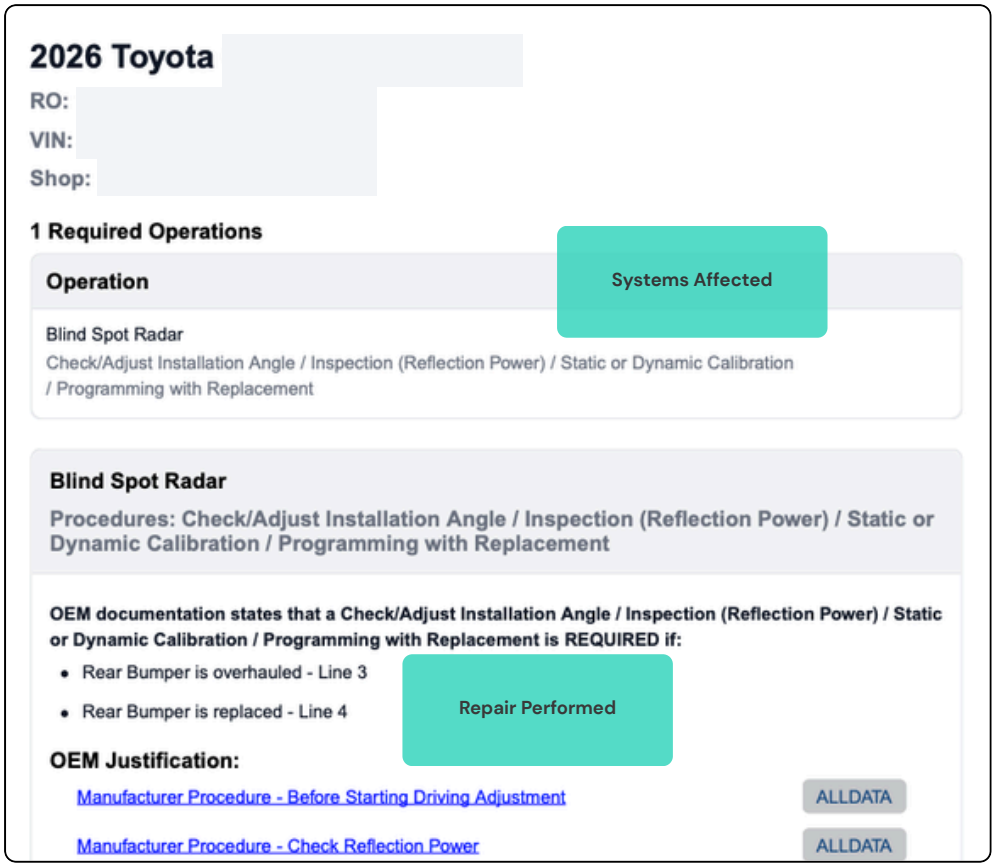
Common Pitfalls to Avoid

- Sending only a target photo.
- Using a generic position statement instead of VIN-specific OEM procedure when available.
- Including unnecessary or disorganized files.
- Leaving out the post-scan report.
- Not clearly explaining what was repaired.

74. The Insurer Support Package

Real-World Example

BLIND SPOT MONITOR STATIC CALIBRATION



2026 Toyota

RO:

VIN:

Shop:

1 Required Operations

Operation
Blind Spot Radar Check/Adjust Installation Angle / Inspection (Reflection Power) / Static or Dynamic Calibration / Programming with Replacement

Systems Affected

Blind Spot Radar

Procedures: Check/Adjust Installation Angle / Inspection (Reflection Power) / Static or Dynamic Calibration / Programming with Replacement

OEM documentation states that a Check/Adjust Installation Angle / Inspection (Reflection Power) / Static or Dynamic Calibration / Programming with Replacement is REQUIRED if:

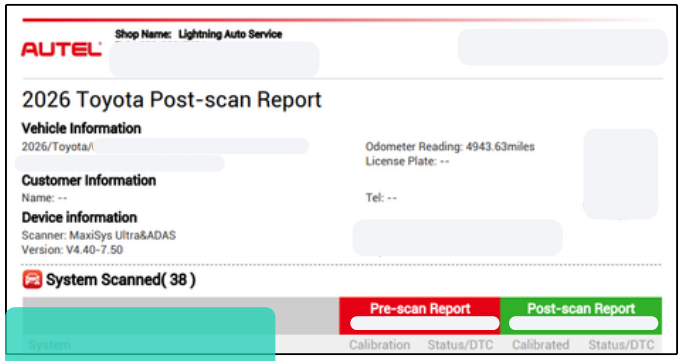
- Rear Bumper is overhauled - Line 3
- Rear Bumper is replaced - Line 4

Repair Performed

OEM Justification:

Manufacturer Procedure - Before Starting Driving Adjustment — ALLDATA

Manufacturer Procedure - Check Reflection Power — ALLDATA



AUTEL Shop Name: Lightning Auto Service

2026 Toyota Post-scan Report

Vehicle Information

2026/Toyota/\

Odometer Reading: 4943.63miles

License Plate: --

Customer Information

Name: --

Tel: --

Device information

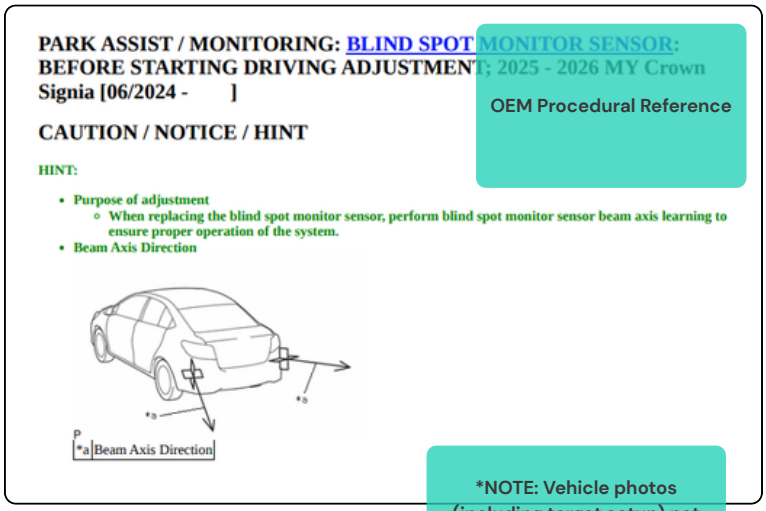
Scanner: MaxiSys Ultra&ADAS

Version: V4.40-7.50

System Scanned(38)

System	Pre-scan Report: Calibration	Pre-scan Report: Status/DTC	Post-scan Report: Calibrated	Post-scan Report: Status/DTC

Post-Scan Report (Completion)



PARK ASSIST / MONITORING: BLIND SPOT MONITOR SENSOR: BEFORE STARTING DRIVING ADJUSTMENT; 2025 - 2026 MY Crown Signia [06/2024 -]

OEM Procedural Reference

CAUTION / NOTICE / HINT

HINT:

- Purpose of adjustment
  - When replacing the blind spot monitor sensor, perform blind spot monitor sensor beam axis learning to ensure proper operation of the system.
- Beam Axis Direction

*a Beam Axis Direction

*NOTE: Vehicle photos (including target setup) not included to protect owner privacy.

75. Glass Company Handoff

The handoff confirms glass part/options, bracket and camera status, cure time, warnings and whether calibration is mobile or static-center appropriate.

KEY ELEMENTS TO CONFIRM

Glass part/options: Verify OEM part number, glass type and options (e.g., acoustic, HUD).
Camera status: Record and confirm camera R&I and condition.
Bracket condition: Inspect new bracket and mounting hardware.
Cure time: Confirm OEM-specified safe-drive-away time and adhesive cure time.
Calibration type: Determine if calibration is mobile or requires a static center and schedule accordingly.

The Glass Handoff Workflow

CONFIRM GLASS. RECORD CAMERA R&I. INSPECT BRACKET. RESPECT CURE. SCHEDULE CALIBRATION.

Confirm glass

Verify OEM part number, glass type and options (e.g., acoustic, HUD, rain/light sensors).

Record camera R&I

Document camera removal and installation, including photos and condition.

Inspect bracket

Confirm new bracket is installed correctly and undamaged. Verify hardware and torque where applicable.

Respect cure

Follow OEM adhesive cure time and safe-drive-away requirements before moving or calibrating the vehicle.

Schedule calibration

Determine if calibration is mobile or requires a static center and schedule at the appropriate time.

Practical Example

WINDSHIELD REPLACEMENT WITH FORWARD CAMERA

OEM part number and options (acoustic).

Camera removed and reinstalled. Photos taken during R&I.

New bracket installed and hardware verified.

Adhesive cure time verified before calibration.

Why This Matters in a Repair Facility

Calibration quality depends on the glass installation and complete information.
A clear handoff ensures:

- The correct OEM glass and options were used.
- The camera and bracket are installed correctly.
- OEM cure time and safe-drive-away requirements are followed
- The right calibration method is used (mobile or static center).
- Fewer comebacks and improved safety.

Common Pitfalls to Avoid

- Arriving before cure or trim completion.
- Leaving cameras unplugged without notification.
- Not verifying OEM glass part number or options.
- Skipping bracket inspection.
- Scheduling calibration too early.

76. Body Shop Handoff

The shop confirms assembly, alignment, structure, parts and access before calibration. Incomplete vehicles create delays and invalid setups.

KEY ELEMENTS TO CONFIRM

Complete assembly: Verify bumper, grille, liners and related components are installed and secure.
Correct alignment: Confirm alignment is complete and within specification.
Repairs finalized: Ensure all related repairs (body, suspension, structural) are completed.
Keys and access: Provide keys, any special access and remove vehicle from the lift if required.
Report warnings: Communicate any known warnings, concerns or conditions.

The Shop Handoff Workflow

CONFIRM ASSEMBLY. ATTACH ALIGNMENT. LIST REPAIRS. PROVIDE KEYS/ACCESS. REPORT WARNINGS.

Confirm assembly

Verify all components related to the calibration are installed (e.g., bumper, grille, liners, sensors) and securely mounted.

Attach alignment

Include a completed alignment sheet showing the vehicle is within OEM specifications.

List repairs

Provide a summary of all related repairs (body, suspension, structural) that may affect calibration and sensor alignment.

Provide keys/access

Ensure all keys, access codes and special instructions are available.
Remove the vehicle from the lift if required

Report warnings

Communicate any known warning lights, DTCs or concerns before calibration begins.

Practical Example

FRONT BUMPER REPLACEMENT - COMPLETE HANDOFF

Assembled bumper

Bumper, grille and liners installed and secure.

Alignment sheet

Front	Left	Right
Camber	-0.6°	-0.6°
Caster	5.8°	5.7°
Toe	0.05°	0.04°
Total Toe	0.09°	

Within OEM Specifications

Completed alignment report attached.

Repair summary

Repairs Performed:
- Front bumper replacement
- Front grille replacement
- Lower air deflector installed
- Front radar sensor reinstalled
- No suspension or structural repairs

Summary of related repairs provided.

Keys and access

Keys and any special access provided.

Why This Matters in a Repair Facility

A proper handoff ensures the vehicle is ready for calibration and prevents delays or invalid setups. It ensures:

- All related repairs are complete.
- The vehicle is properly assembled and aligned
- Required parts and access are available
- Calibrations can be completed correctly the first time.
- Fewer callbacks and improved efficiency.

Common Pitfalls to Avoid

- Sending loose bumper or grille parts.
- Omitting suspension or structural work from request.
- Not providing an alignment report when required.
- Leaving the vehicle on the lift.
- Failing to communicate existing warnings or DTCs.

77. Technician Quality Control Review

A second review compares procedure, setup, result and scan evidence. Independent verification catches wrong target, unit conversion, sequence and documentation errors.

KEY ELEMENTS TO VERIFY

VIN and procedure: Confirm the correct VIN and OEM procedure were used.
Measurements: Verify all measurements are recorded and within specification.
Target setup: Confirm the correct target, placement and alignment were used.
Result: Review scan results and calibration status for a successful completion.
Final scan: Confirm a final scan was completed and free of relevant DTCs.

The Quality Control Workflow

CHECK VIN/PROCEDURE. CHECK MEASUREMENTS. CHECK TARGET. CHECK RESULT. CHECK FINAL SCAN.

Check VIN/procedure

Verify the correct VIN, vehicle configuration and OEM procedure were used.

Check measurements

Confirm all key measurements were taken and within OEM specifications.

Check target

Verify the correct target, tooling and placement were used, and alignment is within specification.

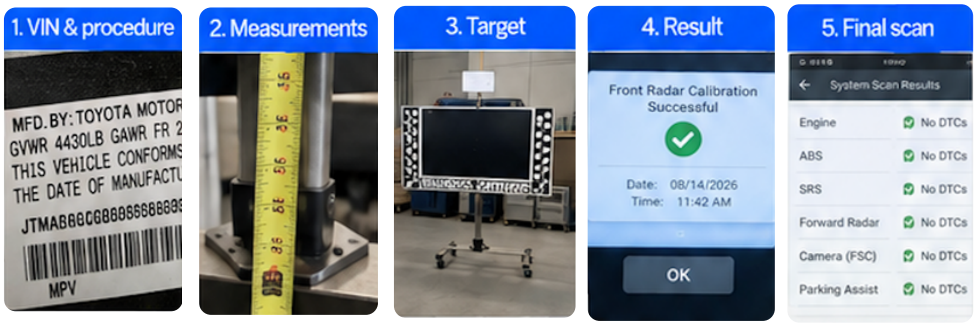
Check result

Review scan results and calibration status to confirm a successful outcome.

Check final scan

Confirm a final scan was completed and free of relevant ADAS-related DTCs.

Practical Example



VIN confirmed and OEM procedure referenced.

Target distance and height measured and within spec.

Correct target, placement and alignment verified.

Calibration completed successfully.

Results of final scan verified.

Why This Matters in a Repair Facility

Independent verification catches mistakes before the vehicle is returned to the customer.
It ensures:

- Correct target, setup and measurements.
- Successful calibration and clean scan.
- Accurate documentation.
- Fewer comebacks and warranty claims.
- Higher quality repairs and customer satisfaction.

Common Pitfalls to Avoid

- Reviewing only whether a PDF report exists.
- Letting the same assumption pass through every stage.
- Missing incorrect target, distance or height.
- Overlooking unit conversions (mm vs. inches).
- Not confirming a final scan was completed.

78. Tool & Target Quality Program

Equipment requires inventory, inspection, software updates, target control and scheduled verification. Damaged or obsolete tools create repeatable errors across many vehicles.

KEY ELEMENTS TO MANAGE

Inventory assets: Maintain a complete inventory of tools, targets and accessories.
Inspect daily: Check condition, cleanliness and functionality.
Verify schedule: Ensure required calibration and verification dates are current.
Control revisions: Use the correct software versions and target configurations.
Quarantine defects: Remove and label damaged or out-of-spec tools and targets.

The Tool/Target Check Workflow

INVENTORY ASSETS. INSPECT DAILY. VERIFY SCHEDULE. CONTROL REVISIONS. QUARANTINE DEFECTS.

Inventory assets

Maintain a complete inventory of tools, targets and accessories (e.g., target boards, tripods, calibration frames, leveling tools).

Inspect daily

Check condition. cleanliness, stability and functionality before use.

Verify schedule

Confirm calibration and verification dates for tools and targets are current and within OEM requirements.

Control revisions

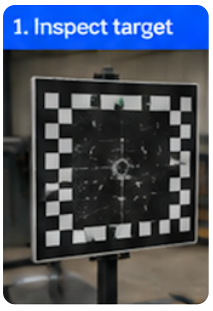
Use the correct software versions, target configurations and setup procedures.

Quarantine defects

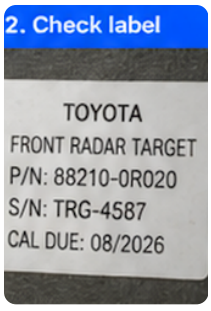
Remove damaged expired or out-of-spec tools and targets from service and label them clearly.

Practical Example

TARGET BOARD INSPECTION AND REPLACEMENT.



Inspect for damage, cleanliness and proper markings.



Verify part number, serial number and calibration/ verification due date.

Ensure the correct target and setup procedure is used.

Replace damaged or out-of-spec targets before use.

Why This Matters in a Repair Facility

A well-maintained tool and target program prevents repeatable errors, reduces comebacks and ensures accurate calibrations.
It ensures:

- Reliable and repeatable results.
- Compliance with OEM requirements.
- Fewer scan and calibration failures.
- Improved efficiency and reduced downtime.
- A safer and higher-quality repair.

Common Pitfalls to Avoid

- Using a damaged target because routines still pass.
- Skipping daily inspection of tools and targets.
- Letting calibration/verification dates expire.
- Using the wrong software version or target configuration.
- Mixing target revisions without proper labeling.

79. Facility Quality Program

The facility controls its zones, lighting, clearance and reflective objects by movement and validation records.

KEY ELEMENTS TO CONTROL

Survey facility: Evaluate bays, floor space. lighting and reflective surfaces.
Mark zones: Define and label calibration areas (e.g., radar, camera, ADAS).
Control lighting: Manage lighting conditions to meet OEM requirements.
Audit housekeeping: Remove reflective objects and maintain a clean environment.
Revalidate: Periodically confirm the space still meets requirements.

The Facility Check Workflow

SURVEY FACILITY. MARK ZONES. CONTROL LIGHTING. AUDIT HOUSEKEEPING. REVALIDATE.

Survey facility

Evaluate bay space, floor condition, ceiling height, lighting and potential reflective surfaces.

Mark zones

Define and label calibration areas (e.g., radar, camera) with clear floor markings and signage.

Control lighting

Ensure lighting levels meet OEM specifications and eliminate direct glare or reflective interference.

Audit housekeeping

Remove or cover reflective objects, keep floors clear and maintain a clean, organized workspace.

Revalidate

Periodically confirm the space, lighting and zones still meet OEM requirements.

Practical Example

CALIBRATION BAY SETUP AND MAINTENANCE

1. Survey facility	2. Mark zones	3. Control lighting	4. Audit housekeeping	5. Revalidate
Check bay size, floor levelness, ceiling height, lighting and surroundings.	Apply floor markings and signage for calibration areas and required clearances.	Verify lighting meets OEM requirements and eliminate reflective glare.	Keep the area free of mirrors, shiny surfaces and unnecessary equipment.	Perform periodic checks to confirm the environment still meets OEM specifications.

Why This Matters in a Repair Facility

A good technician cannot overcome an uncontrolled environment.
A well-maintained facility ensures:

- Accurate and repeatable calibrations.
- Compliance with OEM requirements.
- Fewer calibration failures and reschedules.
- Improved efficiency and throughput.
- A safer, more professional workspace.

Common Pitfalls to Avoid

- Moving cabinets into the radar path.
- Allowing reflective objects (mirrors, tools, glass) in the bay.
- Inconsistent or inadequate lighting.
- Relying on yesterday's bay condition.
- Not revalidating the space after changes (e.g., new equipment).

80. Comeback Investigation

A comeback is investigated from the original trigger through current symptoms without assuming the prior pass was valid or invalid.

KEY ELEMENTS TO INVESTIGATE

Interview customer: Identify the current concern, driving conditions and timeline.
Preserve evidence: Keep scan results, photos and relevant data from the vehicle.
Rescan: Perform a complete system scan to confirm current DTCs and data
Remeasure prerequisites: Verify alignment, mounting, measurements and setup meet specifications.
Correct root cause: Identify and repair the actual cause based on updated information.

The Comeback Investigation Workflow

INTERVIEW CUSTOMER. PRESERVE EVIDENCE. RESCAN. REMEASURE PREREQUISITES. CORRECT ROOT CAUSE.

Interview customer

Understand the current concern, driving conditions and when the issue returns.

Preserve evidence

Save previous and current scan results, photos and other relevant data.

Rescan

Perform a complete system scan to identify current DTCs, data and system status.

Resume prerequisites

Verify alignment, mounting, measurements and setup meet OEM specifications.

Correct root cause

Identify and repair the actual cause based on updated information. Confirm the fix with testing.

Why This Matters in a Repair Facility

Root cause may be repair movement, loose mounting, subsequent alignment, software, environment or missed damage. A thorough investigation prevents repeat comebacks and builds customer trust.

Common Pitfalls to Avoid

- Immediately recalibrating without diagnosis.
- Blaming normal system limitations without reproducing conditions.
- Not preserving original data
- Skipping prerequisite measurements.
- Assuming the original pass was valid or invalid.

Knowledge Check

Explain, in your own words, the appropriate answers to each of the questions below.

1. Explain comeback investigation in your own words. _____
2. Name one situation or condition that can make information inaccurate and explain how to prevent it. _____
3. Name one document or piece of evidence that shows work was correctly done. _____

81. Customer Explanation & Release

The customer receives a plain-language explanation of completed systems, limitations, warning indicators and any unresolved restrictions.

KEY ELEMENTS TO INVESTIGATE

Summarize work: Explain which systems were calibrated, scanned and verified
Explain limitations: Clearly communicate any system limitations or remaining conditions.
Review warnings: Discuss current and potential warning indicators and what they mean.
Disclose exceptions: Explain any unresolved restrictions or conditions.
Provide records: Give the customer a copy of the calibration documentation and any relevant reports.

The Customer Explanation Workflow

SUMMARIZE WORK. EXPLAIN LIMITS. REVIEW WARNINGS. DISCLOSE EXCEPTIONS. PROVIDE RECORDS.

Summarize work

Explain which systems were calibrated, scanned and verified in clear, non-technical language.

Explain limitations

Clearly communicate any system limitations or remaining conditions affecting driver assistance features.

Review warnings

Discuss current or potential warning indicators and what they mean for safe vehicle operation.

Disclose exceptions

Explain any unresolved restrictions, DTCs or systems that could not be calibrated, and the reason why.

Provide records

Give the customer a copy of the calibration report, scan results and any other relevant documentation.

Why This Matters in a Repair Facility

ADAS remains driver assistance. Clear, honest communication builds trust, prevents overconfidence and reduces liability.
A proper customer release ensures they understand what was done, what is limited and what to watch for.

Common Pitfalls to Avoid

- Calling calibration a guarantee against collision.
- Hiding an incomplete feature behind technical wording
- Not explaining remaining limitations or DTCs.
- Failing to provide documentation to the customer.
- Using overly technical language.

Knowledge Check

Explain, in your own words, the appropriate answers to each of the questions below.

1. Explain customer explanation and release in your own words.

2. Identify one condition that can make the information confusing to a customer.

3. Name one document or piece of evidence that proves to the customer the work was completed correctly.

Knowledge Check | Test your understanding of this section's content.

PART IV
CALIBRATION, QUALITY, & DOCUMENTATION

To view and complete the **Knowledge Check** on this section, please visit:

Link to online assessment: https://www.classmarker.com/online-test/start/?quiz=nar6ad9223af0915

Quiz Password: documentation

You may also access the online assessment through this QR code:

PART V

ADAS COMPONENTS & FUNCTIONS

Objective: ADAS names describe functions, not a single piece of hardware. One function. can depend on multiple sensors, controllers, chassis inputs, actuators, software and network messages. The following diagrams explain the physical component, the signal path, the resulting function, common collision-repair vulnerabilities, and the events that must trigger OEM research.

Executive Rules & Decision Logic

ADAS calibration is a measured restoration of a sensing system's relationship to the vehicle. It is not simply "clearing a code," and a successful routine does not prove the underlying repair, wheel alignment, ride height, bracket geometry, target placement or road-test environment was correct.

KEY TAKEAWAYS

Follow the rules: Use build data, inspection, pre-scan and OEM service information.
Use reliable sources: Research the exact VIN and repair event in OEM calibration search and procedures.
Meet prerequisites: Confirm all required conditions before aiming or calibration.
Complete and verify: Perform required routines in the correct order and confirm with post-scan and evidence.
Know when to stop: Do not continue when cirteria conditions are not met.

Universal Decision Gate

Identify exact systems

Use build data, physical inspection, pre-scan and OEM service information. Do not infer equipment solely from a trim badge or dashboard switch.

Research the exact VIN

Use the OEM calibration search and OEM procedures. Public position statements are policy-level evidence, not model-specific instructions.

Correct prerequisites

Verify structure, suspension, alignment/ thrust angle, tire pressure and size, ride height, glass and bracket installation, battery support, DiCs, cargo, fuel level and environmental conditions.

Perform required routines

Complete every required initialization, programming, static aim, dynamic learn, or combination routine in the order specified by the OEM.

Verify completion

Confirm with the OEM-defined completion message, post-scan, road test/system function check when allowed, and a complete evidence package.

Why This Matters

A vehicle involved in collision repair should be screened for ADAS even when no warning lamp is illuminated.

Many systems store faults without a dedicated lamp, and collision forces or component disconnection can create faults that appear unrelated. Honda's post-collision position, for example, rejects dashboard lamps as the sole diagnostic decision method and specifies factory-authorized scanning for its vehicles.

When to stop

Invalid conditions

Stop calibration when active power, communication, chassis, brake, steering. restraint, cameras/radar internal, or mounting-related DCs can invalidate the routine.

Out of specification

Stop when the floor, lighting, target size, distance, target height, level, or clearance does not meet the procedure.

Incorrect/incomplete setup

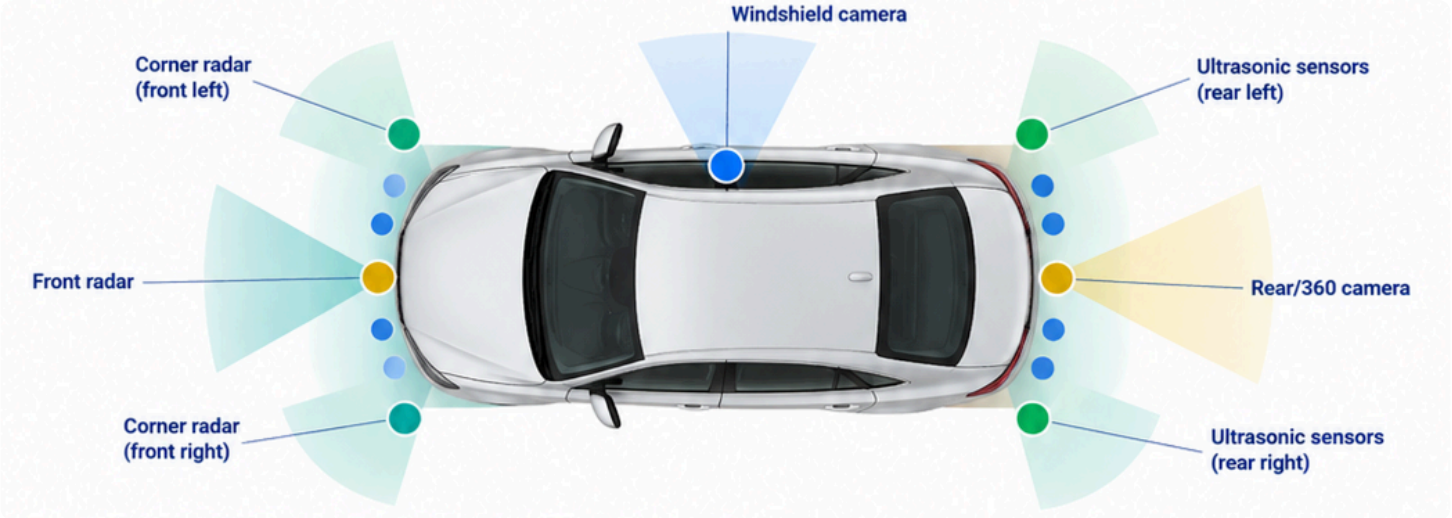
Stop when alignment or ride-height values are outside specification, a sensor/bracket is damaged, the wrong glass/emblem/ bumper material is installed, or software/programming is incomplete.

Document and do not proceed

Document the reason, identify the responsible repair, and do not report the system calibrated.

ADAS Architecture & Sensor Map

Typical sensing zones, actual location, field of view and serviceability differ by vehicle.



Typical sensing zones and functions

Sensor/Input	Common functions	Typical mounting	Repair Risks
Forward camera	Lane departure, lane centering, traffic sign, AEB fusion	Windshield/header or mirror area	Glass type, bracket position, camera removal, windshield angle, pitch/ride height
Forward radar	ACC, forward collision warning, AEB	Grille, emblem, bumper, reinforcement	Cover thickness, paint/repair material, emblem, bracket, yaw/pitch, impact
Corner radar	Blind spot, rear cross traffic, lane change aid	Rear quarter/bumper corners; sometimes front corners	Bumper repair, filler/paint, brackets, quarter damage, sensor swap
Surround cameras	360 view, park assist, automated parking	Grille, mirrors, liftgate, tailgate	Panel/mirror removal, camera replacement, ride height, target layout
Ultrasonic	Parking distance/automated parking	Bumper covers	Orientation, retainers, paint thickness, swapped locations
Chassis inputs	Vehicle path reference	SAS, yaw/acceleration, wheel speed, suspension height	Alignment, steering/suspension repair, module replacement, low voltage

ADAS Architecture & Sensor Map

Typical sensing zones, actual location, field of view and serviceability differ by vehicle.

Fusion Matters

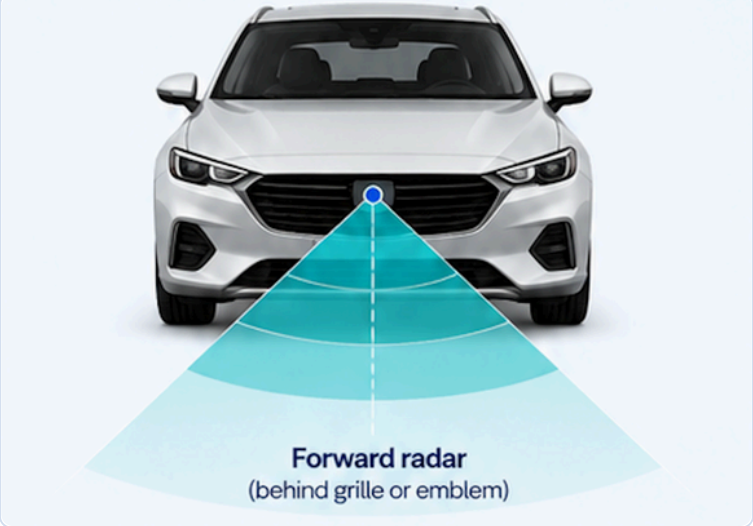
A camera and radar may independently pass calibration yet still disagree because the vehicle centerline, steering angle, wheel alignment, or ride height is wrong. Treat the vehicle geometry and network as part of the sensing system. Calibration planning must include upstream inputs, not only the sensor named on the estimate.

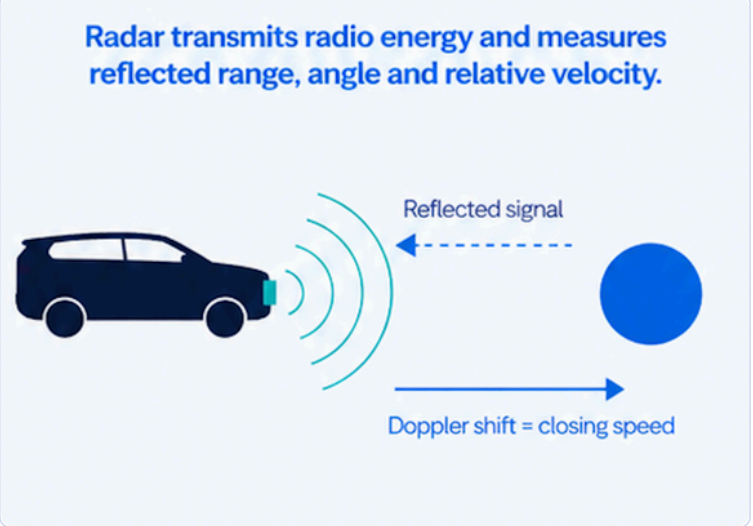
Terms technicians must separate

Term	Meaning
Initialization / reset	Establishes learned zero points, component identity, or baseline values
Programming / coding	Loads software, configuration, variant data, or component identity. It is not calibration.
Static calibration	Uses precisely placed targets, reflectors, boards, lasers, or patterns while stationary.
Dynamic calibration	Learns during a prescribed drive under required speed, road, weather, lane, and traffic conditions. Confirms.
Verification	Routine completion, DTC status, specified data values, and permissible functional checks.

Forward Radar

Radio-frequency ranging and velocity sensing.





What it is, and what it measures

A forward radar is an electronic transceiver commonly mounted behind a grille, emblem, bumper cover or dedicated radome. It emits radio-frequency energy and analyzes reflected signals. From signal timing, phase, angle and Doppler shift, the module estimates object range, lateral position and closing or separating velocity. Long-range radar prioritizes vehicles farther ahead; short- and mid-range units widen near-field coverage.

How the system performs its function

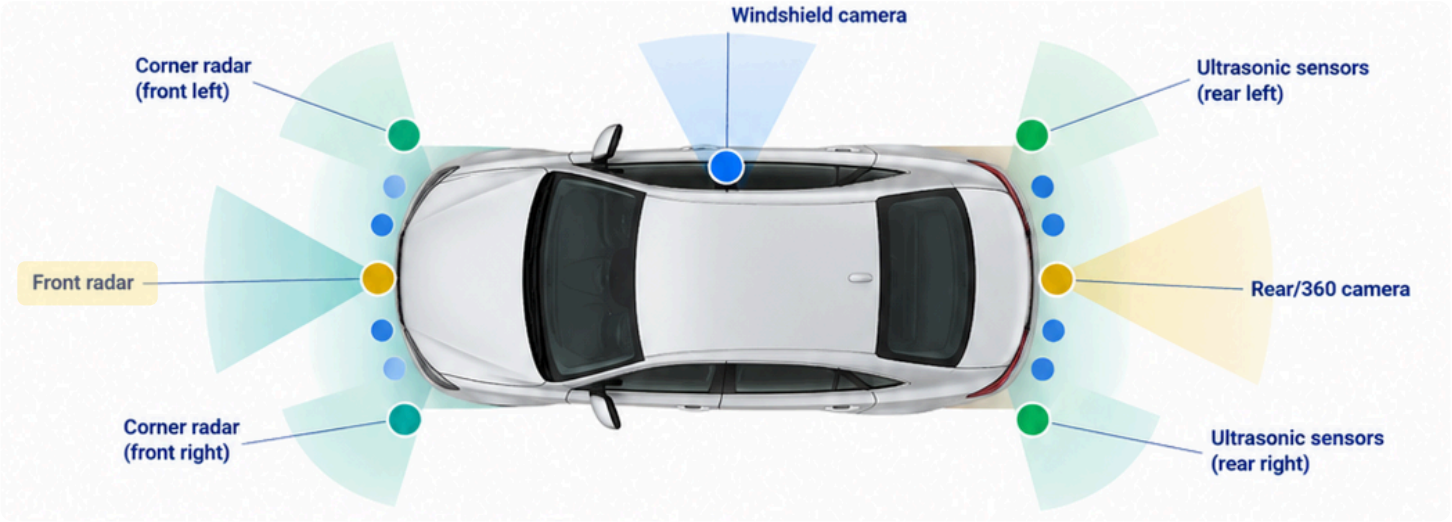
The radar builds and updates object tracks. ACC uses the lead-object track to regulate following distance; FCW/AEB uses relative speed, path prediction and time-to-collision logic. Radar performs well in darkness and can directly measure relative speed, but it does not identify objects as richly as a camera. A small sensor yaw or pitch error shifts the projected track into the wrong lane or changes ground/overhead reflections.

Typical sensing zones and functions

Input/Status	Common functions
Radar returns (targets)	Vehicle speed, yaw rate, steering angle, brake status, camera fusion, selected ACC gap
Module output	Tracked objects, range, azimuth, relative velocity, collision-risk data and requests to powertrain/brake controllers

Forward Radar

Radio-frequency ranging and velocity sensing.



Collision-repair vulnerabilities

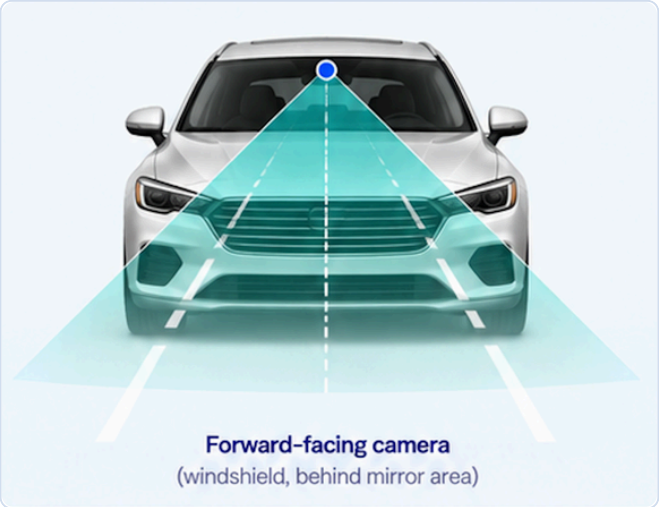
- Bent, twisted or cracked bracket; incorrect fastener stack or torque; hidden rail/reinforcement distortion.
- Radar transmission-zone changes caused by filler, plastic welding, film, excessive paint, metallic pigment, badges, wraps, snow or contamination.
- Wrong emblem, grille, bumper or absorber; sensor installed in the wrong orientation or location.
- Module coding/software mismatch, power/ground/CAN faults, or a successful aim performed before alignment/structure was corrected.

Calibration and research triggers

- Radar removal, loosening, replacement, transfer or programming.
- Front impact, bracket or mounting-structure repair, even when the radar housing looks intact.
- Bumper cover, grille, emblem, radome, absorber or reinforcement work in the beam path.
- Alignment, suspension, ride-height or structural change when listed by the exact OEM procedure.
- Radar DTC, limited-assist message, object-tracking complaint or unexplained AEB/ACC behavior.

Forward-facing Camera

Optical classification and road-geometry sensing.



What it is, and what it measures

A forward camera converts visible light into image frames. Image-processing software identifies lane markings, road edges, vehicles, pedestrians, cyclists, signs, lamps and drivable space. Mono cameras infer depth through scale, motion and learned models; stereo cameras use the offset between two lenses to derive depth more directly.

How the system performs its function

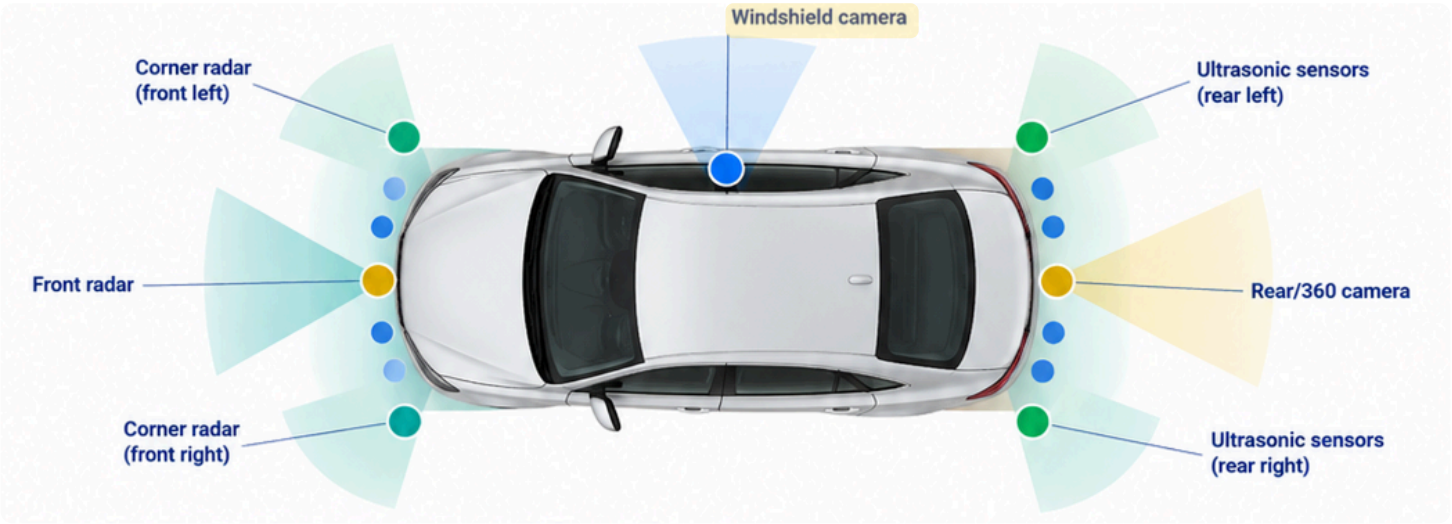
The camera estimates lane curvature, host-vehicle position within the lane, object class and sometimes distance. Lane departure and lane centering rely heavily on camera geometry. AEB and ACC may fuse camera classification with radar range/velocity.
The camera's optical axis must match the vehicle's actual forward path; windshield angle, bracket location, vehicle pitch and target geometry therefore matter.

Typical sensing zones and functions

Input/Status	Common functions
Pixels, exposure, lens temperature, steering/yaw/speed, radar objects, wiper/defogger status	Lane model, object class, traffic-sign data, departure prediction, steering support and fusion confidence

Forward-facing Camera

Optical classification and road-geometry sensing.



Collision-repair vulnerabilities

- Incorrect windshield optical properties, distortion, tint band, frit, camera window or OE option.
- Bracket shifted during glass removal, improperly bonded bracket, camera cover interference or damaged heater.
- Roof/header/A-pillar repair, camera removal, wrong fasteners, lens contamination or fingerprints.
- Incorrect ride height, tire pressure, alignment or static target placement; glare and patterned shop backgrounds.

Calibration and research triggers

- Windshield or camera/bracket replacement, R&I, adjustment or repair.
- Roof, header or A-pillar repair near the camera datum.
- Alignment, suspension, tire or ride-height changes when the procedure lists calibration.
- Camera/module programming, DTC, poor lane performance or collision above minor cosmetic damage.

Ultrasonic Parking Sensors

Short-range acoustic distance measurement.

Rear ultrasonic sensors
(typical locations on bumper)

Multiple bumper sensors form a parking envelope
(front and rear shown)

What it is, and what it measures

An ultrasonic sensor is a piezoelectric transmitter/receiver mounted through a bumper cover. It emits an inaudible sound pulse and measures the round-trip time of the echo. The controller converts.

How the system performs its function

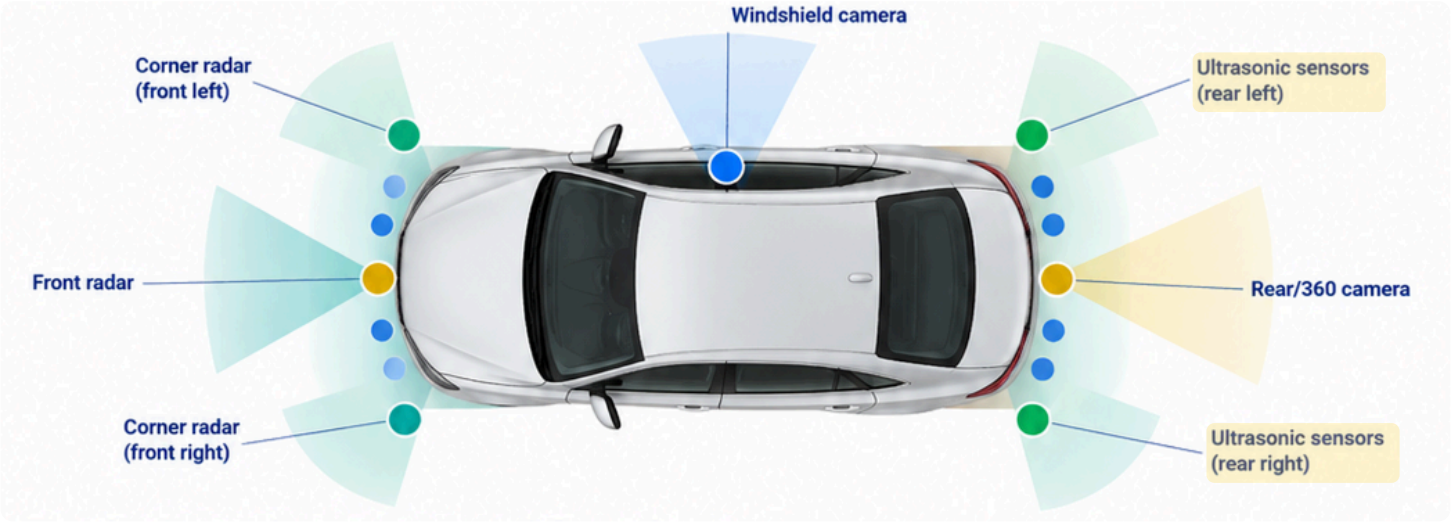
Parking-distance control compares multiple sensor returns, filters noise and reports increasing audible/visual urgency as clearance decreases. Automated parking may use the same array to measure open spaces and monitor obstacles while steering. Performance changes with object angle, softness, size, rain, ice, dirt, temperature and cross-talk between sensors.

Typical sensing zones and functions

Input/Status	Common functions
Echo time/amplitude, temperature compensation, vehicle speed/direction, steering angle, gear selection	Distance zones, park warnings, space measurement, automated-parking obstacle input and braking requests (on equipped vehicles)

Ultrasonic Parking Sensors

Short-range acoustic distance measurement.



Collision-repair vulnerabilities

- Sensor installed upside down or in the wrong numbered opening; damaged retainer or decoupling ring
- Excess paint or filler on the sensor face; painted sensor that must remain unpainted; mismatched part frequency.
- Bumper thickness/shape changed, wiring pinched, sensor flushness wrong or adjacent accessories causing reflection.

Calibration and research triggers

- Sensor or retainer replacement/transfer; bumper cover replacement or repair; refinish of sensor face where permitted
- Harness repair, park-assist DTC, false continuous tone, missed object or automated-parking complaint.
- Initialization, coding or teach-in after controller/sensor replacement when specified; many ultrasonic systems do not use a target-based aim, so follow the exact routine.

LiDAR & Time-of-Flight Sensing

Three-dimensional optical ranging.

What it is, and what it measures

Lidar emits laser pulses and measures their return time to create a three-dimensional point cloud. Scanning or solid-state designs may be mounted in the roof, grille, windshield area or fascia. Lidar can provide precise geometry and range, while cameras provide rich classification and radar provides robust relative velocity.

How the system performs its function

The system groups points into surfaces and objects, tracks movement and contributes to free-space, lane and collision models. Window cleanliness, optical transmission, mounting pose and software alignment are critical. Some systems use lidar primarily for advanced highway assistance; others integrate it with parking or automated-driving functions.

Typical sensing zones and functions

Input/Status	Common functions
Laser returns, camera/radar fusion, vehicle pose, speed, yaw, suspension/ride-height data	3-D point cloud, object geometry, free-space map, range tracks and automated-driving confidence

Collision-repair vulnerabilities

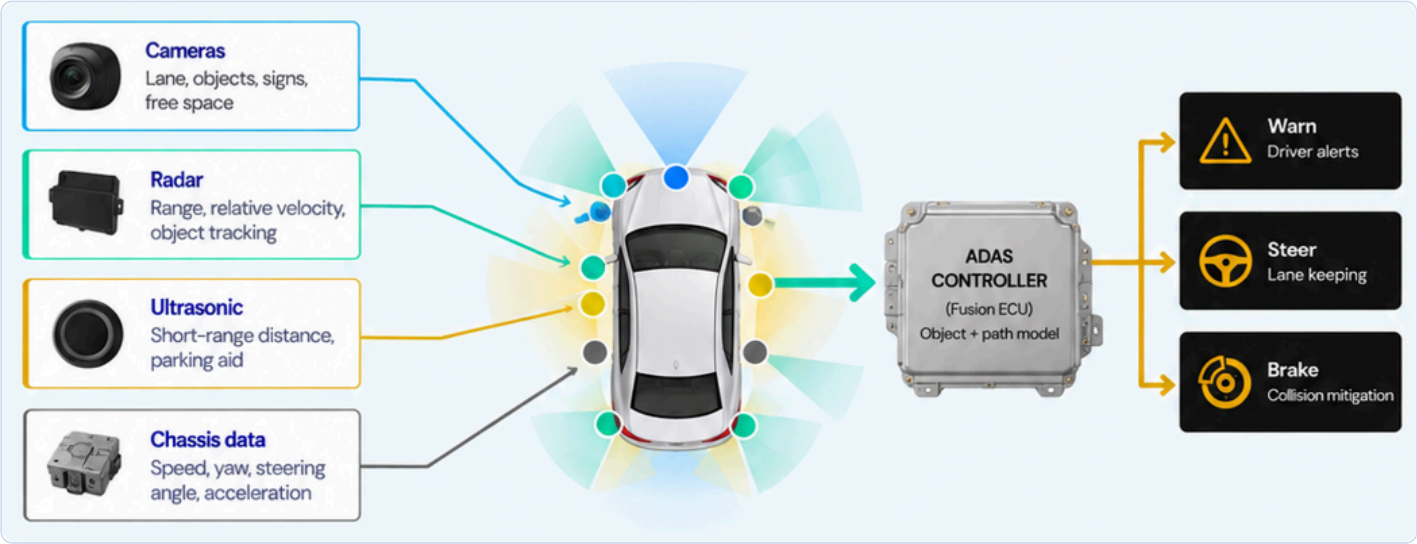
- Scratched, polished, tinted, coated, painted or contaminated optical window; incorrect cleaning method.
- Mounting-angle or body-datum change; fascia/roof/glass replacement; water intrusion or internal damage.
- Laser safety and service-mode requirements; attempting to judge operation by eye.

Calibration and research triggers

- Lidar removal/replacement/programming; repair or replacement of its window/radome or mounting panel.
- Roof, glass, fascia, structure, suspension alignment or ride-height work identified by OEM procedure
- Lidar DTC, driver-assistance unavailable message or collision in the sensing/mounting zone.

Sensor Fusion & ADAS Controller

Combining imperfect sensors into one world model.



What it is, and what it measures

Sensor fusion is software that associates camera detections, radar tracks, lidar points, ultrasonic distances and chassis data with the same physical objects and road geometry. Fusion may occur inside a camera/radar module, a dedicated ADAS domain controller or several networked ECUs.

How the system performs its function

Each sensor contributes strengths and uncertainty. Radar measures relative speed, camera identifies object type and lane context, lidar adds geometry, and chassis sensors define host-vehicle motion. The controller predicts trajectories, assigns confidence and decides whether to inform, warn, steer, reduce torque or brake. A geometrically misaligned sensor can still communicate normally while corrupting the fused model.

Typical sensing zones and functions

Input/Status	Common functions
Camera, radar, lidar and ultrasonic outputs; vehicle speed, wheel speeds, steering angle, yaw/acceleration, maps/GNSS on some systems	Object list, lane and free-space model, risk calculation, warning, steering, powertrain and brake requests

Collision-repair vulnerabilities

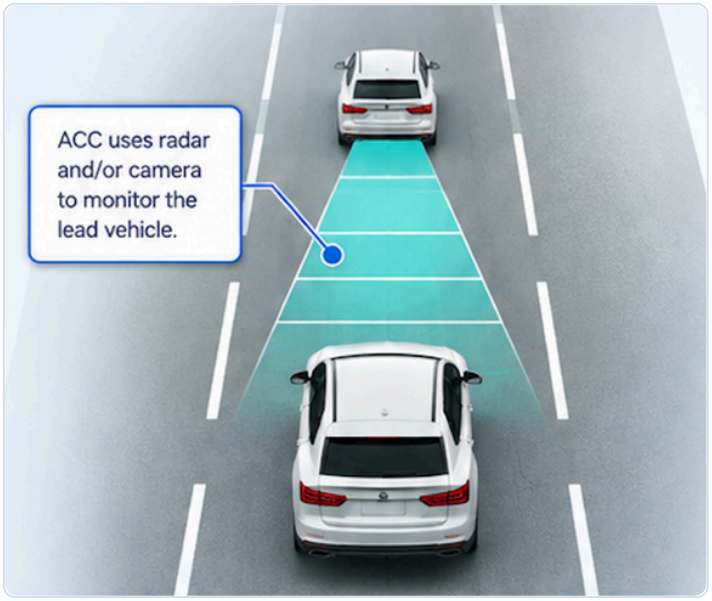
- Calibrating only the sensor named in a DTC while missing related chassis zero points or companion sensors.
- Variant coding, software version or replacement-controller configuration not completed.
- Timestamp/network errors, low voltage, intermittent communication or mismatched tire/ride-height inputs.

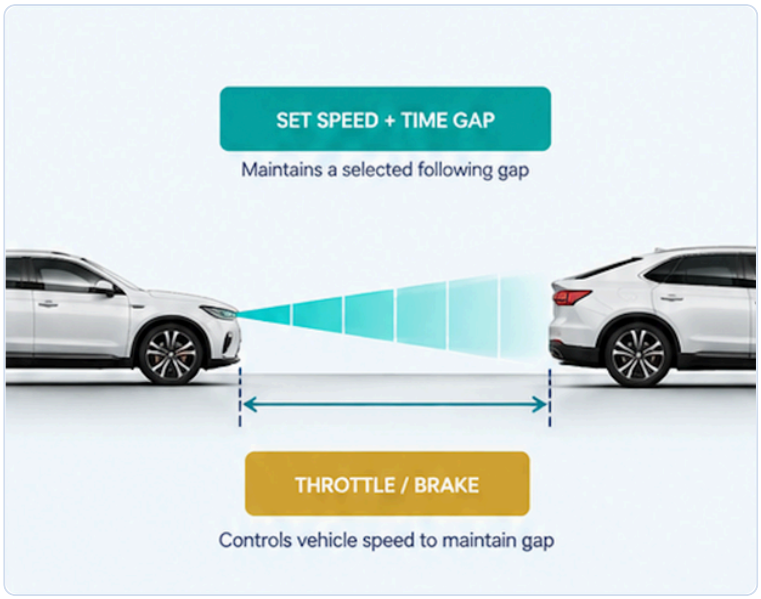
Calibration and research triggers

- Any constituent sensor replacement or calibration; ADAS/domain controller replacement/programming.
- Steering-angle, yaw/acceleration, wheel-speed, suspension-height or brake-system service when linked by procedure.
- Post-collision faults across multiple modules, implausible-data DCs or inconsistent system behavior

Adaptive Cruise Control

Longitudinal speed and gap control.





What it is, and what it measures

ACC is a function, not a sensor. It usually depends on forward radar, forward camera or both, plus vehicle speed, brake and powertrain data. The driver chooses a set speed and following-gap setting. The system then controls propulsion and braking within defined limits.

How the system performs its function

When no lead vehicle is relevant, ACC maintains set speed. When a lead object enters the predicted host path, the controller calculates a target gap, reduces torque and may apply brakes. Stop-and-go systems can control to very low speeds and sometimes restart. Curves, cut-ins, stationary objects, poor visibility and sensor uncertainty change availability and response.

Typical sensing zones and functions

Input/Status	Common functions
Forward object tracks, lane/path model; selected speed/gap; vehicle speed; SAS/yaw; brake/powertrain status	Torque request, gear/downshift request, brake request, gap display, takeover alert and system-status message

Adaptive Cruise Control

Longitudinal speed and gap control.

Collision-repair vulnerabilities

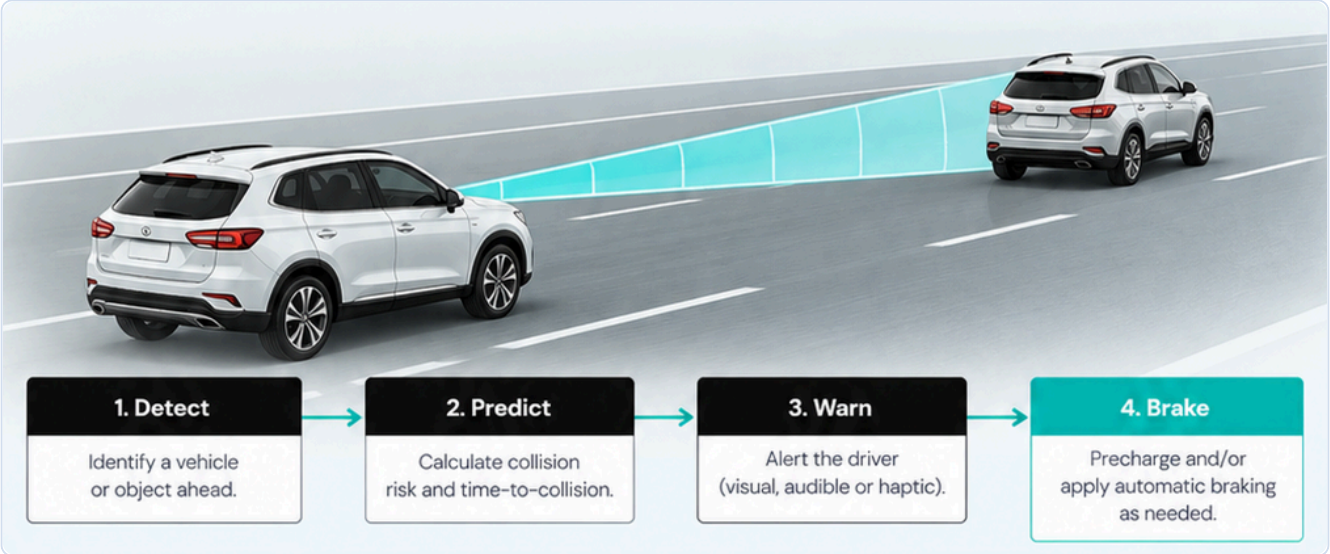
- Radar aimed toward an adjacent lane, causing wrong lead selection; camera path model shifted by windshield/bracket error.
- Brake, ABS, wheel-speed, steering or powertrain faults that inhibit ACC even after radar calibration.
- Incorrect test method: normal ACC limitations should not be confused with calibration failure, but customer complaints require OEM diagnostics.

Calibration and research triggers

- Every trigger applicable to the forward radar/camera used by the vehicle.
- Brake, ABS, wheel-speed, steering, alignment, suspension or controller work listed by OEM procedure.
- ACC unavailable/limited message, unexpected lead selection, gap-control complaint or related DTC.

Forward Collision Warning & Automatic Emergency Braking

Collision prediction, warning and automatic braking.



What it is, and what it measures

FCW and AEB are functions that estimate whether the host vehicle is likely to strike a detected object. Depending on platform, the system may use camera, radar, lidar or sensor fusion. AEB can prefill/precharge the brake system, warn the driver, apply partial braking and escalate toward maximum available braking.

How the system performs its function

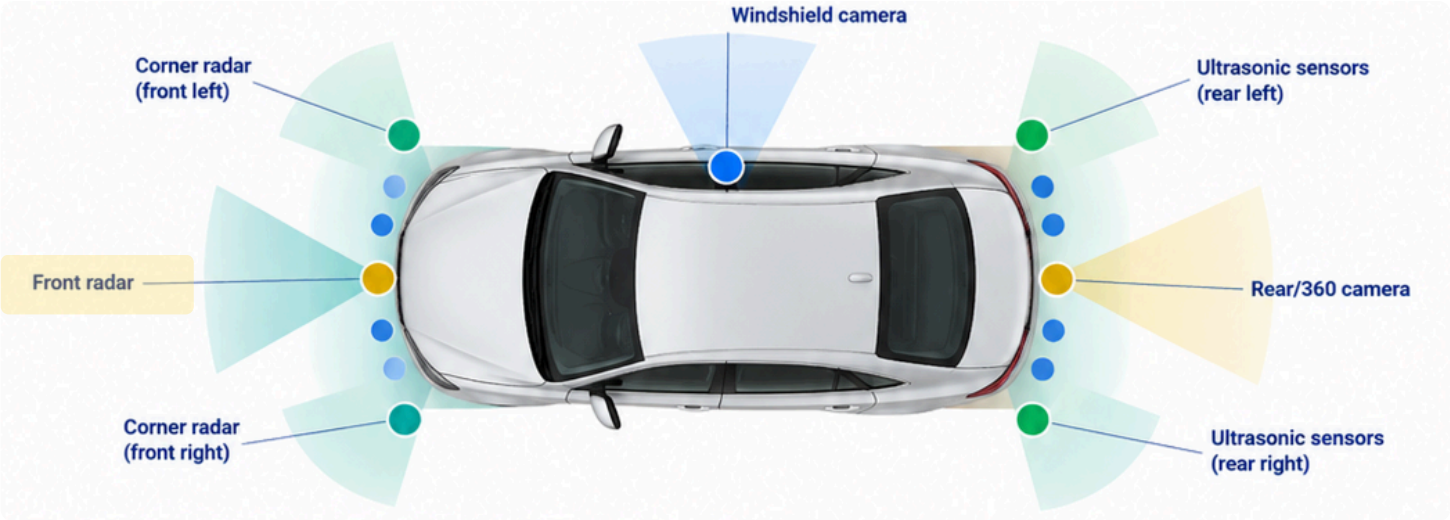
The controller predicts host and object paths, calculates collision probability and time-to-collision, then applies staged thresholds.
Thresholds vary with speed, driver input, object class, roadway context and system confidence. Calibration error can delay intervention, create nuisance warnings or associate an object with the wrong path.

Typical sensing zones and functions

Input/Status	Common functions
Fused object/path data; vehicle speed; driver throttle/brake/steering; ABS/ESC status; seat-belt/restraint data on some plattorms	Visual/audible/haptic warning, brake preparation, torque reduction, automatic braking and event/status data

Forward Collision Warning & Automatic Emergency Braking

Collision prediction, warning and automatic braking.



Collision-repair vulnerabilities

- Treating "no warning lamp" as proof of correct AEB operation.
- Road-testing by intentionally approaching an object; functional testing must use OEM-approved methods and safe equipment.
- Incomplete brake bleed/learn, ABS/SAS/yaw faults, wrong tires or geometry after collision repair.

Calibration and research triggers

- Camera/radar/lidar trigger events; brake/ABS/ESC work; wheel speed, SAS or yaw-sensor work when specified
- Collision in the sensing or braking-system area; AEB/FCW DTC or complaint; programming/software work.

Blind Spot Monitoring & Rear Cross-Traffic Alert

Side/rear object tracking.

What it is, and what it measures

Blind-spot monitoring typically uses short-range corner radar modules behind the rear bumper cover or quarter area. The radars monitor adjacent-lane zones. Rear cross-traffic alert uses the same or related sensors to detect vehicles approaching laterally while reversing.

How the system performs its function

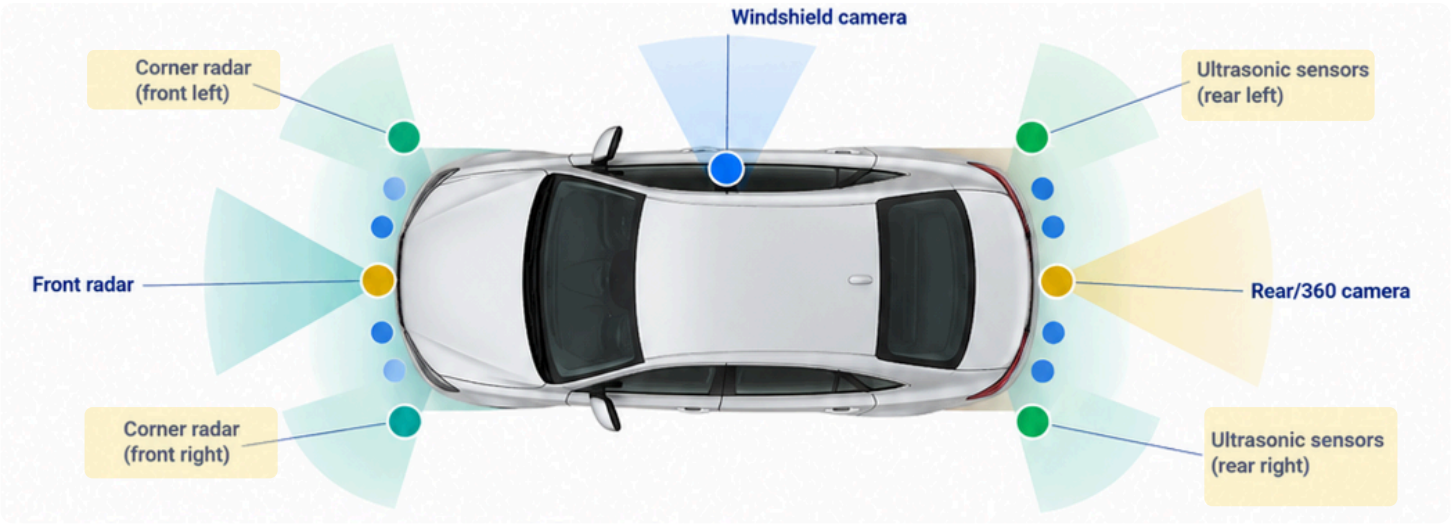
The controller tracks object range, angle and relative velocity, filters stationary roadside objects and decides whether a target is relevant. A mirror indicator warns of occupancy; a stronger alert may occur with a turn signal. Rear cross-traffic braking on some vehicles can request automatic braking.

Typical sensing zones and functions

Input/Status	Common functions
Corner-radar tracks; vehicle speed/direction; turn signal; gear; steering angle; trailer status on some vehicles	Mirror indicator, audible/visual alert, lane-change warning and rear automatic-braking request where equipped

Blind Spot Monitoring & Rear Cross-Traffic Alert

Side/rear object tracking.



Collision-repair vulnerabilities

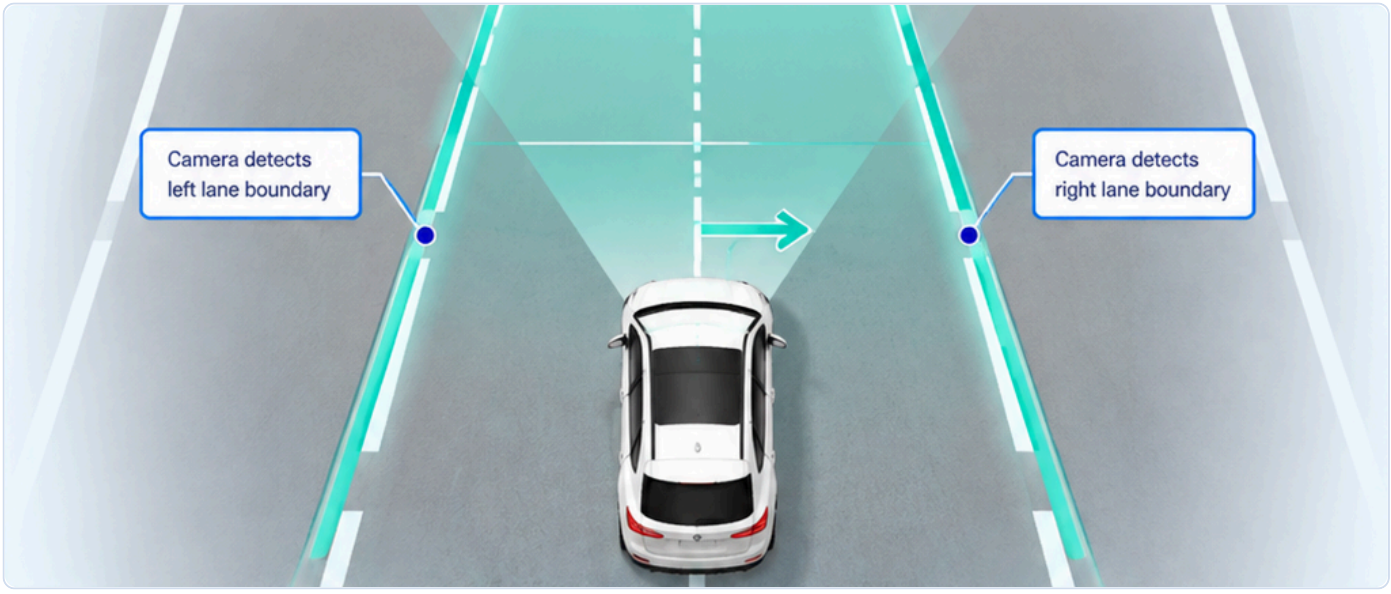
- Plastic repair, filler, weld, metallic paint, wrap, PPF or excessive coating in the radar transmission zone.
- Sensor/bracket installed on wrong side, wrong orientation, incorrect torque or hidden quarter/bumper-mount distortion.
- Trailer hitch, accessory, body kit, snow/ice or cargo affecting coverage.

Calibration and research triggers

- Rear bumper/quarter repair or replacement near the sensor; radar R&I/replacement/programming.
- Rear/side collision, sensor bracket/retainer work, transmission-zone refinishing or BSM DTC/complaint.
- Static target or dynamic learning requirements vary widely; some systems self-learn only after prerequisites are satisfied.

Lane Departure Warning & Lane Keeping

Camera-based lateral path assistance.



What it is, and what it measures

LDW detects an unintended approach to a lane boundary and warns the driver. LKA or lane-centering functions may apply steering torque to help maintain position. These systems primarily use a forward camera, with steering angle, yaw rate and vehicle speed defining the expected path.

How the system performs its function

The camera builds left and right lane models, estimates lane center and compares the vehicle's predicted path to boundaries.
The controller accounts for turn-signal use, driver steering torque, speed, curvature and confidence. Assistance is limited and does not replace driver control.

Typical sensing zones and functions

Input/Status	Common functions
Forward-camera lane model; SAS; yaw; speed; turn signal; driver torque; map data on some systems	Departure warning, steering-torque request, lane-centering status and driver-attention/takeover alert

Lane Departure Warning & Lane Keeping

Camera-based lateral path assistance.

Collision-repair vulnerabilities

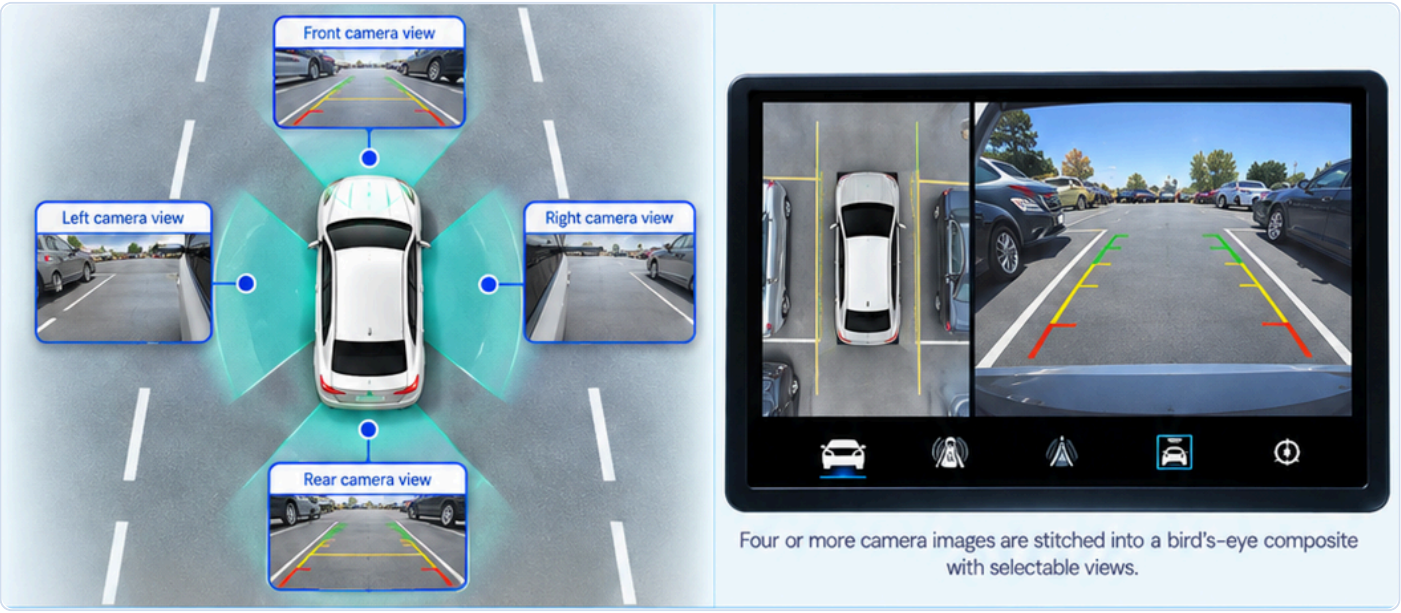
- Windshield/camera pitch error, steering wheel off center, thrust-angle error or uninitialized SAS.
- Poor lane markings, glare, weather and road curvature can create normal availability limitations; distinguish these from repair-caused faults.
- Alignment performed after camera aim can invalidate the relationship established during calibration.

Calibration and research triggers

- Windshield/camera/bracket work; alignment, steering or suspension work listed by the procedure.
- Structural/ride-height changes; SAS, EPS, yaw or camera/controller replacement; lane-system DTC or complaint.

Surround-View & Parking Camera

Geometric image stitching and close-range visibility.



Four or more camera images are stitched into a bird's-eye composite with selectable views.

What it is, and what it measures

A surround-view system uses wide-angle cameras in the grille/front, mirrors/sides and liftgate or trunk/rear. Software corrects lens distortion and projects each image onto a virtual ground plane, then blends overlapping areas into a bird's-eye view.

How the system performs its function

Calibration teaches each camera's pose relative to the vehicle and the ground. If one camera shifts, parking lines and object positions can appear bent, doubled or displaced. The composite is a driver aid and may also feed automated-parking or curb-view functions.

Typical sensing zones and functions

Input/Status	Common functions
Multiple camera images; steering angle; gear/direction; ride height; vehicle dimensions/configuration	Bird's-eye composite, selectable views, dynamic guidelines, curb/trailer view and automated-parking visual input

Collision-repair vulnerabilities

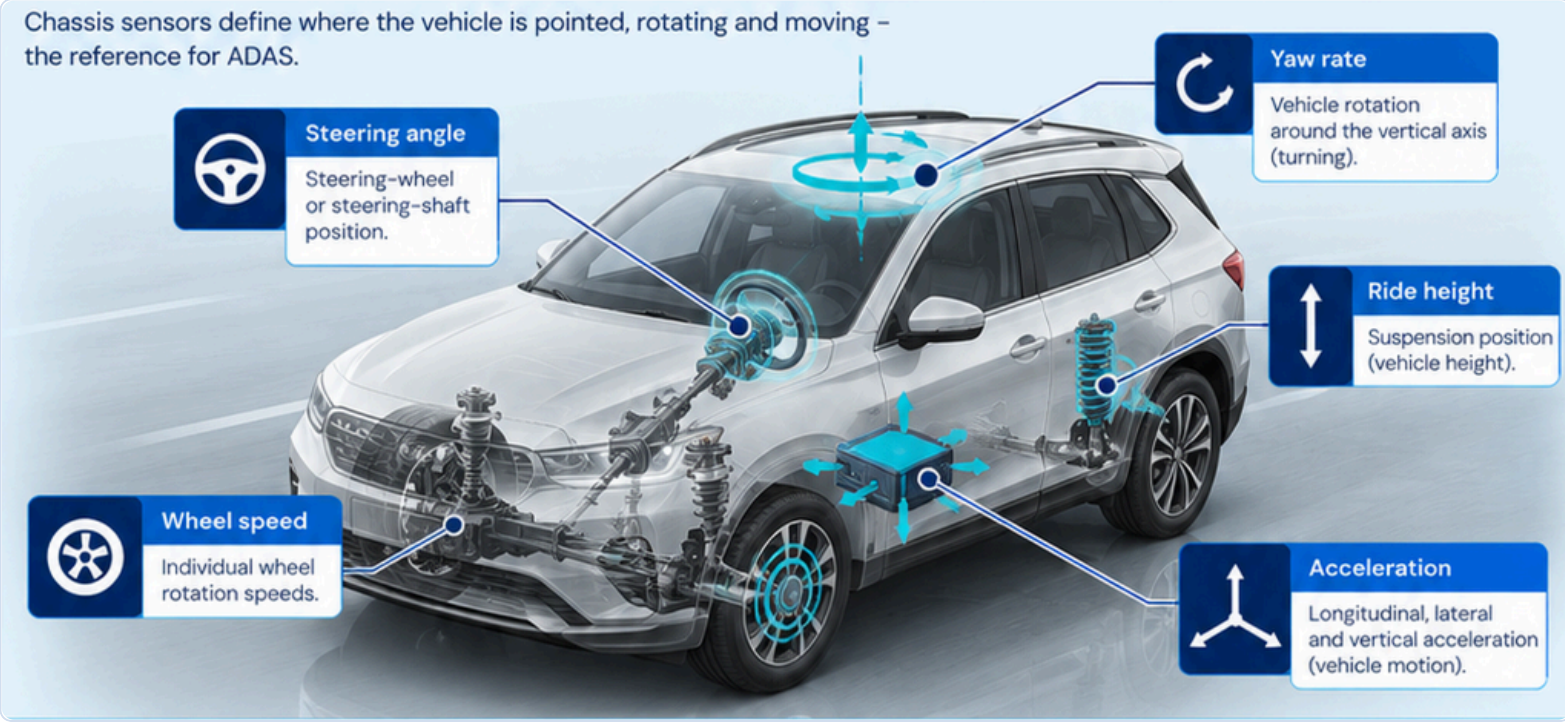
- Mirror replacement or failure to seat at its normal detent; door/hinge adjustment; camera installed in wrong clip position.
- Liftgate, tailgate, grille, bumper or panel misalignment; wrong camera part or incomplete coding.
- Target mats placed on uneven floor, wrong dimensions, glare, target wrinkles or blocked camera overlap.

Calibration and research triggers

- Any surround camera R&I/replacement/programming; mirror, door, grille, liftgate/tailgate or adjacent panel work.
- Ride-height/suspension change when specified; distorted composite, incorrect guidelines, DTC or parking-system complaint.

Steering Angle, Yaw, Acceleration, & Wheel-Speed Inputs

The vehicle-motion reference beneath ADAS.



What it is, and what it measures

The steering-angle sensor measures steering-wheel or steering-shaft position. Yaw and acceleration sensors measure rotation and vehicle acceleration; wheel-speed sensors report individual wheel rotation; ride-height sensors report suspension position.
These inputs tell ADAS where the vehicle is actually pointed and moving.

How the system performs its function

ESC, EPS, ABS and ADAS exchange these values over the network. The controller compares sensor-observed objects and lanes with the predicted host path. Zero-point errors can make a correctly aimed radar or camera appear inconsistent. Calibration sequence matters: mechanical geometry and chassis zero points generally precede final ADAS aiming when the OEM specifies both.

Steering Angle, Yaw, Acceleration, & Wheel-Speed Inputs

The vehicle-motion reference beneath ADAS.

Typical sensing zones and functions

Input/Status	Common functions
SAS; yaw/acceleration; wheel speed; ride height; alignment; brake pressure; EPS torque	Host path, stability intervention, steering support, headlamp leveling and reference data for camera/radar fusion

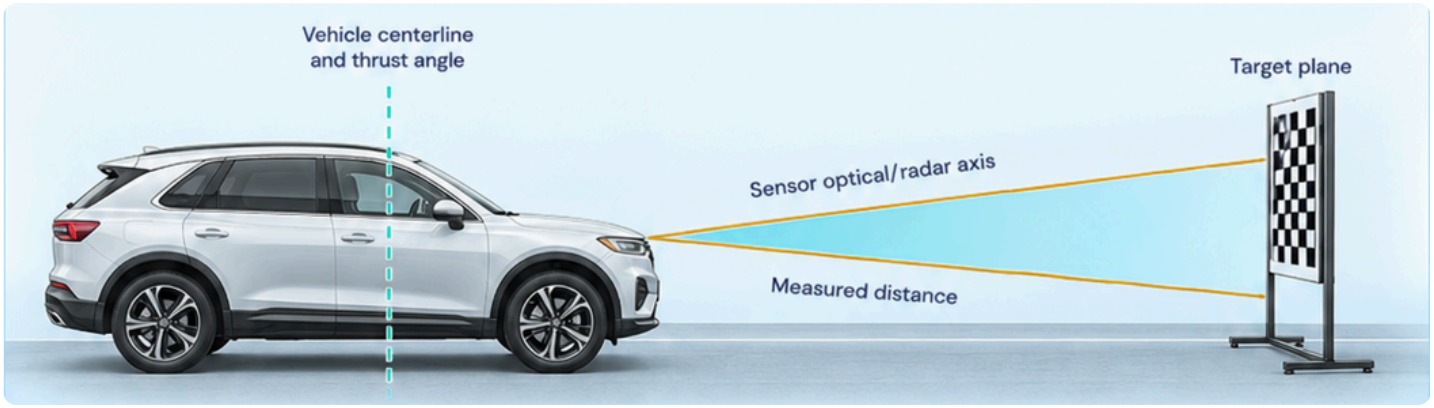
Collision-repair vulnerabilities

- Steering wheel mechanically off center, incorrect toe/thrust angle, mismatched tires or low pressure.
- SAS initialized before alignment; yaw sensor calibrated on a sloped surface; ride-height linkage bent or installed incorrectly.
- Wheel-speed signal faults, low-voltage events, subframe shift or suspension settling after repair.

Calibration and research triggers

- Alignment, steering rack/column/tie rod; subframe, knuckle, strut, control arm, spring or ride-height work
- SAS/yaw/ABS/EPS/module replacement, battery/voltage event when specified, related DTC or collision affecting geometry.

Calibration Types & Prerequisites



Static setup essentials

- **Level floor** within the equipment and OEM tolerance across the vehicle and target area. Record the level survey or facility validation.
- **Correct target part number, target face, size, height, orientation, distance, offset, and centerline.** Never scale a printed target unless expressly permitted.
- **Vehicle straight-ahead reference** established using the OEM/equipment method: wheel centers, thrust line, body centerline, or specified datum.
- **Controlled lighting** with glare, backlighting, shadows, strong patterns, reflective clutter, and moving objects managed as the procedure requires.
- **Sufficient clear floor area**. Verify no metal objects, lifts, stands, cabinets, or vehicles interfere with radar targets or reflection paths.

Dynamic setup essentials

- Road, lane markings, curvature, traffic density, speed range, distance, weather, and daylight **meet the procedure**.
- **Tires, pressure, steering center, alignment, ride height, fuel/cargo, and sensor cleanliness are correct.**
- A second person or safe scan-tool mounting is used when tool interaction would distract the driver. **Obey traffic law; never force conditions.**
- **Record** start/end time, miles, route/conditions, speed window, routine status, interruptions, and final result.

Calibration Types & Prerequisites

Vehicle Prerequisite Checklist

Category	Prerequisites before calibration	Verified
Power/network	Stable 12-V supply; battery test/support as required; complete module communication; relevant DCs diagnosed	☐
Structure/body	Restored; closures, bumper, grille, emblem, brackets, lamps, glass, mirrors and trim fully installed. Correct wheels/tires; pressures.	☐
Chassis / loading	No excess play, ride height, alignment and steering angle within specification. Fuel, cargo, spare, tools, occupants, suspension.	☐
Sensor & software	Mode and ballast match procedure.	☐
Environment	Correct part; undamaged lens/radome; clean; correct fasteners/torque; no unauthorized repair, coating or material. Required programming, coding variant configuration, resets and relearns complete.	☐
Target area	Floor/road light, temperature, weather, clearance and target specifications satisfied.	☐

Knowledge Check | Test your understanding of this section's content.

PART V
ADAS COMPONENTS & FUNCTIONS

To view and complete the **Knowledge Check** on this section, please visit:

Link to online assessment: https://www.classmarker.com/online-test/start/?quiz=m466ad926cb3f2cf

Quiz Password: adascomponents

You may also access the online assessment through this QR code:

PART VI

REFERENCES & GLOSSARY OF TERMS

Objective: Provide technicians, estimators, and repair professionals with reliable research references, OEM service-information resources, and a comprehensive glossary of ADAS terminology. This section supports accurate interpretation of technical procedures, reinforces industry-standard terminology, and helps ensure repair and calibration decisions are based on current, manufacturer-approved information.

Trigger Criteria by Repair Operation

Use this chapter's repair-planning screen. A listed event means research the exact OEM procedure; it does not convert all events into universal calibration

Windshield or camera-area glass

Reason / repairs	What to verify / possible rule
Windshield replacement; camera removal/reinstallation/ replacement; camera bracket movement or bonding: glass specification change (e.g., acoustic/IR); pillar repair; ride-height or suspension change when listed; camera DTC or poor lane-system pertormance.	Expect forward-camera aiming or learning on many vehicles. Verify OE-equivalent glass, bracket geometry, camera cover and heating/defogging components. Never assume "same camera, same bracket" avoids calibration.

Front bumper, grille, emblem, radar area

Reason / repairs	What to verify / possible rule
Radar removal, loosening, replacement, programming or bracket disturbance; bumper cover, grille or emblem replacement/repair; front impact; ride-height/alignment change; radar DTC.	Bumper removal alone is not universally a calibration trigger, but it is always a research trigger when radar or its transmission path (mounting) is marked. Inspect hidden brackets and radome materials; follow paint/film restrictions.

Rear bumper and quarter panels

Reason / repairs	What to verify / possible rule
Rear/corner radar removal; bumper cover replacement or repair; filler/refinish near sensor; trailer or bracket change; rear/side impact; quarter-panel repair; tail-lamp or adjacent structure work.	BSM radar can "see through" a bumper only when the specified material, thickness, finish and geometry are maintained. Cosmetic repair can change attenuation/reflection.

Mirrors, doors, liftgate, tailgate

Reason / repairs	What to verify / possible rule
Mirror/camera replacement or removal; door replacement/ hinge adjustment; liftgate/taiigate/camera work; surround-view camera disturbance.	Investigate side/rear camera aiming and surround-view calibration. Panel fit and mirror detent must be correct before calibration.

Trigger Criteria by Repair Operation

Use this chapter's repair-planning screen. A listed event means research the exact OEM procedure; it does not convert all events into universal calibration

Suspension, steering, alignment,

Reason / repairs	What to verify / possible rule
Alignment; tie rod, rack, knuckle, strut, control arm, subframe, axle or ride-height sensor work; wheel/tire size change; steering angle/yaw module replacement; calibration affecting geometry.	Complete mechanical repair and alignment first. Perform SAS/yaw/acceleration/ ride-height relearns and then camera/radar calibration in OEM sequence.

Structural/body measurement

Reason / repairs	What to verify / possible rule
Rail/apron/core support/strut tower/roof/A-pillar/quarter repair; pull or sectioning; rear sensor datum, unibody/frame correction.	Complete structural measurements first. Sensor brackets may be dimensionally wrong without visible damage. Document final dimensions before calibration.

Module, battery, electrical, software

Reason / repairs	What to verify / possible rule
ADAS ECU/sensor/module replacement; programming/ coding; CAN or power repair; battery disconnect/low voltage when procedure specifies; software campaign.	Programming and calibration are separate line items. Resolve U-codes, voltage faults and configuration errors before aiming

Collision/no visible sensor damage

Reason / repairs	What to verify / possible rule
Any impact above minor cosmetic damage; airbag deployment; ADAS complaint; stored DTC; measure written cosmetics-only repair that creates a sensing zone.	Use build data, scan, inspection and OEM procedure. Absence of a lamp or visible sensor damage is not evidence that calibration is unnecessary.

End-to-End Operating Procedure

1 Intake and Authorization

Capture VIN, vehicle mileage, RO/repair facility, insurer, loss type, impact area, repair stage, requested service, warning messages, keys/access and contact. Confirm authorization for diagnostic time if calibration cannot proceed.

2 Identify Equipment

Decode build information, visually locate cameras/radars, inspect options and telltales, pre-scan all modules, and compare to estimate and teardown. Photograph each relevant sensor and mounting area.

3 Retrieve procedures

Obtain current service information for the VIN. Search every operation: calibration/aiming, initialization, programming, prerequisites, target/tool, sequence, completion criteria and post-repair inspection. Save the procedure title/version/date or licensed reference.

4 Damage Analysis

Trace the impact path and every disturbed component. Inspect brackets, fasteners, radomes, emblems, glass, harnesses, grounds, structure, alignment and chassis inputs. Add missed operations before reassembly.

5 Precondition

Finish body/mechanical/glass work. Correct codes and communication faults. Verify battery, tires, pressure, loading, fuel, ride height, alignment and steering center. Install all trim and covers required by the procedure.

6 Set up

Validate floor/environment. Select exact targets. Establish vehicle centerline/thrust reference. Measure target distance, lateral offset, height, yaw/pitch and target level. Photograph measurements and complete setup.

7 Execute

Run initialization/programming/calibration in the required order. Save scan-tool screen or report. For a dynamic routine, follow all road conditions and document interruptions or inability to meet criteria.

8 Verify

Confirm OEM completion status. Cycle ignition/sleep network as required. Clear permitted DTCs and perform full post-scan. Recheck alignment/data parameters and perform specified road/function check.

9 Document and release

Issue itemized invoice and evidence package. State calibrated systems and methods, procedure reference, equipment, prerequisites, results, exceptions, technician and date. Never use "calibration complete" if any required routine is incomplete.

Calibration Documentation & Insurer File

A defensible file explains why the operation was required, how prerequisites were established, what was performed, and how success was verified. A scan report alone does not show target placement; a target photo alone does not show OEM completion.

Minimum evidence package

Evidence	Minimum content	Included
Repair trigger	Estimate/repair plan line, impact area, part R&I/replacement, glass/suspension/structural/module event, or DTC. Procedure or reference.	☐
OEM authority / pre-scan prerequisites	Position statement cite, model/year or VIN applicability, access data, and licensed source reference.	☐
Setup photos	Battery, tire/pressure, fuel/load, ride height, alignment, structures and relevant DTC resolution.	☐
Calibration result / post-scan verification	Vehicle, sensor, targets, distance/height/centerline measurements, level, laser or unobstructed environment. System/routine, vehicle/dynamic/combined, target/tool, start/end, success message/report and technician.	☐
Invoice narrative	All modules, remaining DTC disposition, required drive/function check, warnings and limitations. Separate diagnostic programming, calibration setup/measurement and sublet lines where supported; avoid duplicate or misleading descriptions.	☐

Suggested Insurer Narrative

Required operation: **[system]** calibration was required following **[specific repair trigger]**. The vehicle was identified as equipped with **[feature/sensor]**. Current OEM service information **[procedure title/reference]** specifies **[calibration/aiming/initialization]** after **[event]**. Prerequisites were verified, the procedure completed using **[method/equipment]**, and completion was confirmed by **[result]** with post-scan attached.

Do Not Overstate

- **Do not** call a generic health check an OEM scan when the OEM requires factory software or a specific tool.
- **Do not** state "passed" when a routine merely launched, when dynamic conditions were not met, or when relevant DTCs remain.
- **Do not** cite a position statement as though it contains model-specific target measurements.
- **Do not** make blanket claims that every bumper removal, battery disconnect, wheel alignment or windshield replacement requires the same operation.

Diagnostics & Failure Analysis

Failure / symptom	Likely cause categories	Required response
Routine will not start	Active DTC; low voltage; wrong variant; incomplete programming; prerequisite data out of range.	Stabilize power, scan all modules, verify configuration and live data; diagnose before retry.
Static aim fails repeatedly	Wrong target; wrong distance/height; bad centerline; floor slope; glare; bracket/sensor damage; alignment/ride height.	Rebuild setup from procedure; inspect geometry and mounting; do not "adjust around" damaged parts.
Dynamic learn never completes	Road/lane/weather/speed conditions; traffic; dirty sensor; alignment/SAS; software; intermittent DTC.	Record conditions and data; select compliant route; correct vehicle faults; follow OEM fallback.
Calibration succeeds but warning remains	Additional routine; drive cycle; unrelated DTC; programming/coding; network issue; sensor internal fault.	Post-scan and follow DTC diagnostics; verify all linked modules and required relearns.
System complaint after release	Incorrect prerequisite or setup; changed ride height/alignment; loose bracket; wrong glass/bumper/emblem; damage or misread.	Stop relying on previous pass report; reinspect, measure, retrieve procedures, diagnose and recalibrate if directed.
No communication	Power/ground; connector/harness; CAN fault; gateway/security access; module failure.	Restore communication first. Calibration cannot validate a module that cannot communicate.

Quality-control escalation

- **Escalate** any result that conflicts with physical measurements, scan data, road behavior, or OEM completion criteria.
- **Quarantine** incorrect or suspect targets and verify target revision/part number.
- **Track repeat failures** by VIN, OEM, system, technician, equipment, bay and cause; correct systemic issues.
- For a questionable sensor or transmission-zone repair, **replace/repair per OEM instructions** rather than using calibration as a test of acceptability.

OEM-by-OEM Reference Profiles

AMERICAN HONDA MOTOR CO.

Brand(s)	Common ADAS Hardware
Acura, Honda	Forward camera, front radar, BSM/corner radar, LaneWatch/side cameras, surround/rear camera

Trigger screen: Windshield/camera work; radar or bracket work; bumper/grille/emblem disturbance; collision/body repair; suspension/ alignment/ride-height changes; module replacement; DTC or complaint. Honda's public post-collision position calls for preliminary and post-repair scans and identifies calibration/aiming needs; use i-HDS and model-specific service information where directed.
Authoritative access: https://techinfo.honda.com/

TOYOTA MOTOR NORTH AMERICA

Brand(s)	Common ADAS Hardware
Toyota, Lexus (legacy Scion)	Forward recognition camera, millimeter-wave radar, BSM radar, panoramic-view cameras, clearance sonar

Trigger screen: Windshield/camera; front radar/grille/emblem; rear bumper/BSM zone; wheel alignment or steering/suspension work when procedure lists it; sensor/ECU replacement; body-height change; collision/DTC. Toyota procedures frequently separate beam-axis adjustment, camera optical-axis learning and utility resets.

Authoritative access: https://techinfo.toyota.com/

GENERAL MOTORS

Brand(s)	Common ADAS Hardware
Buick, Cadillac, Chevrolet, GMC	Front-view camera, long/short-range radar, side blind-zone modules, surround cameras, park sensors, Super Cruise hardware where equipped

Trigger screen: Windshield/camera; fascia/grille/emblem/radar/bracket; BSM-zone bumper/quarter work; alignment/SAS; suspension/ body structure; camera/radar/module replacement; programming; collision or DTC. Follow SPS programming and calibration sequence in Service Information.

Authoritative access: https://www.acdelcotds.com/

OEM-by-OEM Reference Profiles

FORD MOTOR COMPANY

Brand(s)	Common ADAS Hardware
Ford, Lincoln	IPMA camera, cruise-control radar, side obstacle detection, 360 cameras, park sensors, BlueCruise hardware where equipped

Trigger screen: Windshield/IPMA; radar or bracket; bumper/grille/emblem; rear bumper/side radar; mirror/camera; alignment/steering/ suspension; module replacement/programming; structural repair; DTC. Ford routines may include programmable module installation plus camera/radar alignment.

Authoritative access: https://www.motorcraftservice.com/

STELLANTIS

Brand(s)	Common ADAS Hardware
Alfa Romeo, Chrysler, Dodge, Fiat, Jeep, Maserati, Ram	Forward-facing camera, ACC radar, blind-spot modules, surround cameras, park sensors

Trigger screen: Windshield/camera; front fascia/radar/bracket; rear fascia/BSM; wheel alignment/SAS; suspension/ride height; module replacement/coding; collision or DTC. Verify wiTECH routine, target and sequence; brand/platform differences are material.

Authoritative access: https://www.techauthority.com/

NISSAN MOTOR CO.

Brand(s)	Common ADAS Hardware
Nissan, INFINITI	ICC/ADAS radar, lane camera, BSW radars, Around View Monitor cameras, sonar

Trigger screen: Radar replacement/removal or post-repair events; windshield/camera; front/rear bumper and bracket work; AVM camera/panel/mirror work; alignment/SAS; collision/DTC. Nissan/INFINITI publicly prohibit salvage/aftermarket/secondary-market ADAS components and require approved processes/equipment.

Authoritative access: https://www.nissan-techinfo.com/

OEM-by-OEM Reference Profiles

SUBARU OF AMERICA

Brand(s)	Common ADAS Hardware
Subaru	Stereo EyeSight cameras, rear radar detection, mono camera on some platforms, surround/rear cameras

Trigger screen: Windshield/EyeSight cameras: glass bracket/header/roof work; wheel alignment, tire/ride-height or suspension changes when listed; rear bumper/radar work; sensor/module replacement; collision/DTC. EyeSight setup is highly sensitive to glass, camera mounting, vehicle level and target environment.

Authoritative access: https://techinfo.subaru.com/

HYUNDAI MOTOR GROUP

Brand(s)	Common ADAS Hardware
Genesis, Hyundai, Kia	Forward camera, front radar, corner radar, surround cameras, parking sensors

Trigger screen: Windshield/camera; bumper/radar/grille/emblem; rear bumper/corner radar; mirror/door/camera; alignment/SAS/suspension; sensor/module replacement; structural repair; DTC. Use each brand portal and exact platform procedure; similar components do not guarantee identical targets or routines.

Authoritative access: https://www.hyundaitechinfo.com/

MAZDA NORTH AMERICAN OPERATIONS

Brand(s)	Common ADAS Hardware
Mazda	Forward sensing camera, front radar, BSM radar, 360 cameras, ultrasonic sensors

Trigger screen: Windshield/camera; radar/bracket/grille/emblem; bumper/BSM zone; alignment/SAS/suspension; camera/mirror/body work; sensor/module replacement; collision/DTC. Confirm M-MDS requirements and all preliminary inspections.

Authoritative access: https://www.mazdaserviceinfo.com/

OEM-by-OEM Reference Profiles

MITSUBISHI MOTORS

Brand(s)	Common ADAS Hardware
Mitsubishi	FCM camera/radar, BSW/LCA radar, multi-around-view cameras, sonar

Trigger screen: Windshield/camera; front radar/bumper/grille; rear bumper/BSW; camera/panel work; alignment/SAS; sensor/module replacement; collision/DTC. Use model-specific MUT/service procedures.

Authoritative access: https://www.mitsubishi-techinfo.com/

PORSCHE CARS NORTH AMERICA

Brand(s)	Common ADAS Hardware
Porsche	Front camera, ACC radar, lane-change radars, surround cameras, park assist

Trigger screen: Windshield/camera; bumper/radar; alignment/chassis; rear bumper/side radar; camera/panel work; component replacement/programming; collision/DTC.
Use PIWIS and model-specific workshop information.

Authoritative access: https://www.oemlstop.com/porsche

TESLA

Brand(s)	Common ADAS Hardware
Tesla	Forward/repeater/B-pillar/rear cameras, front radar on applicable generations, ultrasonic sensors on applicable vehicles

Trigger screen: Camera, windshield, fascia/bumper, fender/B-pillar/trunk work; suspension/alignment/ride-height; camera/sensor replacement; structural repair; software or DTC. Tesla may use camera calibration drives and service-mode procedures; hardware differs substantially by generation.

Authoritative access: https://service.tesla.com/

OEM-by-OEM Reference Profiles

RIVIAN AUTOMOTIVE

Brand(s)	Common ADAS Hardware
Rivian	Front/side/rear cameras, front/corner radars, ultrasonic sensors

Trigger screen: Glass/camera; fascia/radar; bumper/corner radar; mirror/body/camera; suspension/alignment/ride-height; module replacement; structural repair; DTC. Follow Rivian repair guidelines, service access and calibration workflow; do not generalize Tesla or legacy-OEM methods.

Authoritative access: https://rivian.com/support

LUCID MOTORS

Brand(s)	Common ADAS Hardware
Lucid	DreamDrive cameras, radar, lidar where equipped, surround cameras, ultrasonic sensors

Trigger screen: Glass/camera; fascia/radar/lidar; bumper/corner sensors; body/mirror/camera; suspension/alignment; module replacement/software; collision/DTC. Use Lucid Techinfo and required approved equipment; lidar adds unique contamination, optical and aiming considerations.

Authoritative access: https://techinfo.lucidmotors.com/

GEELY / VOLVO-RELATED U.S. BRANDS

Brand(s)	Common ADAS Hardware
Volvo, Polestar	Platform-dependent cameras, radars, lidar, surround cameras and parking sensors

Trigger screen: Apply brand-specific service information. Never assume shared corporate ownership means identical trigger criteria, targets, tools or completion standards.

Authoritative access: https://www.oem1stop.com/

Position Statement & Repair Information Directory

REMINDER: Public statements are screening references, not substitutes for the OEM repair procedure. Some portals require subscriptions or authorized credentials. Verify links before publication.

OEM / group	Public statements / index	OEM repair information
All participating OEMs	OEM1Stop Statements	OEM1Stop Repair Directory
I-CAR	I-CAR Resources	ADAS repair information
Honda / Acura	OEM1Stop directory	Honda TechInfo
Toyota / Lexus	OEM1Stop directory	Toyota TIS
GM Brands	OEM1Stop directory	ACDelco TDS
Ford / Lincoln	OEM1Stop directory	MotorCraft Service
Stellantis brands	OEM1Stop directory	TechAuthority
Nissan / Infiniti	OEM1Stop directory	Nissan TechInfo
Subaru	OEM1Stop directory	Subaru TechInfo
Hyundai/Genesis	OEM1Stop directory	Hyundai TechInfo
Kia	OEM1Stop directory	Kia TechInfo
Mazda	I-CAR RTS	Mazda Service Info
Mitsubishi	I-CAR RTS	Mitsubishi TechInfo
Volkswagen	OEM1Stop directory	VW erWIN
Audi	OEM1Stop directory	Audi erWIN
BMW / MINI / Rolls-Royce	OEM1Stop directory	BMW TechInfo
Mercedes / smart	OEM1Stop directory	XENTRY
Volvo	OEM1Stop directory	Volvo TechInfo
Jaguar / Land Rover	I-CAR RTS	TOPIx
Porsche	OEM1Stop directory	Porsche workshop info
Tesla	OEM1Stop directory	Tesla Service
Rivian	I-CAR RTS	Rivian Support
Lucid	OEM1Stop directory	Lucid TechInfo

ADAS Repair-Planning Worksheet

Complete before scheduling calibration. Use the current VIN-specific OEM procedure.

FIELD	RECORD / REQUIRED INFORMATION
Vehicle	Year / make / model / VIN / mileage:
Loss and repair	Impact area; estimate lines; R&I / replaced / repaired parts:
Systems identified	Forward camera / radar / BSM / surround / park / night vision / lidar / other:
Research	OEM portal; procedure title/number; revision/date; position statement:
Triggers	Glass / bumper / grille / emblem / radar / camera / mirror / alignment / suspension / structure / module / DTC:
Required operations	Scan / inspect / program / initialize / static / dynamic / combined / verify:
Prerequisites	Battery / tires / fuel-load / ride height / alignment / structure / DTs / environment:
Result	Completed routines; report IDs; remaining faults/exceptions; technician/date:

Static Setup Record

MEASUREMENT	SPECIFIED	ACTUAL / EVIDENCE
Floor level / slope		
Vehicle centerline / thrust reference		
Target part number / face		
Sensor/vehicle-to-target distance		
Target height		
Lateral offset / center		
Target yaw/pitch/level		
Vehicle ride height / loading Lighting / clearance / temperature		

Pre-Scan checklist

- [ ] Connect scan tool and perform a complete vehicle scan to capture all diagnostic trouble codes (DTCs) present.
- [ ] Save and document pre-scan results (include date, time and VIN).
- [ ] Check for module communications issues (no response, low voltage, or network faults).
- [ ] Identify and document inactive, pending and history codes related to ADAS systems.
- [ ] Verify battery voltage is stable (typically 12.4V+ with ignition on, or use a battery maintainer).
- [ ] Print or save the full pre-scan report to the repair file and reference during the repair process.

Glossary of Terms

TERM	DEFINITION
ACC	Adaptive cruise control
AEB	Automatic emergency braking
ADAS	Advanced driver assistance systems
AVM / SVM	Around-view or surround-view monitor
BSM / BSW	Blind-spot monitor or blind-spot warning
DTC	Diagnostic trouble code
FCW	Forward collision warning
FOV	Field of view
Initialization	Reset or learning of a zero point/baseline
LCA	Lane-change assist
LDW / LKA	Lane-departure warning / lane-keeping assist
OEM	Original equipment manufacturer
Radome	Material/cover through which radar energy passes
R&I	Remove and install
SAS	Steering-angle sensor
Static calibration	Stationary procedure using measured targets/reflectors. Road-ready environment required per OEM
Dynamic calibration	Driven learning procedure under prescribed conditions. Rotation of the vehicle and real-world driving required per OEM
Yaw / pitch / roll	Rotation about vertical / lateral / longitudinal axes

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Source-use note: Public information support policy, discovery, and repair planning. Subscription portals may require credentials and contain copyrighted procedures that should be retained according to license terms. Always check for superseded documents and VIN/model-year applicability.

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